
Core Probability

— *EYNTKA* Series —

NATHANAEL CHWOJKO-SRAWLEY

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Contents

Contents	3
Mathematical Preliminaries	i
0.1 Combinatorial Arguments	i
0.1.1 Counting Functions	i
0.1.2 The Binomial and Multinomial Theorems	ii
0.2 Asymptotic Analysis	iv
0.3 Calculus and Inequalities	vi
0.3.1 Taylor Series Expansions	vi
0.3.2 Inequalities and Integrals	vi
0.4 Change of Variables	vii
0.4.1 The Univariate Case	vii
0.4.2 The Multivariate Case (Jacobian)	xi
1 Introduction	1
1.1 Motivation and Basic Terminology	1
1.2 Basic Definitions	3
1.3 Discrete Distributions	14
1.3.1 Countable Expected Value	15
1.3.2 Example of Discrete Distributions	22
1.4 Poisson Properties	28
1.4.1 Poisson Approximation of the Binomial	29

1.5	Expected Value of Functions of Random Variables	32
1.5.1	Independence Revisited	33
1.5.2	Expected Value of Functions of RV's	38
1.6	Inequality With Expected Value	44
1.7	Computation trick: Linearity of Expectation	46
1.7.1	Functions on Finite sets	47
1.7.2	Longest Increasing Subsequence	48
1.7.3	Cliques in Erdős-Rényi graph	50
2	Second Moment: Variation and Correlation	53
2.1	Variance and Covariance	53
2.2	Law of Large Numbers	57
2.3	Bernstein Polynomials	58
2.4	Cliques on Erdős-Rényi Graph	60
3	Exponential Inequalities	63
3.1	Markov and Hoeffding's Inequality	63
3.2	Johnson-Lindenstrauss lemma	69
3.3	Hoeffding-Chernoff Inequality	72
3.4	Azuma's Inequality	78
3.5	Applications of Azuma's Inequality	80
3.5.1	Chromatic Number of Erdős Rényi graph	80
3.5.2	Balls and boxes	81
3.5.3	Max-cut Sparse Graph	82
3.5.4	Hamming Cube	84
4	Gaussian Distribution	87
4.1	Gaussian Distribution on $\mathbb{R}$	89
4.2	Gaussian on $\mathbb{R}^n$	97
4.2.1	General Gaussian distribution on $\mathbb{R}^n$	101
4.3	Central Limit Theorem	103
4.4	Distributions Related to Gaussian	110
5	Markov Chains	115
A	Glossary	117
A.1	Distributions	117

Index	120
--------------	------------

Bibliography	121
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This course introduces students to various topics in mathematical probability theory. Topics include basic concepts (such as probability, random variables, expectations, conditional probability) from a mathematical point of view, examples of distributions and stochastic processes and their properties, convergence results (such as the law of large numbers, central limit theorem, random series, etc.); various inequalities, and examples of applications of probabilistic ideas beyond statistics (for example, in geometry and computer science).

So far, my intuition on probability is that:

1. Define probability rigorously (we define a probability space Ω with the possible events, and the probability of the sum of the probabilities of all the events must be 1)
2. capture events quantitatively (ex. $\{\text{heads, tails}\}$, we can make a function $f : \{\text{heads, tails}\} \rightarrow \{0, 1\}$ that gives a quantitative values). This function will represent an arbitrary choice, and so we call it a *random variable*
3. We will take a lot of time on how to analyze the average value this function will return, or more precisely the *expected value* (given you know the probability of all the events, and you have quantified what each even means with your function, what value do you “expect” to happen)
4. Then, we might ask how many of the events land close to the expected value, that will define the *variance*.
5. Switching gears a bit, we will want to learn how to simplify really complicated problems by learning how to *approximate* them. Since we are working over $\mathbb{R}$, we have an order, and so bounding will usually mean that we can bound value from above and bellow, and so we will define many *inequalities*. For this, I highly recommend you look at the proofs, since they contain many bounding tricks.

Mathematical Preliminaries

In this section, we outline a collection of results from combinatorics, asymptotic analysis, and calculus that will be referenced throughout the book. While some results may appear only once, others represent recurring properties essential for the flow of subsequent arguments.

0.0.1 Note The results gathered here are presented without extensive context. Familiarity with these tools will significantly aid in following the probabilistic derivations in later chapters.

0.1 Combinatorial Arguments

0.1.1 Counting Functions

A fundamental problem in discrete probability is determining the number of functions mapping a finite set A to a finite set B .

Proposition 0.1.1: Counting Functions

Let A and B be finite sets. The number of functions $f : A \rightarrow B$ is given by:

$$|B|^{|A|}$$

Proof:

Write B^A for the set of functions $A \rightarrow B$, and induct on $|A|$, with the convention $z^0 = 1$ for every integer $z \geq 0$. If $|A| = 0$, then $A = \emptyset$ and the empty function is the only element of B^A , so $|B^A| = 1 = |B|^0$.

Suppose the count holds for sets with n elements, and let $|A| = n + 1$. Fix $a \in A$ and put $A' = A \setminus \{a\}$, so $|A'| = n$. The map $B^A \rightarrow B^{A'} \times B$ sending f to the pair $(f|_{A'}, f(a))$ is injective, since $A = A' \cup \{a\}$ and so f is determined by its restriction to A' and its value at a . It is

surjective, since a pair (g, b) is the image of the function equal to g on A' and to b at a . The set $B^{A'} \times B$ is the disjoint union, over $b \in B$, of the $|B|$ sets $B^{A'} \times \{b\}$, each with $|B^{A'}|$ elements. Hence, by the induction hypothesis,

$$|B^A| = |B^{A'}| \cdot |B| = |B|^n \cdot |B| = |B|^{n+1}$$

completing the induction.

0.1.2 Remark: The Base and the Exponent To recover the order of the exponentiation, consider the sets term-by-term:

- If $|A| = 1$ and $|B| = 2$, the single element of A has 2 choices. Total: 2^1 .
- If $|A| = 2$ and $|B| = 1$, both elements of A must map to the single element of B . Total: 1^2 .

The number of choices ($|B|$) is the base, and the number of independent decisions ($|A|$) is the exponent.

0.1.2 The Binomial and Multinomial Theorems

We frequently encounter powers of sums, such as $(x_1 + x_2 + \cdots + x_m)^n$. The expansion of these terms is governed by the Multinomial Theorem.

Theorem 0.1.3: The Binomial Theorem

For any $x, y \in \mathbb{R}$ and $n \in \mathbb{N}$:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$

Proof:

Here $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ for $0 \leq k \leq n$, and $z^0 = 1$ for every $z \in \mathbb{R}$. We induct on n . For $n = 0$ both sides equal 1. Suppose the identity holds for some $n \geq 0$. For $1 \leq k \leq n$,

$$\binom{n}{k-1} + \binom{n}{k} = \frac{n!}{(k-1)!(n-k+1)!} + \frac{n!}{k!(n-k)!} = \frac{(n+1)!}{k!(n+1-k)!} = \binom{n+1}{k}$$

Multiplying the identity for n by $x + y$, and shifting the index of the first resulting sum by one,

$$\begin{aligned} (x + y)^{n+1} &= \sum_{k=0}^n \binom{n}{k} x^{k+1} y^{n-k} + \sum_{k=0}^n \binom{n}{k} x^k y^{n+1-k} \\ &= \sum_{k=1}^{n+1} \binom{n}{k-1} x^k y^{n+1-k} + \sum_{k=0}^n \binom{n}{k} x^k y^{n+1-k} \end{aligned}$$

In the last expression the term x^{n+1} has coefficient $\binom{n}{n} = 1 = \binom{n+1}{n+1}$, the term y^{n+1} has coefficient $\binom{n}{0} = 1 = \binom{n+1}{0}$, and for $1 \leq k \leq n$ the term $x^k y^{n+1-k}$ has coefficient $\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$. So $(x + y)^{n+1} = \sum_{k=0}^{n+1} \binom{n+1}{k} x^k y^{n+1-k}$, completing the induction.

Definition 0.1.4: Multinomial Coefficient

For $n \in \mathbb{N}$ and non-negative integers $k_1, \dots, k_m$ with $k_1 + \dots + k_m = n$, the *multinomial coefficient* is

$$\binom{n}{k_1, \dots, k_m} = \frac{n!}{k_1! k_2! \dots k_m!}$$

The generalization to m variables is as follows:

Theorem 0.1.5: The Multinomial Theorem

Let $x_1, \dots, x_m \in \mathbb{R}$ and $n \in \mathbb{N}$. Then:

$$(x_1 + \dots + x_m)^n = \sum_{k_1 + \dots + k_m = n} \binom{n}{k_1, \dots, k_m} x_1^{k_1} \dots x_m^{k_m}$$

where the summation extends over all non-negative integer solutions to $\sum k_i = n$.

Proof:

We induct on m . For $m = 1$ the only index is $k_1 = n$, and both sides equal x_1^n , since $\binom{n}{n} = n!/n! = 1$ by definition 0.1.4. Let $m \geq 2$ and suppose the theorem holds for $m - 1$ variables and every exponent. Put $s = x_1 + \dots + x_{m-1}$. Theorem 0.1.3, applied to s and x_m , followed by the induction hypothesis applied to each power s^j , gives

$$(s + x_m)^n = \sum_{j=0}^n \binom{n}{j} s^j x_m^{n-j} = \sum_{j=0}^n \sum_{k_1 + \dots + k_{m-1} = j} \binom{n}{j} \binom{j}{k_1, \dots, k_{m-1}} x_1^{k_1} \dots x_{m-1}^{k_{m-1}} x_m^{n-j}$$

With $k_m = n - j$, the binomial coefficient of theorem 0.1.3 and definition 0.1.4 give

$$\binom{n}{j} \binom{j}{k_1, \dots, k_{m-1}} = \frac{n!}{j! k_m!} \cdot \frac{j!}{k_1! \dots k_{m-1}!} = \binom{n}{k_1, \dots, k_m}$$

As j runs over $0, \dots, n$ and $(k_1, \dots, k_{m-1})$ over the non-negative integer solutions of $k_1 + \dots + k_{m-1} = j$, the tuple $(k_1, \dots, k_{m-1}, n - j)$ runs exactly once over the non-negative integer solutions of $k_1 + \dots + k_m = n$, because $j = n - k_m$ is determined by the tuple. Hence the double sum is the sum in the statement, completing the induction.

0.1.6 Remark: The Multinomial Coefficient Counts Position Assignments Consider the product $\underbrace{(x_1 + \dots + x_m) \dots (x_1 + \dots + x_m)}_{n \text{ times}}$.

Forming a term $x_1^{k_1} \dots x_m^{k_m}$ requires choosing x_1 from k_1 of the parentheses, x_2 from k_2 of them, and so on. The multinomial coefficient counts the number of ways to partition the n positions into groups of sizes $k_1, \dots, k_m$.

0.2 Asymptotic Analysis

When analyzing limiting distributions, the exact values of constants often matter less than their growth rates.

Theorem 0.2.1: Stirling's Formula

For large n , the factorial function is approximated by:

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

meaning that the ratio of the two sides tends to 1 as $n \rightarrow \infty$.

Proof:

Step 1: a bound on a logarithm. For $0 < x < 1$,

$$2x < \log \frac{1+x}{1-x} < 2x + \frac{2x^3}{3(1-x^2)}$$

On $[0, 1)$ put $u(x) = \log(1+x) - \log(1-x) - 2x$ and $v(x) = 2x + \frac{2x^3}{3(1-x^2)} - \log(1+x) + \log(1-x)$. By [2, The Derivative of the Logarithm], [2, The Chain Rule] and [2, The Rules of Differentiation], both are differentiable, with

$$u'(x) = \frac{1}{1+x} + \frac{1}{1-x} - 2 = \frac{2x^2}{1-x^2}$$

and, since the derivative of $x^3/(1-x^2)$ is $(3x^2 - x^4)/(1-x^2)^2$,

$$v'(x) = 2 - \frac{2}{1-x^2} + \frac{2(3x^2 - x^4)}{3(1-x^2)^2} = \frac{-6x^2 + 6x^4 + 6x^2 - 2x^4}{3(1-x^2)^2} = \frac{4x^4}{3(1-x^2)^2}$$

Both derivatives are positive on $(0, 1)$, and $u(0) = v(0) = 0$. For $0 < x < 1$, [2, The Mean Value Theorem] on $[0, x]$ gives points $c, c' \in (0, x)$ with $u(x) = u'(c)x > 0$ and $v(x) = v'(c')x > 0$, which is the claim.

Step 2: $n!/(n^{n+1/2}e^{-n})$ converges to a positive limit. For $n \geq 1$ put $d_n = \log n! - (n + \frac{1}{2}) \log n + n$. Since $\log(n+1)! = \log n! + \log(n+1)$,

$$d_n - d_{n+1} = (n + \frac{1}{2}) (\log(n+1) - \log n) - 1 = (n + \frac{1}{2}) \log \frac{n+1}{n} - 1$$

With $x = 1/(2n+1)$ one has $(1+x)/(1-x) = (n+1)/n$ and $n + \frac{1}{2} = 1/(2x)$, so Step 1 gives

$$0 < d_n - d_{n+1} < \frac{x^2}{3(1-x^2)} = \frac{1}{3((2n+1)^2 - 1)} = \frac{1}{12n(n+1)} = \frac{1}{12n} - \frac{1}{12(n+1)}$$

Hence (d_n) is decreasing and $(d_n - \frac{1}{12n})$ is increasing, so $d_n > d_n - \frac{1}{12n} \geq d_1 - \frac{1}{12} = d$ for every n . By [2, Bounded Monotone Sequences Converge], $d_n \rightarrow d$ for some $d \in \mathbb{R}$. The exponential is continuous, being differentiable ([2, The Exponential Is Its Own Derivative], [2, Differentiability Implies Continuity]), so

$$r_n := \frac{n!}{n^{n+1/2}e^{-n}} = e^{d_n} \rightarrow C := e^d > 0$$

Step 3: the Wallis integrals. For $k \geq 0$ put $W_k = \int_0^{\pi/2} \sin^k t \, dt$. Then $W_0 = \pi/2$, and $W_1 = \cos 0 - \cos(\pi/2) = 1$ by [2, The Fundamental Theorem of Calculus, Second Form] with antiderivative $-\cos$ ([2, The Derivatives of the Cosine and the Sine]), using $\cos(\pi/2) = 0$ from [2, The Number π]. For $k \geq 2$, apply [2, Integration by Parts] with $F = \sin^{k-1}$, whose derivative is $(k-1)\sin^{k-2}\cos$, and $G = -\cos$, whose derivative is $\sin$. The boundary term FG vanishes at 0 because $\sin 0 = 0$ and at $\pi/2$ because $\cos(\pi/2) = 0$, so, using $\cos^2 = 1 - \sin^2$ ([2, The Pythagorean Identity]),

$$W_k = (k-1) \int_0^{\pi/2} \sin^{k-2} t \cos^2 t \, dt = (k-1)(W_{k-2} - W_k)$$

that is, $W_k = \frac{k-1}{k}W_{k-2}$. Induction on n then gives

$$W_{2n} = \frac{(2n)!}{4^n(n!)^2} \cdot \frac{\pi}{2} \quad \text{and} \quad W_{2n+1} = \frac{4^n(n!)^2}{(2n+1)!}$$

because the right-hand sides equal W_0 and W_1 at $n = 0$, and passing from $n-1$ to n multiplies them by $\frac{2n-1}{2n}$ and by $\frac{2n}{2n+1}$ respectively, as the recursion does.

On $[0, \pi/2]$ the cosine is nonnegative ([2, The Number π]), so the sine, whose derivative it is, is increasing there by [2, Monotonicity From the Sign of the Derivative]. Hence $0 = \sin 0 \leq \sin t \leq 1$ for $t \in [0, \pi/2]$, the upper bound by [2, The Pythagorean Identity]. Then $\sin^{k+1} t \leq \sin^k t$ on $[0, \pi/2]$, and [2, Order Bounds and Additivity over a Subdivision Point] gives $W_{k+1} \leq W_k$. So $W_{2n+1} \leq W_{2n} \leq W_{2n-1}$ for $n \geq 1$, and dividing by $W_{2n+1} > 0$,

$$1 \leq \frac{W_{2n}}{W_{2n+1}} \leq \frac{W_{2n-1}}{W_{2n+1}} = \frac{2n+1}{2n}$$

By [2, The Squeeze Theorem], $W_{2n}/W_{2n+1} \rightarrow 1$.

Step 4: the constant. By Step 3, and then by writing $(2n)! = r_{2n}(2n)^{2n+1/2}e^{-2n}$ and $n! = r_n n^{n+1/2}e^{-n}$ with r_n from Step 2,

$$\frac{W_{2n}}{W_{2n+1}} = \frac{\pi}{2} \cdot \frac{(2n+1)((2n)!)^2}{16^n(n!)^4} = \pi \cdot \frac{2n+1}{n} \cdot \frac{r_{2n}^2}{r_n^4}$$

As $n \rightarrow \infty$, $(2n+1)/n \rightarrow 2$, $r_n \rightarrow C$, and $r_{2n} \rightarrow C$ by [2, Subsequences of a Sequence with a Limit], so by [2, The Algebra of Limits] the right-hand side tends to $2\pi/C^2$. By Step 3 the left-hand side tends to 1, so $C^2 = 2\pi$, and $C = \sqrt{2\pi}$ because $C > 0$. Finally $\sqrt{2\pi n}(n/e)^n = \sqrt{2\pi} n^{n+1/2}e^{-n}$, so

$$\frac{n!}{\sqrt{2\pi n}(n/e)^n} = \frac{r_n}{\sqrt{2\pi}} \rightarrow 1$$

as claimed.

Definition 0.2.2: Big-O Notation

We say that $f(x) = O(g(x))$ as $x \rightarrow \infty$ if there exist constants $C > 0$ and $x_0 \in \mathbb{R}$ such that:

$$|f(x)| \leq C|g(x)| \quad \text{for all } x \geq x_0$$

0.3 Calculus and Inequalities

0.3.1 Taylor Series Expansions

Taylor approximations are ubiquitous in deriving probability bounds. The following expansions around $x = 0$ are standard:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots \quad (|x| < 1)$$

These expansions provide the useful inequality $1+x \leq e^x$ for all $x \in \mathbb{R}$.

0.3.2 Inequalities and Integrals

Theorem 0.3.1: Cauchy-Schwarz Inequality

For real numbers $x_1, \dots, x_n$ and $y_1, \dots, y_n$:

$$\left(\sum_{i=1}^n x_i y_i \right)^2 \leq \left(\sum_{i=1}^n x_i^2 \right) \left(\sum_{i=1}^n y_i^2 \right)$$

For integrals, let $J \subseteq \mathbb{R}$ be an interval and let $f, g: J \rightarrow \mathbb{R}$ be square-integrable, meaning that f and g are Riemann integrable on every closed bounded subinterval of J and that $\int_J f(x)^2 dx$ and $\int_J g(x)^2 dx$ converge, as improper integrals at each endpoint of J that J does not contain. Then $\int_J f(x)g(x) dx$ converges and

$$\left(\int_J f(x)g(x) dx \right)^2 \leq \left(\int_J f(x)^2 dx \right) \left(\int_J g(x)^2 dx \right)$$

Proof:

Both inequalities have the form $B^2 \leq AC$ for three real numbers with $A \geq 0$ and

$$0 \leq \lambda^2 A - 2\lambda B + C \quad \text{for every } \lambda \in \mathbb{R}$$

and this condition implies $B^2 \leq AC$. If $A > 0$, the choice $\lambda = B/A$ gives $0 \leq C - B^2/A$, that is, $B^2 \leq AC$. If $A = 0$ and $B \neq 0$, the choice $\lambda = (C+1)/(2B)$ gives $\lambda^2 A - 2\lambda B + C = -1 < 0$, which is impossible, so $B = 0$ and $B^2 = 0 = AC$.

For the sums, put $A = \sum_i x_i^2$, which is nonnegative, $B = \sum_i x_i y_i$ and $C = \sum_i y_i^2$. For every $\lambda \in \mathbb{R}$,

$$0 \leq \sum_{i=1}^n (\lambda x_i - y_i)^2 = \lambda^2 A - 2\lambda B + C$$

by expanding each square.

For the integrals, if J is a single point all three integrals are 0. Otherwise let $c < d$ be the endpoints of J , possibly $c = -\infty$ or $d = +\infty$, and fix m with $c < m < d$. For a function h that

is Riemann integrable on the closed bounded subintervals of J , the integral $\int_J h$ is $\int_c^m h + \int_m^d h$, where each term is an ordinary integral if its endpoint lies in J and otherwise an improper integral in the sense of [2, Improper Integral]. The tests of [2, Convergence Tests for Improper Integrals] are stated at $+\infty$ and at a finite left endpoint. They hold at $-\infty$ and at a finite right endpoint as well, by reflection. A partition $p = x_0 < \cdots < x_k = q$ of $[p, q]$ corresponds to the partition $-x_k < \cdots < -x_0$ of $[-q, -p]$, and the suprema and infima of $u \mapsto h(-u)$ on the subintervals of the second are those of h on the corresponding subintervals of the first. So the Darboux sums agree, and by [2, The Integrability Criterion], $h \in \mathcal{R}[p, q]$ exactly when $u \mapsto h(-u)$ lies in $\mathcal{R}[-q, -p]$, with the same integral. A limit as an endpoint is approached becomes the limit as the reflected endpoint is approached, so an improper integral of h at $-\infty$ or at a finite right endpoint is an improper integral of $u \mapsto h(-u)$ at $+\infty$ or at a finite left endpoint.

On each closed bounded $[s, t] \subseteq J$, the functions f^2 , g^2 and fg are Riemann integrable by [2, Products, Absolute Values and Pointwise Extrema], and $|fg| \leq \frac{1}{2}(f^2 + g^2)$ because $(|f| - |g|)^2 \geq 0$. At each endpoint excluded from J , the convergence tests, in the reflected form where needed, applied with the convergent integral of $\frac{1}{2}(f^2 + g^2)$, show that the integral of $|fg|$ converges, and then that the integral of fg converges. So $A = \int_J f^2$, $B = \int_J fg$ and $C = \int_J g^2$ are real numbers.

Choose sequences (s_k) and (t_k) in J with $s_k < m < t_k$, $s_k \rightarrow c$ and $t_k \rightarrow d$, constant at an endpoint that lies in J . For each $h \in \{f^2, fg, g^2\}$, $\int_{s_k}^{t_k} h = \int_{s_k}^m h + \int_m^{t_k} h$ by [2, Order Bounds and Additivity over a Subdivision Point], and this tends to $\int_J h$. Indeed, given $\epsilon > 0$, the definition of each improper integral, through [2, One-Sided Limits] at a finite endpoint and [2, Limits at Infinity] at an infinite one, gives a set of points near the endpoint on which the partial integral lies within ϵ of its limit, and s_k and t_k lie in these sets for all large k . For $\lambda \in \mathbb{R}$ the function $(\lambda f - g)^2 = \lambda^2 f^2 - 2\lambda fg + g^2$ is nonnegative, and so is f^2 , so [2, Linearity of the Integral] and [2, Order Bounds and Additivity over a Subdivision Point] give

$$0 \leq \lambda^2 \int_{s_k}^{t_k} f^2 - 2\lambda \int_{s_k}^{t_k} fg + \int_{s_k}^{t_k} g^2 \quad \text{and} \quad 0 \leq \int_{s_k}^{t_k} f^2$$

By [2, Limits Preserve Weak Inequalities], $0 \leq \lambda^2 A - 2\lambda B + C$ and $A \geq 0$. In both cases the first paragraph gives $B^2 \leq AC$, as claimed.

0.4 Change of Variables

A core technique in probability theory is transforming random variables. The mathematical engine behind these transformations is the Change of Variables formula for integrals.

0.4.1 The Univariate Case

For functions of a single variable, this is equivalent to the u -substitution method from calculus, but formulated for integrals over domains rather than directed intervals.

Theorem 0.4.1: Change of Variables (1D)

Let $\varphi : U \rightarrow V$ be a continuously differentiable bijection between open intervals $U, V \subseteq \mathbb{R}$. If $f : V \rightarrow \mathbb{R}$ is an integrable function, meaning that f is Riemann integrable on every closed bounded subinterval of V and the improper integral $\int_V f(y) dy$ converges, then the improper integral on the right below converges and:

$$\int_V f(y) dy = \int_U f(\varphi(x)) |\varphi'(x)| dx$$

Proof:

Write $U = (a, b)$ and $V = (c, d)$ with $-\infty \leq a < b \leq +\infty$ and $-\infty \leq c < d \leq +\infty$. For a point $m \in V$, the improper integral $\int_V f$ converges when $\int_c^m f$ and $\int_m^d f$ converge in the sense of [2, Improper Integral], and it is their sum, which does not depend on m by [2, Order Bounds and Additivity over a Subdivision Point]. The integral over U is defined with a point of U in the role of m .

Step 1: φ is strictly monotone. On the set $T = \{(x, y) \in U \times U : x < y\}$ put $D(x, y) = \varphi(y) - \varphi(x)$, which is never 0 because φ is injective. If $D(p) > 0 > D(q)$ for some $p, q \in T$, then the points $p + \lambda(q - p)$ with $\lambda \in [0, 1]$ lie in T , because U is an interval and the inequality $x < y$ is preserved by averaging, so $\lambda \mapsto D(p + \lambda(q - p))$ is a continuous function on $[0, 1]$ that changes sign, and [2, The Intermediate Value Theorem] gives a zero of D , which is impossible. So D has one sign on T , that is, φ is strictly increasing or strictly decreasing.

Step 2: the formula on a closed bounded interval. Let $a < s < t < b$, let J be the closed interval with endpoints $\varphi(s)$ and $\varphi(t)$, and put $g = f \circ \varphi$ and $w = |\varphi'|$ on $[s, t]$. Then $gw \in \mathcal{R}[s, t]$ and

$$\int_s^t f(\varphi(x)) |\varphi'(x)| dx = \int_J f(y) dy$$

By Step 1 and [2, The Continuous Image of a Closed Bounded Interval], φ maps $[s, t]$ bijectively onto $J \subseteq V$. So f is integrable on J , hence bounded there, say $|f| \leq K$, and then $|g| \leq K$. The function w is continuous on $[s, t]$, hence uniformly continuous by [2, Continuity on a Closed Bounded Interval Is Uniform]. Let $\epsilon > 0$. Choose $\delta > 0$ with $|w(x) - w(x')| < \epsilon$ whenever $|x - x'| < \delta$, and, by [2, The Integrability Criterion], a partition Q_0 of J with $\bar{S}(Q_0, f) - \underline{S}(Q_0, f) < \epsilon$, where $\bar{S}(P, h)$ and $\underline{S}(P, h)$ denote the upper and lower Darboux sums that [2, Upper and Lower Darboux Sums] writes $U(P, h)$ and $L(P, h)$, renamed here because U is the interval of the statement. Let $P = \{x_0 < x_1 < \dots < x_n\}$ be a partition of $[s, t]$ of mesh less than δ that contains the preimage under φ of every point of Q_0 . The points $\varphi(x_0), \dots, \varphi(x_n)$ form a partition Q of J that refines Q_0 , and its subintervals are the intervals J_i with endpoints $\varphi(x_{i-1})$ and $\varphi(x_i)$, with $\varphi([x_{i-1}, x_i]) = J_i$. Writing $M_i = \sup_{J_i} f$ and $m_i = \inf_{J_i} f$, the supremum and infimum of g on $[x_{i-1}, x_i]$ are therefore M_i and m_i . By [2, The Mean Value Theorem] there is $\xi_i \in (x_{i-1}, x_i)$ with $|J_i| = |\varphi(x_i) - \varphi(x_{i-1})| = w(\xi_i)(x_i - x_{i-1})$. For $x \in [x_{i-1}, x_i]$ we have $|x - \xi_i| < \delta$ and $w(\xi_i) \geq 0$, so

$$m_i w(\xi_i) - K\epsilon \leq g(x)w(\xi_i) + g(x)(w(x) - w(\xi_i)) = g(x)w(x) \leq M_i w(\xi_i) + K\epsilon$$

Taking the supremum and the infimum over $[x_{i-1}, x_i]$, multiplying by $x_i - x_{i-1}$ and summing over i , and using $\sum_i M_i w(\xi_i)(x_i - x_{i-1}) = \sum_i M_i |J_i| = \bar{S}(Q, f)$ and $\sum_i m_i w(\xi_i)(x_i - x_{i-1}) = \sum_i m_i |J_i| = \underline{S}(Q, f)$,

$$\underline{S}(Q, f) - K\epsilon(t - s) \leq \underline{S}(P, gw) \leq \bar{S}(P, gw) \leq \bar{S}(Q, f) + K\epsilon(t - s)$$

Since Q refines Q_0 , [2, Refinement Moves the Darboux Sums Together] gives $\bar{S}(Q, f) - \underline{S}(Q, f) < \epsilon$, so $\bar{S}(P, gw) - \underline{S}(P, gw) < \epsilon(1 + 2K(t - s))$. As ϵ is arbitrary, $gw \in \mathcal{R}[s, t]$ by [2, The Integrability Criterion], and the same theorem places both $\int_s^t gw$ and $\int_J f$ in the interval $[\underline{S}(Q, f) - K\epsilon(t - s), \bar{S}(Q, f) + K\epsilon(t - s)]$, whose length is less than $\epsilon(1 + 2K(t - s))$. Hence the two integrals are equal.

Step 3: passage to U and V . Suppose first that φ is increasing. Fix $u_0 \in U$ and put $m = \varphi(u_0)$. Then φ maps $(a, u_0]$ onto $(c, m]$ and $[u_0, b)$ onto $[m, d)$. Let $\epsilon > 0$. By the definitions of the improper integrals $\int_c^m f$ and $\int_m^d f$ ([2, Improper Integral], read through [2, One-Sided Limits] at a finite endpoint and [2, Limits at Infinity] at an infinite one), there are $y_1 \in (c, m)$ and $y_2 \in (m, d)$ with $|\int_y^m f - \int_c^m f| < \epsilon$ for $c < y < y_1$ and $|\int_m^y f - \int_m^d f| < \epsilon$ for $y_2 < y < d$. For $a < s < \varphi^{-1}(y_1)$ and $\varphi^{-1}(y_2) < t < b$ we have $c < \varphi(s) < y_1$ and $y_2 < \varphi(t) < d$, so Step 2, applied on $[s, u_0]$ and on $[u_0, t]$, gives

$$\left| \int_s^{u_0} f(\varphi(x)) |\varphi'(x)| dx - \int_c^m f \right| < \epsilon \quad \text{and} \quad \left| \int_{u_0}^t f(\varphi(x)) |\varphi'(x)| dx - \int_m^d f \right| < \epsilon$$

By the same two definitions, this says that both halves of $\int_U f(\varphi(x)) |\varphi'(x)| dx$ converge, to $\int_c^m f$ and to $\int_m^d f$, whose sum is $\int_V f$. If φ is decreasing, then $\tilde{\varphi}(x) = \varphi(-x)$ is an increasing continuously differentiable bijection of $(-b, -a)$ onto V , with $|\tilde{\varphi}'(x)| = |\varphi'(-x)|$ by [2, The Chain Rule]. The reflection $x \mapsto -x$ carries integrals of $f(\tilde{\varphi}(x)) |\tilde{\varphi}'(x)|$ over subintervals of $(-b, -a)$ to integrals of $f(\varphi(x)) |\varphi'(x)|$ over the reflected subintervals of (a, b) , as in the proof of theorem 0.3.1, together with their limits. So the increasing case, applied to $\tilde{\varphi}$, gives the formula for φ , as claimed.

0.4.2 Note In standard calculus substitution, we often write $\int_a^b$. Here, we integrate over the set V . The absolute value term $|\varphi'(x)|$ accounts for the fact that volume (length) is always positive, even if the function φ is decreasing (reversing orientation).

Proposition 0.4.3: The Gaussian Integral

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

Proof:

Step 1: two inequalities. For every real y , $e^y \geq 1 + y$. Indeed $h(y) = e^y - 1 - y$ has $h'(y) = e^y - 1$ by [2, The Exponential Is Its Own Derivative], which is negative for $y < 0$ and positive for $y > 0$ because the exponential is strictly increasing with $e^0 = 1$ ([2, The Exponential Is a Strictly Increasing Bijection onto $(0, \infty)$]). For $y \neq 0$, [2, The Mean Value Theorem] on the interval between y and 0 gives a c strictly between them with $h(y) - h(0) = h'(c)y$, and $h'(c)$ and y have the same sign, so $h(y) \geq h(0) = 0$. With $y = -x^2$ and with $y = x^2$ this gives $0 \leq 1 - x^2 \leq e^{-x^2}$ for $0 \leq x \leq 1$ and $1 + x^2 \leq e^{x^2}$ for every x . Raising to a power $n \geq 1$ preserves inequalities between nonnegative numbers, so

$$(1 - x^2)^n \leq e^{-nx^2} \quad (0 \leq x \leq 1) \quad \text{and} \quad e^{-nx^2} \leq (1 + x^2)^{-n} \quad (x \in \mathbb{R})$$

Step 2: convergence and scaling. Since $(x-1)^2 \geq 0$, $e^{-x^2} \leq e^{1-2x}$, and $\int_0^X e^{1-2x} dx = \frac{e}{2}(1 - e^{-2X}) \leq \frac{e}{2}$ for $X > 0$ by [2, The Fundamental Theorem of Calculus, Second Form]. So $G = \int_0^\infty e^{-x^2} dx$ converges by [2, Convergence Tests for Improper Integrals]. For $n \geq 1$, theorem 0.4.1 with $\varphi(x) = \sqrt{n}x$ from $(0, \infty)$ onto $(0, \infty)$ gives $\int_0^\infty e^{-nx^2} \sqrt{n} dx = G$, that is, $\int_0^\infty e^{-nx^2} dx = G/\sqrt{n}$. Also $(1+x^2)^{-n} \leq \min(1, x^{-2})$ and $\int_1^X x^{-2} dx = 1 - 1/X \leq 1$, so $\int_0^\infty (1+x^2)^{-n} dx$ converges by the same tests.

Step 3: Wallis integrals. Let $W_k = \int_0^{\pi/2} \sin^k t dt$ as in Step 3 of the proof of theorem 0.2.1, where $W_k = \frac{k-1}{k} W_{k-2}$ for $k \geq 2$, $W_{k+1} \leq W_k$ and $W_{2n}/W_{2n+1} \rightarrow 1$ were shown. Substitution ([2, Substitution]) with $x = \cos t$ on $[0, \pi/2]$, using $\cos' = -\sin$, $\cos 0 = 1$, $\cos(\pi/2) = 0$ and $1 - \cos^2 t = \sin^2 t$ ([2, The Derivatives of the Cosine and the Sine], [2, The Number π], [2, The Pythagorean Identity]), gives

$$\int_0^1 (1-x^2)^n dx = \int_0^{\pi/2} \sin^{2n} t \sin t dt = W_{2n+1}$$

The cosine is positive on $[0, \pi/2)$ by [2, The Number π], so the sine, whose derivative it is, is strictly increasing there by [2, Monotonicity From the Sign of the Derivative], and $\sin t > \sin 0 = 0$ for $0 < t < \pi/2$. So $\cot = \cos/\sin$ is continuously differentiable on $(0, \pi/2)$, with $\cot' = -1/\sin^2 = -(1 + \cot^2) < 0$ by [2, The Rules of Differentiation] and the Pythagorean identity. It is strictly decreasing, it tends to $+\infty$ as $t \rightarrow 0^+$ because $\cos t \rightarrow 1$ and $\sin t \rightarrow 0$ with $\sin t > 0$, and it tends to $\cos(\pi/2)/\sin(\pi/2) = 0$ as $t \rightarrow (\pi/2)^-$. By [2, The Intermediate Value Theorem] it maps $(0, \pi/2)$ onto $(0, \infty)$. Theorem 0.4.1 with $\varphi = \cot$, and $1 + \cot^2 t = 1/\sin^2 t$, give

$$\int_0^\infty \frac{dx}{(1+x^2)^n} = \int_0^{\pi/2} \frac{1 + \cot^2 t}{(1 + \cot^2 t)^n} dt = \int_0^{\pi/2} \sin^{2n-2} t dt = W_{2n-2}$$

where the integral over $(0, \pi/2)$ given by the theorem is an ordinary integral, because its integrand is continuous on $[0, \pi/2]$ and the integral of such a function depends continuously on its endpoints ([2, The Fundamental Theorem of Calculus, First Form]).

Step 4: the limit. By Steps 1 to 3, [2, Order Bounds and Additivity over a Subdivision Point] and the comparison part of the convergence tests, for every $n \geq 1$

$$W_{2n+1} \leq \int_0^1 e^{-nx^2} dx \leq \frac{G}{\sqrt{n}} \leq W_{2n-2}$$

The recursion $(k+1)W_{k+1} = kW_{k-1}$ gives $(k+1)W_{k+1}W_k = kW_kW_{k-1}$ for $k \geq 1$, so by induction $(k+1)W_{k+1}W_k = W_1W_0 = \frac{\pi}{2}$ for every $k \geq 0$. With $k = 2n$ and with $k = 2n-2$,

$$n W_{2n+1}^2 = \frac{n}{2n+1} \cdot \frac{\pi}{2} \cdot \frac{W_{2n+1}}{W_{2n}} \quad \text{and} \quad n W_{2n-2}^2 = \frac{n}{2n-1} \cdot \frac{\pi}{2} \cdot \frac{W_{2n-2}}{W_{2n-1}}$$

The ratio W_{2n+1}/W_{2n} tends to 1, as the reciprocal of the ratio shown to tend to 1 in Step 3 of the proof of theorem 0.2.1, and $1 \leq W_{2n-2}/W_{2n-1} \leq W_{2n-2}/W_{2n} = \frac{2n}{2n-1}$, so W_{2n-2}/W_{2n-1} tends to 1 by [2, The Squeeze Theorem]. By [2, The Algebra of Limits] both right-hand sides tend to $\pi/4$, and by continuity of the square root ([2, The n -th Root Function Is Continuous]), $\sqrt{n} W_{2n+1}$ and $\sqrt{n} W_{2n-2}$ tend to $\sqrt{\pi}/2$. The Squeeze Theorem applied to the display gives $G = \sqrt{\pi}/2$. Finally, theorem 0.4.1 with $\varphi(x) = -x$ from $(-\infty, 0)$ onto $(0, \infty)$ gives $\int_{-\infty}^0 e^{-x^2} dx = G$, so the integral over $\mathbb{R}$ is $2G = \sqrt{\pi}$, as claimed.

0.4.2 The Multivariate Case (Jacobian)

When moving to multiple dimensions, the derivative $\varphi'(x)$ is replaced by the determinant of the Jacobian matrix. This determinant captures the local scaling factor of the transformation.

Definition 0.4.4: The Jacobian Matrix

Let $\varphi : U \rightarrow V$ be a differentiable map where $U, V \subseteq \mathbb{R}^n$. For a point $\mathbf{x} = (x_1, \dots, x_n)$, the Jacobian matrix $J_\varphi(\mathbf{x})$ is the $n \times n$ matrix of partial derivatives:

$$J_\varphi(\mathbf{x}) = \begin{pmatrix} \frac{\partial \varphi_1}{\partial x_1} & \dots & \frac{\partial \varphi_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial \varphi_n}{\partial x_1} & \dots & \frac{\partial \varphi_n}{\partial x_n} \end{pmatrix}$$

Theorem 0.4.5: Change of Variables (General)

Let $\varphi : U \rightarrow V$ be a diffeomorphism (a bijection that is differentiable with a differentiable inverse) between open sets in $\mathbb{R}^n$. For an integrable function $f : V \rightarrow \mathbb{R}$:

$$\int_V f(\mathbf{y}) d\mathbf{y} = \int_U f(\varphi(\mathbf{x})) |\det J_\varphi(\mathbf{x})| d\mathbf{x}$$

Proof Key ideas:

The proof is not given here: it needs a theory of integration on $\mathbb{R}^n$, the Lebesgue integral or Riemann integration over Jordan-measurable sets, together with the inverse function theorem, and this book develops neither. An invertible linear map A carries a cube Q onto a parallelepiped of volume $|\det A|$ times the volume of Q , which is the formula for $\varphi = A$ and $f = 1$ on $A(Q)$. On a small cube around $\mathbf{x}$, the map φ is close to the affine map $\mathbf{z} \mapsto \varphi(\mathbf{x}) + J_\varphi(\mathbf{x})(\mathbf{z} - \mathbf{x})$, so it multiplies volumes of small cubes by approximately $|\det J_\varphi(\mathbf{x})|$, and summing over a fine partition of U into cubes and refining the partition gives the formula. In the statement, *integrable* means Lebesgue integrable. When φ and φ^{-1} are continuously differentiable, the theorem is [4, Theorem 2.47]. This covers the linear changes of variables used in this book, and the polar map $(r, \theta) \mapsto (r \cos \theta, r \sin \theta)$ from $(0, \infty) \times (0, 2\pi)$ onto $\mathbb{R}^2$ with the closed ray $\{(x, 0) : x \geq 0\}$ deleted, a set of measure zero that does not change integrals over $\mathbb{R}^2$. For a differentiable bijection as stated, it is [5, Theorem 7.26], applied to the positive and negative parts of f .

0.4.6 Remark: The Jacobian Determinant as a Local Volume Scaling For a linear map represented by a matrix A , the determinant $\det(A)$ measures how much the map scales volume: the unit cube maps to a parallelepiped of volume $|\det(A)|$. For a general non-linear map φ , the Jacobian matrix $J_\varphi(\mathbf{x})$ is the *best linear approximation* of the function near the point $\mathbf{x}$. Therefore, $|\det J_\varphi(\mathbf{x})|$ describes the local expansion or contraction of volume at that specific point.

Example 0.4.7: Polar Coordinates

Consider the transformation from polar to Cartesian coordinates, $\varphi(r, \theta) = (r \cos \theta, r \sin \theta)$. The Jacobian matrix is:

$$J = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}$$

The determinant is $r \cos^2 \theta - (-r \sin^2 \theta) = r(\cos^2 \theta + \sin^2 \theta) = r$. This recovers the familiar area element $dx \, dy = r \, dr \, d\theta$.

Introduction

1.1 Motivation and Basic Terminology

The following example is from Panchenko as a motivation to why we should study probability. The essence of this example is that we can sometimes answer existential questions using statistical means:

Example 1.1.1: Motivation

Let $n \geq 2$, and fix a choice of n vector on the unit sphere

$$v_1, v_2, \dots, v_n \in S^{n-1}$$

or equivalently n vectors from $\mathbb{R}^n$ where $\|v_i\| = 1$ for all i . Then among all linear combinations of the form

$$v = v_1 \pm v_2 \pm \dots \pm v_n$$

does there exists one such that $\|v\| \leq \sqrt{n}$? What about $\|v\| \geq \sqrt{n}$?

Notice that if all the vectors are orthogonal, then we will form a “n”-triangle in $\mathbb{R}^n$, and so $\|v\| = \sqrt{n}$, but it is not necessarily the case that they are orthogonal.

The way we solve this is the following: Consider the vector space $\{-1, 1\}^n$ (i.e. $(\mathbb{Z}/2\mathbb{Z})^n$ as an n dimensional $\mathbb{Z}/2\mathbb{Z}$ vector-space). Consider any $\epsilon \in \{-1, 1\}^n$ and define the function

$$v(\epsilon) = \epsilon_1 v_1 + \epsilon_2 v_2 + \dots + \epsilon_n v_n$$

note that the choice of the v_i 's are fixed, and so v is fixed, meaning we only need to keep track of the choice of ϵ . The question now becomes if we can find an ϵ such that $\|v(\epsilon)\| \leq \sqrt{n}$. In fact, let's modify this question a bit: it is equivalent to ask where there exists an ϵ such that $\|v(\epsilon)\|^2 \leq n$. We do this since working with a square-root can be cumbersome, and getting rid of it does not change the question. To proceed with the question, remember that if we compute the average

of a set of numbers then there must exist at least one number that's higher than or equal to the average (equivalently lower than or equal to the average). To this end, take

$$\frac{1}{2^n} \sum_{\epsilon \in \{-1,1\}^n} \|v(\epsilon)\|^2 = \frac{1}{2^n} \sum_{\epsilon \in \{-1,1\}^n} \|\epsilon_1 v_1 + \cdots + \epsilon_n v_n\|^2 = \frac{1}{2^n} \sum_{\epsilon \in \{-1,1\}^n} \sum_{i,j=1}^n \epsilon_i \epsilon_j \langle v_i, v_j \rangle$$

then we can split the equation in the following way

$$\frac{1}{2^n} \sum_{\epsilon \in \{-1,1\}^n} \sum_{i,j=1}^n \epsilon_i \epsilon_j \langle v_i, v_j \rangle = \sum_{i,j=1}^n \langle v_i, v_j \rangle \frac{1}{2^n} \left(\sum_{\epsilon \in \{-1,1\}^n} \epsilon_i \epsilon_j \right)$$

at this point, we will split the some in when $i = j$ and $i \neq j$. If $i = j$, then $\epsilon_i \epsilon_j = \epsilon_i \epsilon_i = 1$, so we get

$$\frac{1}{2^n} \sum_{\epsilon \in \{-1,1\}^n} \epsilon_i \epsilon_j = \frac{2^n}{2^n} = 1$$

if $i \neq j$, we can observe that the terms cancel out. For every choice of the other signs, the pair (ϵ_i, ϵ_j) takes on the four values $(1, 1), (1, -1), (-1, 1), (-1, -1)$ exactly the same number of times. The sum of the products of these pairs is $1 - 1 - 1 + 1 = 0$. More formally, we can factor the summation:

$$\sum_{\epsilon \in \{-1,1\}^n} \epsilon_i \epsilon_j = \left(\sum_{\epsilon_i \in \{-1,1\}} \epsilon_i \right) \left(\sum_{\epsilon_j \in \{-1,1\}} \epsilon_j \right) 2^{n-2} = (0)(0)2^{n-2} = 0$$

Thus, only the terms where $i = j$ survive. We get

$$\frac{1}{2^n} \sum_{\epsilon \in \{-1,1\}^n} \|v(\epsilon)\|^2 = \sum_{i=1}^n \langle v_i, v_i \rangle (1) + \sum_{i \neq j} \langle v_i, v_j \rangle (0) = \sum_{i=1}^n \|v_i\|^2$$

Since each vector is a unit vector, $\|v_i\|^2 = 1$, and so the sum is exactly n .

Thus, we get

$$\frac{1}{2^n} \sum_{\epsilon \in \{-1,1\}^n} \|v(\epsilon)\|^2 = n$$

Since the average of the *squared* norms is n , by the logic earlier there must be at least one ϵ such that $\|v(\epsilon)\|^2 \leq n$ and at least one such that $\|v(\epsilon)\|^2 \geq n$. Consequently, taking the square root (a monotonic increasing function and hence preserving inequalities), there exists an ϵ such that $\|v(\epsilon)\| \leq \sqrt{n}$ and one such that $\|v(\epsilon)\| \geq \sqrt{n}$.

We can then ask a more subtle question: How is the length $\|v(\epsilon)\|$ distributed? It turns out that due to *concentration of measure*, the vast majority of sign choices result in a vector with norm very close to $\sqrt{n}$. While the average argument guarantees the existence of combinations with $\|v(\epsilon)\| \leq \sqrt{n}$ and $\|v(\epsilon)\| \geq \sqrt{n}$, it does not show how far from the average we can go. In many applications, the goal is precisely to find the rare “outlier” assignments that deviate significantly from this average behavior. We will return to the techniques for finding such deviations in chapter 3.

This shows the power of probability. Probability is an old study that pre-dates some common set-theory vocabulary. Therefore, there is an equivalence of vocabulary:

notation	set theory	probability
X	set, space	Universe, Probability space
$x \in X$	element	outcome
$A \subseteq X$	subset	event
$f : X \rightarrow \mathbb{R}$	function	random variable
$\frac{1}{m(X)} \int_X f$	average	expected value

1.2 Basic Definitions

In the previous example, the set $\Omega = \{-1, 1\}^n$ was the probability space. Notice that every element in the probability space was associated with some “chance” or “probability” of choosing that element. More generally, if we choose some subset (i.e. event) from Ω , we can give a probability of that event happening. This idea of capturing the chances of an outcome or an event¹ is captured by a special function $\mathbb{P} : P(\Omega) \rightarrow \mathbb{R}$ called a *Probability*, where $P(\Omega)$ is the powerset of Ω . If you have taken Real Analysis, this is called the *probability measure*, since it is an example of a particular measure that will follow a couple of properties we will list below that will make it align with our intuition of probability.

Every probability space *must* have an associated probability function. In the previous example, we sneakily used the probability $\mathbb{P}(\{\epsilon\}) \stackrel{!}{=} \mathbb{P}(\epsilon) = \frac{1}{2^n}$ where $\stackrel{!}{=}$ is a common abuse of notation for event’s with only 1 outcome. For a general event $A \subseteq \Omega$ in the previous example, we have:

$$\mathbb{P}(A) = \frac{\#(A)}{2^n}$$

this particular probability is called the *uniform measure* since every outcome was weighed equally.

What properties should $\mathbb{P}$ have? Let’s take for example the roll of a dice where each side has equal probability. Should we include the chance that the dice gets stuck in a corner, so that the probability of any of the faces showing up is less than 1? Should we have a special variable called “fluke” which discounts that possibility? If one of the outcomes affect another (let’s say you rolled the dice onto a sticky surface, so now one of the faces will stick to the floor more when you role the dice again). Should the probability take this into account, or should it be the responsibility of another function that takes into account dependent events? Throughout the centuries, the expected properties of what this probability function $\mathbb{P}$ must follow were greatly philosophized. Eventually, a Russian mathematician named Kolmogorov came up with the accepted properties a probability function needs to follow:

¹we can think of an outcome as an event with a single outcome

Definition 1.2.1: Probability [Measure]

Let Ω be a set and $P(\Omega)$ its power set^a. Let $\mathbb{P}$ be a function from a probability space $P(\Omega) \rightarrow \mathbb{R}$. Then if $\mathbb{P}$ satisfies:

1. For all $A \subseteq \Omega$, $\mathbb{P}(A) \in [0, 1]$
2. $\mathbb{P}(\Omega) = 1$
3. If $A_1, A_2 \subseteq \Omega$ are disjoint then

$$\mathbb{P}(A_1 \sqcup A_2) = \mathbb{P}(A_1) + \mathbb{P}(A_2)$$

then $\mathbb{P}$ is called a *probability [measure]*

^aor more precisely, the set of all measurable subsets of Ω

There is in fact a couple more caveats when it comes to defining a probability that appear if Ω is infinite (in particular at least uncountably infinite). It will turn out that these rules will contradict each other², and so we would also restrict from the powerset $P(\Omega)$ to the measurable subsets $\mathcal{M}(\Omega)$. The details of this restriction is covered in a course on measure theory, and will never be a problem for us as in its a technicality dealt with measure there never to be thought of again like formal set theory to most mathematicians. Simply know that the only *unmeasurable sets* that we [currently] know of were always purposefully esoteric to find a counter-example, and never came up as an actual problem when doing probability.

Until section 1.3, Ω shall always be finite.

Example 1.2.2: probabilities

Prove the following:

1. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
2. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
3. $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$
4. $\mathbb{P}(\bigcup_{i=1}^n A_i) \leq \sum_{i=1}^l \mathbb{P}(A_i)$

If you have done some measure theory, these are essentially the same proofs for finite measures (since $\mathbb{P}(X) = \mu(X) < \infty$)

This now gives us the language to talk about the chances of something happening.

The next piece of the puzzle in the picture of probability is a way to capture the idea of asking “probability questions” on a space. Each “probability question” will depend on a numerical values to the outcomes or events in a space. As an example, imagine you have 10 coins labeled x_1 to x_{10} with each side having a 1 or 0 (or any two arbitrary value per coin). We might ask ourselves the following question: if we choose n coins, what’s the sum of all the values? Mathematically, we can

²see Vitali sets

set $\Omega = P(\{x_1, \dots, x_{10}\})$ then define

$$f : \Omega \rightarrow \mathbb{R} \quad f(A) = \text{sum of the value}$$

Using this, we can ask some questions like:

1. What is the average value you get by picking any possible random outcome?
2. How many outcomes are “close” to the average? Is it spread out? Can we bound “most” close to the average?

These two questions will be central to the beginning of our exploration of probability.

Let us start with capturing the values assigned to elements of the set Ω . If we want to know the average number of a dice roll. The probability will capture the fact that each face has a $\frac{1}{6}$ chance of happening, but not that the faces have values ranges from 1, ..., 6. To capture this idea, we use a different notion called a *random variable*.

Definition 1.2.3: Random Variable

A random variable is a function $X : \Omega \rightarrow \mathbb{R}$ that gives a numerical value to events in Ω^a .

In the infinite case where $(\Omega, \mathcal{F}, \mathbb{P})$ is a measure space, X is a measurable function $(\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (\mathbb{R}, \mathcal{B})$ where $\mathcal{B}$ is the usual σ -Borel algebra.

^aJust like for $\mathbb{P}$, we are technically restricting to $\mathcal{M}(\Omega)$

Since the codomain is the real numbers, we take on the usual definition of

$$(XY)(a) = X(a)Y(a) \quad (X + Y)(a) = X(a) + Y(a)$$

At the beginning, it might be confusing that random variable's are denoted with X or Y . This will be more intuitive later one when we treat these functions as variables and sum over or take the product of random variables. The word “random” here is that we are finding the numerical value of a “random event”. The image of a random variable is called the *weight* or the *bias*, depending on what data the random variable is representing.

Example 1.2.4: Random Variable

Let's say $\Omega = \{a, b, c, d\}$ with the probability of choosing any event of Ω being $1/4$ ($\mathbb{P}(x) = 1/4$, $x \in \Omega$). Define $\epsilon : \Omega \rightarrow \mathbb{R}$, $\epsilon(a) = \epsilon(b) = 1$ and $\epsilon(c) = \epsilon(d) = -1$. Then we have:

$$\mathbb{P}(\{\omega \in \Omega \mid \epsilon(\omega) = 1\}) = 1/2$$

and similarly if $\epsilon(\omega) = -1$.

We can define a different random variable on Ω : let $Z : \Omega \rightarrow \mathbb{R}$, $Z(a) = 2$, $Z(b) = -2$, $Z(c) = 3$, $Z(d) = -3$. Then

$$\mathbb{P}(\{\omega \in \Omega \mid Z(\omega) = 2\}) = 1/4$$

and similarly for all other possible values. The sets in both cases are very commonly abbreviated as:

$$\{\epsilon = 1\} = \{\omega \in \Omega \mid \epsilon(\omega) = 1\}$$

so that we would write

$$\mathbb{P}(\{\epsilon = 1\}) = 1/2$$

or even be even shorter with our notation and write:

$$\mathbb{P}(\epsilon = 1) = 1/2$$

This final notation will be very common when using random variables, as it really holds the key information about how the random variable interacts with the probability.

It is a very common convention in probability to omit Ω as much as possible when working with random variables (as we have already demonstrated in example 1.2.4), either relying on the context to give us the probability space, or more generally not really caring about the particular probability space; it does not matter if it's 6 face of a die or 6 cosmic events, if each have a random variable defined identically with the same probability (ex. each face of the die and each cosmic event have the same probability, and the random variable gives either number of the face or the intensity of the cosmic event from 1-6), then what we really care about is the result from the random variable and it's interaction with the probability, and not the probability space. In fact, the probability space can be different in size and we still can define the same random variables. Let's say there were 12 cosmic events, each with the same probability, and two of those events have intensity 1, two of those events have intensity 2, and so on. Then a random variable that captures the intensity will be indistinguishable to the one with the dice roll.

Example 1.2.5: Event with new Notation

Taking the same Ω and $\mathbb{P}$ as from the motivating example (example 1.1.1). Define

$$B = \{\pi_1 = 1\}$$

Then $\mathbb{P}(B) = \frac{2^{n-1}}{2} = \frac{1}{2}$

As mentioned earlier, one of the most sought-after information we try to get from probability is the average. Average is dependent on how we numerically represent the values of the probability space, which should motivate the following definition:

Definition 1.2.6: [Finite] Expected Value

Let X be a random variable and $(\Omega, \mathbb{P})$ be a probability space. Then the expected value for the random variable X is:

$$\mathbb{E}X = \sum_{\omega \in \Omega} X(\omega)\mathbb{P}(\omega)$$

The expected value is also called the *mean* of X .

Note that this is well-defined since Ω is finite. Later chapters give conditions under which the expected value is well-defined. In particular, in chapter 4 we work with continuous random variables, and there the expected value becomes an integral. Sometimes, the notation for the discrete and continuous case is used interchangeably:

$$\mathbb{E}X = \int_{\Omega} X(\omega)d\mathbb{P}(\omega)$$

The way I like to think of the expected value is that it associated a number to each function representing what value we should get if we ran every possibility.

Example 1.2.7: Expected Value: Indicator Function

let Ω be any probability with $\mathbb{P}$ as it's probability function. Define the following random variable:

$$I_A \stackrel{!}{=} I(\omega \in A) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \notin A \end{cases}$$

this random variable is called the *indicator function*. The $\stackrel{!}{=}$ equality is an abuse of notation to mean the function I is “indicating” over the set A , since the indicator function is very common.

The expected value is:

$$\mathbb{E}(I_A) = \sum_{\omega \in \Omega} I_A(\omega) \mathbb{P}(\omega) = \sum_{\omega \in A} \mathbb{P}(\omega) = \mathbb{P}(A)$$

showing that the expected value of I_A is the probability of the event of A ! Take a moment to think about this case to have an idea of what a random variable is capturing

Proposition 1.2.8: Properties of Expected Value

Let f and g be random variables on $(\Omega, \mathbb{P})$ and $|\Omega| < \infty$. Then

1. $\mathbb{E}(f + g) = \mathbb{E}f + \mathbb{E}g$
2. If $f \geq 0$, then $\mathbb{E}f \geq 0$
3. More generally, if $f \leq g$ then $\mathbb{E}f \leq \mathbb{E}g$

Proof:

For (1), the pointwise identity $(f + g)(\omega) = f(\omega) + g(\omega)$ and definition 1.2.6 give

$$\mathbb{E}(f + g) = \sum_{\omega \in \Omega} (f(\omega) + g(\omega)) \mathbb{P}(\omega) = \sum_{\omega \in \Omega} f(\omega) \mathbb{P}(\omega) + \sum_{\omega \in \Omega} g(\omega) \mathbb{P}(\omega) = \mathbb{E}f + \mathbb{E}g$$

where splitting the sum termwise is valid because Ω is finite.

For (2), property (1) of definition 1.2.1, applied to the singleton $\{\omega\}$, gives $\mathbb{P}(\omega) \geq 0$ for every $\omega \in \Omega$. If $f(\omega) \geq 0$ for every ω , then each term $f(\omega)\mathbb{P}(\omega)$ is a product of nonnegative numbers, so

$$\mathbb{E}f = \sum_{\omega \in \Omega} f(\omega) \mathbb{P}(\omega) \geq 0$$

For (3), suppose $f(\omega) \leq g(\omega)$ for every $\omega \in \Omega$. The pointwise difference $g - f : \Omega \rightarrow \mathbb{R}$ is again a random variable, and $(g - f)(\omega) \geq 0$ for every ω , so part (2) gives $\mathbb{E}(g - f) \geq 0$. Applying part (1) to the random variables f and $g - f$,

$$\mathbb{E}g = \mathbb{E}(f + (g - f)) = \mathbb{E}f + \mathbb{E}(g - f) \geq \mathbb{E}f$$

The next concept we shall point out is that of *independence*. When defining probability, I mentioned in the paragraph above the definition that we have to somehow deal with events affecting one another. Similarly, we have to know how to deal with events that do not affect one another, that are independent of another. The former we shall return to. For the latter, notice that we were already sneakily using the notion of independent events in the motivating example. In particular, the choice of adding or subtracting one vector did not affect the choice of another. The way we will capture this idea of independence is through the use of random variables. Let $\Omega = \{-1, 1\}^n$ be a finite set of vectors in a vector-space. Notice that we can define random variables

$$\pi_i : \Omega \rightarrow \mathbb{R}, \pi_i(\epsilon) = \epsilon_i$$

where every random variable π_i captures the idea of the sign of the i th vector. Often, we abuse notation and write ϵ_i as the random variable itself. Given these random variables, we can define an outcome from them (or more generally, an event). For some choice of signs $a_1, a_2, \dots, a_n$, we can define:

$$A = \{\epsilon \in \Omega \mid \pi_1(\epsilon) = a_1, \dots, \pi_n(\epsilon) = a_n\}$$

or, to abuse notation a bit, we write

$$A = \{\pi_1 = a_1, \dots, \pi_n = a_n\}$$

where we omit a reference to Ω or its elements. We may then wonder whether computing the probability of A can be split among each random variable. If this is possible, this is called *independence*:

Definition 1.2.9: [Discrete] Independence

Let $(\Omega, \mathbb{P})$ be a probability space and let $X_1, \dots, X_n$ be discrete random variables. They are said to be *independent* if

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n \mathbb{P}(X_i = x_i)$$

for all x_i in the range of X_i . If the above equation does not hold, then these variables are said to be *dependent*.

The discrete definition above relies on the probability of outcomes occurring at specific points (masses). However, for continuous random variables (like those with Gaussian distributions), $\mathbb{P}(X = x) = 0$ for every x , making the product formula $0 = 0$ trivially true but useless.

Definition 1.2.10: Joint Distribution Measure

Let $X_1, \dots, X_n$ be random variables mapping $(\Omega, \mathcal{F}, \mathbb{P})$ to measurable spaces $(S_i, \mathcal{S}_i)$. Let $(S, \mathcal{S})$ denote the product space $\prod S_i$ equipped with the product σ -algebra. The *joint measure* (or *joint distribution measure*) is the composition $\mu_{\mathbf{X}} = \mathbb{P} \circ \mathbf{X}^{-1}$. For any measurable set $E \in \mathcal{S}$:

$$\mu_{\mathbf{X}}(E) := \mathbb{P}(\mathbf{X}^{-1}(E)) = \mathbb{P}(\{\omega : (X_1(\omega), \dots, X_n(\omega)) \in E\})$$

For those motivated by the naturality of category arguments, joint distribution is the measure $\mu_{\mathbf{X}}$ on $(S, \mathcal{S})$ defined as the pushforward of $\mathbb{P}$ under the random vector $\mathbf{X} = (X_1, \dots, X_n)$. Explicitly, it is the unique measure that makes the following diagram of set-functions commute:

$$\begin{array}{ccc}
\mathcal{S} & \xrightarrow{\mathbf{X}^{-1}} & \mathcal{F} \\
& \searrow \mu_{\mathbf{X}} & \downarrow \mathbb{P} \\
& & [0, 1]
\end{array}$$

It is also useful to see the definition in non-vectorized notation:

$$\mu_{(X_1, \dots, X_n)}(E) := \mathbb{P}(\{\omega : (X_1(\omega), \dots, X_n(\omega)) \in E\})$$

Definition 1.2.11: [General] Independence

Let $X_1, \dots, X_n$ be random variables. They are said to be *independent* if their joint distribution is exactly the product measure of their marginal distributions. That is,

$$\mu_{(X_1, \dots, X_n)} = \mu_{X_1} \times \dots \times \mu_{X_n}$$

Equivalently, in terms of σ -algebras, the sub- σ -algebras $\sigma(X_i) = \{X_i^{-1}(B) : B \in \mathcal{B}_{\mathbb{R}}\}$ generated by the variables must be independent classes.

The definition above is equivalent to saying that for any “rectangle” $A_1 \times \dots \times A_n$ in the product space:

$$\underbrace{\mathbb{P}\left(\bigcap_{i=1}^n X_i^{-1}(A_i)\right)}_{\text{Intersection in } \Omega} = \mu_{(X_1, \dots, X_n)}(A_1 \times \dots \times A_n) \stackrel{\text{indep}}{=} (\mu_{X_1} \times \dots \times \mu_{X_n})(A_1 \times \dots \times A_n) = \underbrace{\prod_{i=1}^n \mathbb{P}(X_i^{-1}(A_i))}_{\text{Product of Numbers}}$$

This highlights that independence is the specific structural condition where the joint measure (induced by Ω) factorizes into the product measure (constructed from marginals).

Example 1.2.12: Independence and Dependence

1. Let $\Omega = [0, 1] \times [0, 1]$ and let the probability be the integral over the area^a. We can let f and g be continuous^b functions from $[0, 1] \rightarrow \mathbb{R}$. You can imagine them being random variables on Ω by extending them to $[0, 1] \times [0, 1]$ by mapping those points to zero. Then we know from calculus that

$$\iint_{[0,1]^2} f(x)g(y)dx dy = \int_{[0,1]} f(x)dx \int_{[0,1]} g(y)dy$$

in other words, the given the “length” of one side, it is independent of the “length” of the other; each contribute independently to the area.

2. (Finite Independent Example) Let $\Omega = \{1, 2, 3, 4\}$ and let the probability be the counting measure normalized by $\frac{1}{4}$. Let $X = \mathbb{I}_{\{1,2\}}$ and $Y = \mathbb{I}_{\{1,3\}}$. The range of both variables is $\{0, 1\}$.

To show independence, we must verify the product rule for all four pairs $(x, y) \in \{0, 1\}^2$:

$$\mathbb{P}(X = 1, Y = 1) = \mathbb{P}(\{1\}) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(X = 1)\mathbb{P}(Y = 1)$$

$$\mathbb{P}(X = 1, Y = 0) = \mathbb{P}(\{2\}) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(X = 1)\mathbb{P}(Y = 0)$$

$$\mathbb{P}(X = 0, Y = 1) = \mathbb{P}(\{3\}) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(X = 0)\mathbb{P}(Y = 1)$$

$$\mathbb{P}(X = 0, Y = 0) = \mathbb{P}(\{4\}) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(X = 0)\mathbb{P}(Y = 0)$$

Since the joint distribution factors for every point in the codomain, the variables are independent.

3. (Finite Dependent Example) Let our sample space be three points on a lattice: $\Omega = \{(0, 0), (1, 0), (0, 1)\}$. Let $\mathbb{P}$ be the uniform measure, so each point has mass $1/3$. Let X be the first coordinate and Y be the second. First, we find the marginal probabilities for the value 1:

$$\begin{aligned}\mathbb{P}(X = 1) &= \mathbb{P}(\{(1, 0)\}) = \frac{1}{3} \\ \mathbb{P}(Y = 1) &= \mathbb{P}(\{(0, 1)\}) = \frac{1}{3}\end{aligned}$$

Now we test for independence at the point $(1, 1)$. This point is not in our support Ω , so its joint probability is zero:

$$\mathbb{P}(X = 1, Y = 1) = \mathbb{P}(\emptyset) = 0$$

However, the product of the marginals is non-zero:

$$\mathbb{P}(X = 1)\mathbb{P}(Y = 1) = \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$$

Since $0 \neq 1/9$, the variables are dependent.

4. (Continuous Dependent Example) Let Ω be the triangle defined by $\{(x, y) \in [0, 1]^2 : x + y \leq 1\}$. Let the joint PDF be $f_{X,Y}(x, y) = 2 \cdot \mathbb{I}_{\{x+y \leq 1\}}$. The marginals are $f_X(x) = 2(1 - x)$ and $f_Y(y) = 2(1 - y)$ for $x, y \in [0, 1]$.

To show $f_{X,Y} \neq f_X f_Y$ in the L^1 sense, consider the set $E = [0.8, 0.9] \times [0.8, 0.9]$. This set has Lebesgue measure $\mu(E) = 0.01 > 0$.

For any $(x, y) \in E$, we have $x + y \geq 1.6 > 1$, so the joint density is identically zero on E :

$$\iint_E f_{X,Y}(x, y) \, dx \, dy = 0$$

However, for the product of marginals on E :

$$\iint_E f_X(x)f_Y(y) \, dx \, dy = \left(\int_{0.8}^{0.9} 2(1 - x) \, dx \right) \left(\int_{0.8}^{0.9} 2(1 - y) \, dy \right) > 0$$

Since the integrals over a set of positive measure disagree, the functions cannot be equal almost everywhere.

5. (Infinite Independent Example) Let $\Omega = \mathbb{R}^2$ and let the measure be the standard Gaussian measure γ . If we define random variables $X(x, y) = x$ and $Y(x, y) = y$, their independence follows from the property of the exponential function:

$$\frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}} = \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \right) \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \right)$$

in other words, the joint density factors perfectly into a function of x times a function of y , satisfying the Fubini condition for product measures.

^anote that this is an example where we would limit to measurable sets instead of the powerset of $[0, 1]$

^bmore generally, measurable

Using new Vocabulary and Change of Variables

In the new notation, the step of example 1.1.1 showing that $\frac{1}{2^n} \sum_{\epsilon} \epsilon_i \epsilon_j = 0$ for $i \neq j$ reads

$$\mathbb{E} \pi_i \pi_j \stackrel{\text{indep.}}{=} \mathbb{E} \pi_i \mathbb{E} \pi_j = 0$$

To show this, we simply plug into the equation:

$$\mathbb{E} \pi_i \pi_j = \sum_{\epsilon \in \Omega} \pi_i(\epsilon) \pi_j(\epsilon) \mathbb{P}(\epsilon)$$

This can be hard to compute, so we will re-write this in a different way. This is a very common strategy in probability; many times, writing the same sum in a different way will produce much simpler summation, sometimes drastically simpler (as we will see with distributions in the next section).

The process by which we change the summation is called *change of variables*. To do this, we will partition Ω through the random variable in the following way:

$$A_{a_i, a_j} = \{\epsilon \in \Omega \mid \pi_i(\epsilon) = a_i, \pi_j(\epsilon) = a_j\}$$

Notice that

$$\Omega = \bigsqcup_{a_i, a_j} A_{a_i, a_j}$$

Thus, we can split up our sum which was over every element $\epsilon \in \Omega$ in the following way:

$$\begin{aligned} \mathbb{E}(\pi_i \pi_j) &= \sum_{\epsilon \in \Omega} \pi_i(\epsilon) \pi_j(\epsilon) \mathbb{P}(\epsilon) \\ &= \sum_{a_i, a_j = \pm 1} a_i a_j \sum_{\epsilon \in A_{a_i, a_j}} \mathbb{P}(\epsilon) \\ &= \sum_{a_i, a_j = \pm 1} a_i a_j \mathbb{P}(A_{a_i, a_j}) \\ &= \sum_{a_i, a_j = \pm 1} a_i a_j \mathbb{P}(\pi_i = a_i, \pi_j = a_j) \end{aligned}$$

We have now re-written this sum to look like another expected value. In fact, this *is* an expected value over a different probability space! We will get back to this shortly, let us first finish the computation. By independence, we get

$$\sum_{a_i, a_j = \pm 1} a_i a_j \mathbb{P}(\pi_i = a_i, \pi_j = a_j) = \sum_{a_i = \pm 1} a_i \mathbb{P}(\pi_i = a_i) \sum_{a_j = \pm 1} a_j \mathbb{P}(\pi_j = a_j)$$

Furthermore, with $A_{a_i} = \{\epsilon \in \Omega \mid \pi_i(\epsilon) = a_i\}$, the same grouping gives

$$\begin{aligned} \mathbb{E}(\pi_i) &= \sum_{\epsilon \in \Omega} \pi_i(\epsilon) \mathbb{P}(\epsilon) \\ &= \sum_{a_i = \pm 1} a_i \sum_{\epsilon \in A_{a_i}} \mathbb{P}(\epsilon) \\ &= \sum_{a_i = \pm 1} a_i \mathbb{P}(A_{a_i}) \\ &= \sum_{a_i = \pm 1} a_i \mathbb{P}(\pi_i = a_i) \end{aligned}$$

showing that “independence” equality does indeed hold! For the last equality, we can proceed with the same reasoning we gave in the example. We can split the sum into the ones that equal -1 and those that equal to 1 . There are an equal number in both cases, giving us:

$$(+1)\frac{1}{2} + (-1)\frac{1}{2} = 0$$

finishing the proof, but perhaps more clearly with this change of variables.

As we mentioned, we can think of this new equation as changing the probability space we working over. This new probability space is given by a random variable we are using to calculate our results. To best illustrate this, take π_i from the previous example. Then the image of π_i is $\{-1, 1\}$. Let $\Omega' = \{-1, 1\}$ be our new probability space, and let our new probability measure on Ω' be:

$$\mathbb{P}'(\omega') = \mathbb{P}(\{\omega \in \Omega \mid \pi_i(\omega) = \omega'\})$$

This new probability measure is often called the *distribution* (sometimes called the *law*) of the random variable π_i (more generally the random variable X), or the *image measure of $\mathbb{P}$ by the map π_i* (more generally by the map X). For example, Given π_i , we have the new values of:

$$\mathbb{P}'(1) = \frac{1}{2} \quad \mathbb{P}'(-1) = \frac{1}{2}$$

This new probability space is called the *sample space*. I like to think of it as the sample of *data* that we can get given an outcome, event. A bit confusingly, some statisticians called the original probability space Ω a sample space as well. It might be useful to think of the “default” random variable as $X : \Omega \rightarrow \Omega$, $X(\omega) = \omega$ (i.e. the identity), even though it does not necessarily map to $\mathbb{R}$ and the domain is not the powerset.

Notice that when we did a change of variables, and calculated the estimate over what now can be thought of as Ω' , we have a “new” random variable which is the identity function! In the previous example, we would have $\Omega' = \{-1, 1\}$, and our new probability space would be $(\Omega', \mathbb{P}')$! We give a name to this new probability:

Definition 1.2.13: Image Measure

Let $(\Omega, \mathbb{P})$ be a probability space and let X be a random variable. Define $\Omega' = X(\Omega)$ and let $\mathbb{P}'(x) = \mathbb{P}(X = x)$ for $x \in \Omega'$. Then Ω' is called the *sample space* and $\mathbb{P}'$ is called the *image measure* of Ω' with respect to X .

Many notations exist to represent the new probability on a sample space, including:

$$\mathbb{P}'(a_i) = \mathbb{P}(\pi_i = a_i) = \mathbb{P}(\pi_i^{-1}(a_i)) = \mathbb{P} \circ \pi_i^{-1}(a_i) = \pi_i \# \mathbb{P}(a_i)$$

that last one is read as “ π_i push-forward $\mathbb{P}$ ”.

We sum up our discoveries on expected values with change of variables the following proposition:

Proposition 1.2.14: [Finite] Expected Value Change of Variables

Let $(\Omega, \mathbb{P})$ be a finite probability space and let f be a random variable on it. Let Ω' and $\mathbb{P}'$ be the sample space and image measure of the map f . Then the expected value for the random variable f can be represented as :

$$\mathbb{E}f = \sum_{\omega' \in \Omega'} \omega' \mathbb{P}'(\omega')$$

which is called the *change of variables formula for the expectation*

Proof:

By definition 1.2.13, $\Omega' = f(\Omega)$ and $\mathbb{P}'(\omega') = \mathbb{P}(f = \omega')$. The sets $f^{-1}(\omega') = \{\omega \in \Omega : f(\omega) = \omega'\}$ with $\omega' \in \Omega'$ are pairwise disjoint, and their union is Ω , because each $\omega \in \Omega$ lies in exactly the one with $\omega' = f(\omega)$. For a subset $A \subseteq \Omega$, $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\omega)$. For $A = \emptyset$ both sides are 0, because property (3) of definition 1.2.1 with $A_1 = A_2 = \emptyset$ gives $\mathbb{P}(\emptyset) = 2\mathbb{P}(\emptyset)$, and the inductive step on $|A|$ is property (3) applied to $A \setminus \{\omega\}$ and $\{\omega\}$ for a point $\omega \in A$. Since every sum below is finite, it may be regrouped along this partition, and since $f(\omega) = \omega'$ for $\omega \in f^{-1}(\omega')$, definition 1.2.6 gives

$$\mathbb{E}f = \sum_{\omega \in \Omega} f(\omega) \mathbb{P}(\omega) = \sum_{\omega' \in \Omega'} \sum_{\omega \in f^{-1}(\omega')} \omega' \mathbb{P}(\omega) = \sum_{\omega' \in \Omega'} \omega' \mathbb{P}(f = \omega') = \sum_{\omega' \in \Omega'} \omega' \mathbb{P}'(\omega')$$

as claimed.

A word of warning to avoid confusion. We will often apply functions to random variables. In general, for a random variable X , if we apply a function to it, say $f(X)$, then $\mathbb{P}(X = x)$ does not change *if the bounds of the sum are the same* (i.e. if the sample space remains the same). Grouping the points ω by the value $X(\omega)$, as in the proof of proposition 1.2.14, gives

$$\mathbb{E}f(X) = \sum_{x \in X(\Omega)} f(x) \mathbb{P}(X = x)$$

If we change the sample space to be $f(X(\Omega))$, then proposition 1.2.14, applied to the random variable $f(X)$, gives

$$\mathbb{E}f(X) = \sum_{y \in f(X(\Omega))} y \mathbb{P}(f(X) = y)$$

Each of these can be useful in their own way.

Notice too that we can now simply define a random variable and its interaction with the probability without referencing the probability space since the random variable will define a sample space:

Example 1.2.15: Practice

Let N be a non-negative random variable such that:

$$\mathbb{P}(N > 0) = e^{-1} \quad \mathbb{E}N = e^{10}$$

Find a random variable with this property (in particular, notice that you can define one that “works”, and then the same space will become the probability space)

Exercise 1.2.1

1. Let Ω be any probability space with $|\Omega| \geq 2$. Let $A, B \subseteq \Omega$ where $\mathbb{P}(A) > 0$, $\mathbb{P}(B) > 0$. If $A \cap B = \emptyset$, does that imply they are independent?

1.3 Discrete Distributions

In this section, we will start by focusing on the discrete case, that is, where Ω is either finite or countable (i.e. they can be indexed by the natural numbers). When working in the discrete case, we will define the probability of any given outcome:

$$\mathbb{P}(\omega_n) = p_n \quad 0 \leq p_n \leq 1$$

with the restriction that

$$\sum_{i=1}^{\infty} \mathbb{P}(\omega_i) = 1$$

to insure that $\mathbb{P}(\Omega) = 1$. On a discrete space these values determine $\mathbb{P}$: for every $A \subseteq \Omega$,

$$\mathbb{P}(A) = \sum_{a \in A} \mathbb{P}(a)$$

To see this, list the points of A as $a_1, a_2, \dots$. For each N , the additivity of definition 1.2.1, applied $N - 1$ times, and $\mathbb{P} \geq 0$ give $\sum_{j \leq N} \mathbb{P}(a_j) = \mathbb{P}(\{a_1, \dots, a_N\}) \leq \mathbb{P}(A)$. So the partial sums of the series of nonnegative terms $\sum_{a \in A} \mathbb{P}(a)$ are bounded by $\mathbb{P}(A)$, and by [2, Nonnegative Terms and Bounded Partial Sums] the series converges with $\sum_{a \in A} \mathbb{P}(a) \leq \mathbb{P}(A)$. Applied to A^c , this gives $\sum_{a \in A^c} \mathbb{P}(a) \leq \mathbb{P}(A^c)$. Each partial sum of $\sum_i \mathbb{P}(\omega_i)$ is a sum over finitely many points of A plus a sum over finitely many points of A^c . Finitely many points of a set lie among the first N points of any listing of it, for some N , so the first sum is at most $\sum_{a \in A} \mathbb{P}(a)$ and the second at most $\sum_{a \in A^c} \mathbb{P}(a)$. Letting the partial sums tend to 1 ([2, Limits Preserve Weak Inequalities]),

$$1 = \sum_{i=1}^{\infty} \mathbb{P}(\omega_i) \leq \sum_{a \in A} \mathbb{P}(a) + \sum_{a \in A^c} \mathbb{P}(a) \leq \mathbb{P}(A) + \mathbb{P}(A^c) = 1$$

Both inequalities are therefore equalities, and since $\sum_{a \in A} \mathbb{P}(a) \leq \mathbb{P}(A)$ and $\sum_{a \in A^c} \mathbb{P}(a) \leq \mathbb{P}(A^c)$, each of these is an equality as well. The sum does not depend on the order in which the points of

A are listed, by [2, Absolutely Convergent Series Are Rearrangement Invariant]. Another way of writing this equation would be to use the indicator function we saw before:

$$\mathbb{P}(A) = \sum_{i=1}^{\infty} \mathbb{P}(\omega_i) I(\omega_i \in A)$$

which as we saw before is $\mathbb{E}(I_A)$. Using this fact, all the properties of probabilities in example 1.2.2 will be satisfied.

1.3.1 Countable Expected Value

As mentioned earlier, the expected value might need a couple of conditions to be well-defined; since $X : \Omega \rightarrow \mathbb{R}$ can have both positive and negative values, we must be careful on the convergence condition. In particular, since we are now taking infinite sums, we will have to start being careful with the order of the elements over which we sum (ex. take the series $1 - 1/2 + 1/3 - \dots \pm 1/n \mp \dots$, and re-arrange it to sum to any number $r \in \mathbb{R}$). Because of this, we give the following restriction:

Definition 1.3.1: [Countable] expected value

Let Ω be a discrete probability space with probability measure $\mathbb{P}$ and let X be a random variable. Then the expected value of X is the same way as the finite case:

$$\mathbb{E}X = \sum_{\omega \in \Omega} X(\omega) \mathbb{P}(\omega)$$

but with the additional condition that

$$\mathbb{E}|X| = \sum_{\omega \in \Omega} |X(\omega)| \mathbb{P}(\omega) < \infty$$

i.e. the series as absolutely convergent. Otherwise, we say that $\mathbb{E}X$ is undefined. An expected value with this property is called *integrable*

In these notes, we will consider $\mathbb{E}X = \infty$ as an acceptable value. This means that the first thing you have to check when working with an expected value of a random variable over a discrete probability space is whether or not it's defined. In the following section, we will prove some technical lemmas which help us in computing or proving the existence of an expected value, and we will show a couple of different ways we can view the expected value (something very useful to get a better grasp of the nature of the expected value)

Since we are working with countable probability spaces, these lemmas will be very handy to keep in mind:

Lemma 1.3.2: Equivalent Convergence Condition

Let $(a_i)_{i \geq 1}$ be a sequence of non-negative values. Then

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n a_i = \sup_{\substack{F \subseteq \{1,2,3,\dots\} \\ |F| \text{ finite}}} \sum_{i \in F} a_i$$

where both sides lie in $[0, +\infty]$.

Proof:

Put $s_n = \sum_{i=1}^n a_i$, and let $S \in [0, +\infty]$ be the supremum on the right, taken in the extended real line ([2, The Extended Real Line]). Since $s_{n+1} - s_n = a_{n+1} \geq 0$, the sequence (s_n) is increasing. If it is bounded above, it converges to $\sup_n s_n$ by [2, Bounded Monotone Sequences Converge]. If it is not, then for every M some s_N exceeds M , and so does every s_n with $n \geq N$, so $s_n \rightarrow +\infty = \sup_n s_n$. In both cases $\lim_n s_n = \sup_n s_n$.

For $\leq$, each s_n is the sum over the finite set $F = \{1, \dots, n\}$, so $s_n \leq S$ for every n , and hence $\sup_n s_n \leq S$.

For $\geq$, let F be a finite set of indices. If $F = \emptyset$ its sum is $0 \leq s_1$. Otherwise put $n = \max F$. Then $F \subseteq \{1, \dots, n\}$, and since every $a_i \geq 0$, $\sum_{i \in F} a_i \leq s_n \leq \sup_k s_k$. Taking the supremum over F gives $S \leq \sup_n s_n$. So $\lim_n s_n = \sup_n s_n = S$, as we sought to show.

For a series of nonnegative terms, the right-hand side of lemma 1.3.2 does not refer to the order of the terms, so every rearrangement has the same sum. An absolutely convergent series has the same property by [2, Absolutely Convergent Series Are Rearrangement Invariant]. So the order in which the points of Ω are listed does not affect $\mathbb{E}X$.

Lemma 1.3.3: Summing by Blocks

Let T be a countable set, let $c: T \rightarrow \mathbb{R}$, and let T be the union of pairwise disjoint sets T_k , indexed by k in a countable set K . For a countable set S , write $\sum_{t \in S} c_t$ for the sum of the series $\sum_j c_{s_j}$, where $s_1, s_2, \dots$ lists the points of S without repetition, a finite sum if S is finite.

1. If $c_t \geq 0$ for every $t \in T$, then, with both sides in $[0, +\infty]$,

$$\sum_{t \in T} c_t = \sum_{k \in K} \sum_{t \in T_k} c_t$$

2. If $\sum_{t \in T} |c_t| < \infty$, then for each $k \in K$ the series $\sum_{t \in T_k} c_t$ converges absolutely, the series over $k \in K$ of these sums converges absolutely, and the identity in (1) holds.

In both parts the sums do not depend on the lists chosen.

Proof:

A series of nonnegative terms has the same sum in $[0, +\infty]$ for every listing of its terms, since by lemma 1.3.2 that sum is the supremum of the sums over finite sets of terms, and an absolutely convergent series has the same sum for every listing by [2, Absolutely Convergent Series Are Rearrangement Invariant]. This gives the last sentence of the statement. It also shows that if $c \geq 0$, then for every countable $S \subseteq T$

$$\sum_{t \in S} c_t = \sup \left\{ \sum_{t \in F} c_t : F \subseteq S \text{ finite} \right\}$$

and the same holds for a countable family of numbers in $[0, +\infty]$, whose sum is $+\infty$ if one of them is.

1. Write $s_k = \sum_{t \in T_k} c_t$ and $\Sigma = \sum_{t \in T} c_t$, both in $[0, +\infty]$. A finite set $F \subseteq T$ meets only finitely many of the T_k , say those with k in a finite set G , so

$$\sum_{t \in F} c_t = \sum_{k \in G} \sum_{t \in F \cap T_k} c_t \leq \sum_{k \in G} s_k \leq \sum_{k \in K} s_k$$

by the supremum formula, applied first to each T_k and then to the family (s_k) . Taking the supremum over F gives $\Sigma \leq \sum_k s_k$. For the reverse inequality it suffices to show $\sum_{k \in G} s_k \leq \Sigma$ for every finite nonempty $G \subseteq K$. If $\Sigma = +\infty$ this holds. Otherwise every $s_k \leq \Sigma$ is finite. Let $\epsilon > 0$, and for $k \in G$ choose a finite $F_k \subseteq T_k$ with $\sum_{t \in F_k} c_t > s_k - \epsilon/|G|$. The sets F_k are disjoint, so

$$\sum_{k \in G} s_k < \sum_{k \in G} \sum_{t \in F_k} c_t + \epsilon = \sum_{t \in F'} c_t + \epsilon \leq \Sigma + \epsilon$$

where $F' = \bigcup_{k \in G} F_k$. Since ϵ is arbitrary, $\sum_{k \in G} s_k \leq \Sigma$.

2. Put $c_t^+ = \max(c_t, 0)$ and $c_t^- = \max(-c_t, 0)$, so that $c_t = c_t^+ - c_t^-$ and $0 \leq c_t^\pm \leq |c_t|$. Part (1) applied to $|c|$ gives $\sum_k \sum_{t \in T_k} |c_t| = \sum_{t \in T} |c_t| < \infty$, so each $\sum_{t \in T_k} |c_t|$ is finite and these numbers have a finite sum over k . Part (1) applied to c^+ and to c^- gives $\sum_{t \in T} c_t^\pm = \sum_k \sum_{t \in T_k} c_t^\pm$, where every sum is finite because it is at most the corresponding sum of $|c_t|$. Subtracting, [2, The Algebra of Series and the Triangle Inequality](1), used for the series over T , over each T_k and over K , gives $\sum_{t \in T} c_t = \sum_k \sum_{t \in T_k} c_t$. Finally $|\sum_{t \in T_k} c_t| \leq \sum_{t \in T_k} |c_t|$ by part (3) of the same proposition, so the series over k converges absolutely by [2, The Comparison Test], as claimed.

Corollary 1.3.4: Rearranging a Series as a Double Series

For any bijection $\pi : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, where $\mathbb{N} = \{1, 2, 3, \dots\}$,

$$\sum_{n=1}^{\infty} c_n = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} c_{\pi(n,m)}$$

if either $c_n \geq 0$ for all $n \in \mathbb{N}$, with both sides in $[0, +\infty]$, or $\sum_n |c_n| < \infty$, or $\sum_n \sum_m |c_{\pi(n,m)}| < \infty$.

Proof:

Since π is a bijection, the sets $T_n = \{\pi(n, m) : m \in \mathbb{N}\}$ for $n \in \mathbb{N}$ are pairwise disjoint with union $\mathbb{N}$, and $\pi(n, 1), \pi(n, 2), \dots$ lists T_n without repetition. If every $c_k \geq 0$, lemma 1.3.3(1) with these blocks is the formula. If $\sum_k |c_k| < \infty$, part (2) of the same lemma gives it. If $\sum_n \sum_m |c_{\pi(n,m)}| < \infty$, part (1) applied to $|c_k|$ shows that $\sum_k |c_k| = \sum_n \sum_m |c_{\pi(n,m)}| < \infty$, which is the previous case, as claimed.

Proposition 1.3.5: Countable Additivity

Let $(\Omega, \mathbb{P})$ be a discrete probability space, and let A_k , for k in a countable set K , be pairwise disjoint events. Then

$$\mathbb{P}\left(\bigcup_{k \in K} A_k\right) = \sum_{k \in K} \mathbb{P}(A_k)$$

Proof:

By the identity at the start of section 1.3, $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\omega)$ for every $A \subseteq \Omega$, a sum of nonnegative terms. Lemma 1.3.3(1), applied to $T = \bigcup_k A_k$ with the blocks A_k and $c_\omega = \mathbb{P}(\omega)$, gives $\mathbb{P}(\bigcup_k A_k) = \sum_k \sum_{\omega \in A_k} \mathbb{P}(\omega) = \sum_k \mathbb{P}(A_k)$, as claimed.

Proposition 1.3.6: Linearity of Expected Value

Let X and Y be random variables, and let $\mathbb{E}X, \mathbb{E}Y$ be defined (i.e. $\mathbb{E}|X| < \infty$ and $\mathbb{E}|Y| < \infty$). Then for all $a, b \in \mathbb{R}$ the expected value $\mathbb{E}(aX + bY)$ is defined, and

$$\mathbb{E}(aX + bY) = a\mathbb{E}X + b\mathbb{E}Y$$

Proof:

List the points of the discrete space Ω as $\omega_1, \omega_2, \dots$, a finite or infinite list, and put $Z = aX + bY$. When the list is finite, the series below are finite sums, and each step holds by the triangle inequality and the distributive law. By definition 1.3.1, $\mathbb{E}X = \sum_k X(\omega_k)\mathbb{P}(\omega_k)$ and $\mathbb{E}Y = \sum_k Y(\omega_k)\mathbb{P}(\omega_k)$, both series converging absolutely, and by [2, Absolutely Convergent Series Are Rearrangement Invariant] the values do not depend on the list chosen. Since $\mathbb{P}(\omega_k) \geq 0$, the triangle inequality gives $|Z(\omega_k)|\mathbb{P}(\omega_k) \leq |a| |X(\omega_k)|\mathbb{P}(\omega_k) + |b| |Y(\omega_k)|\mathbb{P}(\omega_k)$. Summing over $k \leq N$, and bounding the partial sums of the two convergent series of nonnegative terms by their

sums ([2, Nonnegative Terms and Bounded Partial Sums]), the partial sums of $\sum_k |Z(\omega_k)|\mathbb{P}(\omega_k)$ are at most $|a|\mathbb{E}|X| + |b|\mathbb{E}|Y|$. By the same proposition this series converges, so $\mathbb{E}|Z| < \infty$ and $\mathbb{E}Z$ is defined. Finally, [2, The Algebra of Series and the Triangle Inequality](1) gives

$$\mathbb{E}Z = \sum_k (aX(\omega_k) + bY(\omega_k))\mathbb{P}(\omega_k) = a \sum_k X(\omega_k)\mathbb{P}(\omega_k) + b \sum_k Y(\omega_k)\mathbb{P}(\omega_k) = a\mathbb{E}X + b\mathbb{E}Y$$

as claimed.

Proposition 1.3.7: Countable Change of Variables

Let X be a random variable on a discrete probability space $(\Omega, \mathbb{P})$ with $\mathbb{E}X$ well-defined, and let $a_1, a_2, \dots$ list the values of X without repetition. Then the change of variables formula applies to $\mathbb{E}X$, that is, the series $\sum_n a_n \mathbb{P}(X = a_n)$ converges absolutely and

$$\mathbb{E}X = \sum_{\omega \in \Omega} X(\omega)\mathbb{P}(\omega) = \sum_{n \geq 1} a_n \mathbb{P}(X = a_n) = \sum_{n \geq 1} a_n \mathbb{P}'(a_n)$$

where $\mathbb{P}'$ is the image measure of definition 1.2.13.

Proof:

The sets $X^{-1}(a_n) = \{X = a_n\}$ are pairwise disjoint with union Ω . Since $\mathbb{E}|X| < \infty$, lemma 1.3.3(2), applied to $c_\omega = X(\omega)\mathbb{P}(\omega)$ with these blocks, shows that the series of block sums converges absolutely and that

$$\mathbb{E}X = \sum_{n \geq 1} \sum_{\omega \in X^{-1}(a_n)} X(\omega)\mathbb{P}(\omega) = \sum_{n \geq 1} a_n \sum_{\omega \in X^{-1}(a_n)} \mathbb{P}(\omega) = \sum_{n \geq 1} a_n \mathbb{P}(X = a_n)$$

The middle step factors the constant a_n out of each block sum ([2, The Algebra of Series and the Triangle Inequality](1)), and the last step is the identity at the start of section 1.3. Finally $\mathbb{P}(X = a_n) = \mathbb{P}'(a_n)$ by definition 1.2.13, as claimed.

Yet another way of calculating the expected value is by calculating what happened at possible “tail end” without weighing the value of the probability:

Definition 1.3.8: Tail Probability

Let X be a random variable on a discrete probability space $(\Omega, \mathbb{P})$, and let $t \in \mathbb{R}$. The *tail probability* of X at t is $\mathbb{P}(X \geq t)$. By proposition 1.3.5, applied to the disjoint events $\{X = k\}$,

$$\mathbb{P}(X \geq t) = \sum_{k \geq t} \mathbb{P}(X = k)$$

where the sum runs over the values k of X with $k \geq t$.

The tail probability tends to 0:

$$\lim_{t \rightarrow \infty} \mathbb{P}(X \geq t) = 0$$

Indeed, list the values of X as $a_1, a_2, \dots$ and let $\epsilon > 0$. By proposition 1.3.5, $\sum_n \mathbb{P}(X = a_n) =$

$\mathbb{P}(\Omega) = 1$, so there is an N with $\sum_{n>N} \mathbb{P}(X = a_n) < \epsilon$. For $t > \max(a_1, \dots, a_N)$ the sum giving $\mathbb{P}(X \geq t)$ has only terms with $n > N$, so $\mathbb{P}(X \geq t) < \epsilon$.

Using tail probabilities, we have yet another way of writing the expected value:

Proposition 1.3.9: Expected Value as Tail Probability

Let X be a random variable on a discrete probability space $(\Omega, \mathbb{P})$, with $\mathbb{E}X$ well-defined. If $X(\Omega) \subseteq \mathbb{N}$, then

$$\mathbb{E}X = \sum_{m=1}^{\infty} \mathbb{P}(X \geq m)$$

If $X(\omega) \geq 0$ for every $\omega \in \Omega$, whether or not the image of X is contained in $\mathbb{N}$, then we can write

$$\mathbb{E}X = \int_0^{\infty} \mathbb{P}(X \geq t) dt$$

In Analysis, this is called the “layer-cake representation”.

Proof:

Since Ω is countable, so is the image $X(\Omega)$. List its elements without repetition as $a_1, a_2, \dots$ and put $p_n = \mathbb{P}(X = a_n)$. By proposition 1.3.7, $\mathbb{E}X = \sum_n a_n p_n$.

The case $X(\Omega) \subseteq \mathbb{N}$. Listing the values in increasing order, dropping the term $0 \cdot \mathbb{P}(X = 0)$ if 0 is a value, and inserting the terms $k \mathbb{P}(X = k) = 0$ for the integers $k \geq 1$ that X does not take, gives $\mathbb{E}X = \sum_{k \geq 1} k \mathbb{P}(X = k)$, a convergent series of nonnegative terms. Let $M \geq 1$. For $1 \leq m \leq M$, definition 1.3.8 gives $\mathbb{P}(X \geq m) = \sum_{k=m}^{M-1} \mathbb{P}(X = k) + \mathbb{P}(X \geq M)$, where the finite sum is empty when $m = M$. Summing over m , each $\mathbb{P}(X = k)$ with $1 \leq k \leq M-1$ is counted once for each $m \leq k$, so

$$\sum_{m=1}^M \mathbb{P}(X \geq m) = \sum_{k=1}^{M-1} k \mathbb{P}(X = k) + M \mathbb{P}(X \geq M)$$

By definition 1.3.8 again, $0 \leq M \mathbb{P}(X \geq M) = \sum_{k \geq M} M \mathbb{P}(X = k) \leq \sum_{k \geq M} k \mathbb{P}(X = k)$, and the right-hand side equals $\mathbb{E}X - \sum_{k=1}^{M-1} k \mathbb{P}(X = k)$, which tends to 0 as $M \rightarrow \infty$. Hence $\sum_{m=1}^M \mathbb{P}(X \geq m) \rightarrow \mathbb{E}X$, which is the first formula. In the array below, column k holds k copies of $\mathbb{P}(X = k)$, and row m , counted from the bottom, sums to $\mathbb{P}(X \geq m)$. The first formula adds the array up by rows, and $\mathbb{E}X = \sum_k k \mathbb{P}(X = k)$ adds it up by columns.

$$\begin{array}{cccc} & & & \ddots \\ & & & \mathbb{P}(X = 3) & \cdots \\ & & \mathbb{P}(X = 2) & \mathbb{P}(X = 3) & \cdots \\ \mathbb{P}(X = 1) & \mathbb{P}(X = 2) & \mathbb{P}(X = 3) & \cdots \end{array}$$

The case $X \geq 0$. Every a_n is nonnegative. When the list $a_1, a_2, \dots$ is finite, with r terms, put $a_n = 0$ and $p_n = 0$ for $n > r$, which changes none of the sums below. By definition 1.3.8, for $t \geq 0$,

$$F(t) := \mathbb{P}(X \geq t) = \sum_{n: a_n \geq t} p_n = \sum_{n \geq 1} p_n I(a_n \geq t)$$

and in particular $\sum_n p_n = \mathbb{P}(X \geq 0) = 1$. Fix $T > 0$, and for $N \geq 1$ let $G_N(t) = \sum_{n=1}^N p_n I(a_n \geq t)$ on $[0, T]$. For every $t \in [0, T]$,

$$0 \leq F(t) - G_N(t) = \sum_{n>N} p_n I(a_n \geq t) \leq \sum_{n>N} p_n$$

and the right-hand side, a tail of the convergent series $\sum_n p_n$, tends to 0 as $N \rightarrow \infty$. So $G_N \rightarrow F$ uniformly on $[0, T]$. On $[0, T]$ the function $t \mapsto I(a_n \geq t)$ is the indicator of $[0, c]$ with $c = \min(a_n, T)$, and its integral is c : if $c = T$ it is constant, and if $c < T$ its lower and upper Darboux sums over the partition with points $0, c + \delta, T$, together with c when $c > 0$, for $0 < \delta < T - c$, are c and $c + \delta$, so [2, The Integrability Criterion] applies. By [2, Linearity of the Integral], $G_N \in \mathcal{R}[0, T]$ with $\int_0^T G_N = \sum_{n=1}^N p_n \min(a_n, T)$, and [2, Uniform Convergence and Integration] gives $F \in \mathcal{R}[0, T]$ with

$$\int_0^T F(t) dt = \lim_{N \rightarrow \infty} \sum_{n=1}^N p_n \min(a_n, T) = \sum_{n \geq 1} p_n \min(a_n, T) =: \Phi(T)$$

By [2, Improper Integral], the second formula says that $\Phi(T) \rightarrow \mathbb{E}X$ as $T \rightarrow \infty$. Since $0 \leq \min(a_n, T) \leq a_n$, comparing partial sums and applying [2, Limits Preserve Weak Inequalities] gives $\Phi(T) \leq \sum_n p_n a_n = \mathbb{E}X$. Let $\epsilon > 0$, choose N with $\sum_{n>N} p_n a_n < \epsilon$, and let $T \geq \max(a_1, \dots, a_N)$. Since every term of $\Phi(T)$ is nonnegative, $\Phi(T) \geq \sum_{n=1}^N p_n \min(a_n, T) = \sum_{n=1}^N p_n a_n > \mathbb{E}X - \epsilon$. So $|\Phi(T) - \mathbb{E}X| < \epsilon$ for all such T , completing the proof.

Example 1.3.10

1. Show that for any nonnegative random variable $X \geq 0$ and any real $n > 0$ with $\mathbb{E}X^n < \infty$,

$$\mathbb{E}X^n = \int_0^\infty n t^{n-1} \mathbb{P}(X \geq t) dt$$

and that if X is any random variable with $\mathbb{E}|X|^n < \infty$, then:

$$\mathbb{E}|X|^n = \int_0^\infty n t^{n-1} \mathbb{P}(|X| \geq t) dt$$

[hint: apply proposition 1.3.9 to X^n , which gives an integral of $\mathbb{P}(X^n \geq s)$, and change variables with $s = t^n$ by theorem 0.4.1]

2. Show that if $c, \epsilon > 0$ and $t_0 \geq 0$ satisfy

$$\mathbb{P}(|X| \geq t) \leq \frac{c}{t^{2+\epsilon}} \quad \text{for all } t \geq t_0$$

then $\mathbb{E}X^2 < \infty$.

1.3.2 Example of Discrete Distributions

In this section, we will go over some common distributions and their representations. In particular, we will review

1. Bernoulli Distribution
2. Product Bernoulli Distribution
3. Binomial Distribution Distribution
4. Poisson Distribution

Each of these distributions will be a quantification of a “probability question”, and essentially all of them will can be formulated as a question about coin flips!

We will also explore some examples of where these distribution are useful (like the Erdős-Rényi random graph).

Bernoulli Distribution

Perhaps the most common probabilistic scenario is that of a coin flip: whether an outcome will or will not happen:

Definition 1.3.11: Bernoulli Distribution

Let $\Omega = \{0, 1\}$. Then for some $p \in [0, 1]$

$$\mathbb{P}(0) = p, \quad \mathbb{P}(1) = 1 - p$$

Then $\mathbb{P}$ is called the *Bernoulli Distribution* and is denoted $B(p)$.

Notice that since the probability space is in $\mathbb{R}$, we are justified in calling the probability measure a distribution. The Bernoulli Distribution is essentially representing a coin toss. In fact, essentially all the following distributions can be thought of as asking different types of questions about outcome of repeated coin tosses (as we will show)! In a sense, this is the foundational scenario for statistics. If $p = 1/2$, then we say it’s a fair coin, and an unfair coin if $p \neq 1/2$.

As we will soon be encountering, we are often more interested in random variables that take on the role of such a distribution. In a sense, our random variable will now represent a “question” we are asking about our probability space:

Definition 1.3.12: Bernoulli Random Variable

Let $(\Omega, \mathbb{P})$ be a probability space, and let $p \in [0, 1]$. Then if X is a random variable on Ω such that

$$\mathbb{P}(X = 1) = p \quad \mathbb{P}(X = 0) = 1 - p$$

Then X is called a *Bernoulli Random Variable* and is denoted $X \sim B(p)$.

For every distribution we will introduce a random variable equivalent will be given too (i.e. the sample space and image measure will be the desired distribution). If $\mathbb{P}$ is an X distribution (where

X is the name of a distribution, like Bernoulli) then the identity function $X(\omega) = \omega$ is always a X random variable.

Example 1.3.13: expected value of $X \sim B(p)$

Since $p + 1 - p = 1$, we only need to sum over 1 and 0. Since this is a finite random variable, there is no special convergence condition that needs to be checked, and so:

$$\mathbb{E}X = 1 \cdot p = 0 \cdot (1 - p) = p$$

A single Bernoulli Distribution gives us the value for a single coin toss, but this might be limiting us in what we can measure, in particular, in the fact that we are often measuring many events at once and want to know the probability of a particular result given the, say, n coins. For that, we introduce the product of Bernoulli Distributions and Variables:

Definition 1.3.14: Product of Bernoulli

Let $p_1, \dots, p_n \in [0, 1]$. And let $\Omega = \{0, 1\}^n$. Then if the probability measure is defined as

$$\mathbb{P}(x_1, x_2, \dots, x_n) = \prod_{i=1}^n p_i^{x_i} (1 - p_i)^{1-x_i}$$

Then $\mathbb{P}$ is called the *product Bernoulli Distribution* and is denoted $\bigotimes_i^n B(p_i)$.

This is indeed a well-defined probability measure. Notice that

$$\sum_{x \in \{0,1\}^n} \prod_{i=1}^n p_i^{x_i} (1 - p_i)^{1-x_i} = \prod_{i=1}^n \sum_{x \in \{0,1\}^n} p_i^{x_i} (1 - p_i)^{1-x_i} = \prod_{i=1}^n 1 = 1$$

and the rest of the conditions become easy to check.

Since we are working over a “multi-dimensional” universe, we will sometimes want to have a random variable for “each dimension”. We can define a random variable for each component, though note that we must be careful on how to make it only work on 1 component, since a random variable on Ω^n must have as domain Ω^n . This might get notationally cumbersome, so to help with notation, we introduce a new concept:

Definition 1.3.15: Random Vector

Let $(\Omega^n, \mathbb{P})$ be a probability space. Then a *random vector* $X : \Omega^n \rightarrow \mathbb{R}^n$, $X = (X_1, \dots, X_n)$ is an n -tuple of random variables $X_i : \Omega^n \rightarrow \mathbb{R}$, and if $A \subseteq \Omega^n$ is an event, and $A_i = \pi_i(A)$ where π_i is the i th coordinate map, then:

$$\mathbb{P}(X \in A) = \mathbb{P}(\{x \mid X_1 \in A_1, \dots, X_n \in A_n\})$$

Note that it need not be that it is the same Ω every time, however it is convenient at the beginning of probability theory to stick to this special case to get a grasp of notions like independence or defining multiple random variables on the same space. We will only much later cover examples where we might have different spaces (in particular, products of discrete spaces such as $\{0, 1\}$ with $\mathbb{R}$ appear in section 4.3).

Example 1.3.16: Product Of Bernoulli Random Vector

If $X = (X_1, X_2, \dots, X_n)$ is a random variable where each component is the random variable X_i , then if

$$\mathbb{P}(X = x) = \prod_{i=1}^n p_i^{x_i} (1 - p_i)^{1-x_i}$$

then $X \sim \bigotimes_i^n B(p_i)$. We will call X the *random vector*. By the way we have defined the product of Bernoulli variables, it is tempting to say that by definition:

$$\mathbb{P}(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) = \prod_i^n \mathbb{P}(X_i = x_i)$$

and so the variables $X_1, X_2, \dots, X_n$ are independent. However, we have to be more careful than that. In particular, notice that $B(p_i)$ not defined on $\{0, 1\}^n$ but on $\{0, 1\}$. Therefore, we must be more careful with our formulation. To simplify the proof for the moment, let's say $(\{0, 1\}^n, \mathbb{P})$ is a probability space with $\mathbb{P}$ being the product Bernoulli Distribution. The natural definition of the Bernoulli random variable on the i th coordinate X_i as a random variable on $\{0, 1\}^n$ would be

$$X_i : \Omega \rightarrow \mathbb{R} \quad X_i(x_1, x_2, \dots, x_i, \dots, x_n) = x_i$$

i.e. the identity function on the i th coordinate. Then the sample space of X_i is equivalent to $\{0, 1\}$ and the $X_i \sim B(p)$! To see this, simply go through the computations remembering that the countable probability measure is additive. For notational simplicity of the change of variables, let $A_i = \{\omega \in \Omega \mid X_i(\omega) = x_i\}$. Then:

$$\begin{aligned} \mathbb{P}(X_i = x_i) &= \sum_{a \in A_i} \mathbb{P}(X_i = a) \\ &= \sum_{a \in A_i} \prod_{i=1}^n p_i^{a_i} (1 - p_i)^{1-a_i} \\ &\stackrel{!}{=} p_i^{x_i} (1 - p_i)^{1-x_i} \sum_{a \in A} \prod_{j=1}^n p_j^{a_j} (1 - p_j)^{1-a_j} \\ &= p_i^{x_i} (1 - p_i)^{1-x_i} \prod_{j \neq i} \underbrace{\sum_{a_j \in \{0,1\}} p_j^{a_j} (1 - p_j)^{1-a_j}}_1 \\ &= p_i^{x_i} (1 - p_i)^{1-x_i} \end{aligned}$$

showing us that our earlier naïve formula for independence was in fact accurate.

Before moving on to our next distribution, a quick word must be set if $p_1 = p_2 = \dots = p_n$ are all equal. Then each random variable $X_1, X_2, \dots, X_n$ will be identically distributed. Furthermore, since they are all independent, they are all *independently identically distributed* random variables (or *i.i.d* variables for short). When this is the case, the equation in definition 1.3.14 simplifies to:

$$\mathbb{P}(X = x) p^{\sum_{i=1}^n x_i} (1 - p)^{n - \sum_{i=1}^n x_i}$$

To emphasize the difference, we usually denote such a distribution with $B(p)^{\otimes n}$. This random variable will be much more common in these notes thanks to it's application to the following important

example:

Example 1.3.17: Erdős-Rényi Random Graph

Let $p \in [0, 1]$ represent the probability of a coin landing on heads, and let $V = \{v_1, v_2, \dots, v_n\}$ be a set of elements called vertices. Then for each possible pair (v_i, v_j) , we will flip a coin, and if it lands on heads, we add that edge to our graph. If it lands on tails, we do not add the edge. The resulting randomly generated graph is denoted $G(n, p)$.

Since each coin toss was independent and identical, then the chances of getting any one graph is the product of i.i.d. Bernoulli variables

$$(X_1, X_2, \dots, X_n) \sim B(p)^{\otimes m} \quad m = \binom{n}{2}$$

where m represents the possible number of vertices (these vertices are undirected).

These graphs have multiple application in sociology in modeling networks of people given some appropriate conditions or simplifications. We will reference back to this graph multiple times when developing new tools to see what properties we can find from it!

Binomial Distribution

Let's say we have a coin with probability of landing on heads is p . If we flip the coin n times, what is the probability that we get k heads? To represent this problem, take $\Omega = \{0, 1\}^n$ to be the n coin flips. We want our probability measure $\mathbb{P}$ to measure the amount of heads we land on. As an exercise, think though why the following formula represents this scenario

Definition 1.3.18: Binomial Random Variable

Let $(\Omega, \mathbb{P})$ be a probability space, and let X be a random variable on Ω . Then if

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \text{ if } k > n?$$

then we call X a *Binomial Random Variable*, and is denoted $X \sim B(n, p)$.

As an exercise, can you define the Binomial distribution; you have to define an explicit Ω and $\mathbb{P}$. The name of the formula is inspired by the binomial formula:

$$(a + b)^n = \sum_{k=1}^n \binom{n}{k} a^k b^{n-k}$$

We can in fact use this formula to show that this is a well-defined distribution, since

$$\sum_{k=1}^n \binom{n}{k} p^k (1-p)^{n-k} = (p + 1 - p)^n = 1^n = 1$$

Given the definition, this random distribution should very naturally connect to the Bernoulli random variable, and indeed it does. If $X_1, X_2, \dots, X_n$ are i.i.d. Bernoulli Random Variables, let

$$S_n = X_1 + X_2 + \dots + X_n$$

The sample space of S_n is $\{0, 1, \dots, n\}$. Next, we will show $S_n \sim B(n, p)$. First, notice that the set $\{S_n = k\}$ can be broken down into $\coprod \{X = (X_1, X_2, \dots, X_n) = x\}$ for all $x \in \{0, 1\}^n$ that give the value of k . Then each $X = x$ is just a Product Bernoulli random variable, so

$$\mathbb{P}(X = x) = p^k(1-p)^{n-k}$$

Since there are $\binom{n}{k}$ possibilities, by countable additivity, we get:

$$\mathbb{P}(S_n = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

showing that $S_n \sim B(n, p)$.

Example 1.3.19: expected value of Binomial

Let $X \sim B(n, p)$. What is $\mathbb{E}(X)$?

Solution There are multiple ways of approaching this question. As usual in probability, it is important to have a good intuition to be able to spot an easy resolution. In this case, by what we have just proved, $X \sim S_n$ for appropriate i.i.d. Bernoulli Random variables $X_1, X_2, \dots, X_n$. Then by proposition 1.3.6, the expected value is linear, so

$$\mathbb{E}X = \mathbb{E}S_n = \mathbb{E}(X_1 + X_2 + \dots + X_n) = \mathbb{E}X_1 + \mathbb{E}X_2 + \dots + \mathbb{E}X_n = np$$

▼

If each trial has r possible outcomes, with probabilities $p_1, \dots, p_r$ and $\sum_i p_i = 1$, then in n independent trials the probability of k_1 outcomes of the first kind, $\dots$, k_r outcomes of the r -th kind, where $k_1 + \dots + k_r = n$, is

$$\frac{n!}{k_1! \dots k_r!} p_1^{k_1} \dots p_r^{k_r}$$

These numbers sum to $(p_1 + \dots + p_r)^n = 1$ by theorem 0.1.5, and they define the *multinomial distribution*.

Geometric Distribution

Toss a coin with probability $p > 0$ of heads until the first head appears. The number N of tosses takes values in $\{1, 2, \dots\}$, apart from the outcome that no head ever appears. That outcome has probability 0: it lies in the event that the first n tosses are all tails, of probability $(1-p)^n$, for every n , and $(1-p)^n \rightarrow 0$. So it can be omitted.

Definition 1.3.20: Geometric Distribution

Let $(\Omega, \mathbb{P})$ be a probability space, let $p \in (0, 1]$, and let N be a random variable. Then if

$$\mathbb{P}(N = n) = (1-p)^{n-1}p \quad (n = 1, 2, \dots)$$

we call N a *Geometric Random Variable*

The geometric random variables is not seen as often and so it usually does not have a special symbol like $B(p)$ or $B(n, p)$ associated to it. Some authors sometimes use $X \sim T(n)$ to represent a Geometric

Random Variable. Like the binomial random variable, this random variable can also be represented in terms of the Bernoulli random variable. In particular, If $\{N = k\}$ is our set, then we can define k i.i.d. Bernoulli random variables,

$$\{N = k\} = \{X_1 = 0, X_2 = 0, \dots, X_k = 1\}$$

Then using the fact that the Bernoulli are independent, we see how this is identical to the geometric random variable.

Example 1.3.21: Geometric Expected Value

1. Prove that the geometric distribution is indeed a probability measure
2. What is the expected value of the geometric distribution? The answer is:^a
3. What is $\mathbb{E}X^2$? The answer is ^b

^a1 over the probability of hitting heads

^b $(2 - p)/p^2$

Poisson Distribution

The Poisson distribution is one of the harder distributions to motivate compared to the other ones. It however comes up all over the place

Here are some of my favourite uses:

1. The number of call per hour a call center receives: $\mathbb{P}(X = k)$ is Poisson
2. The number of expected restaurant clients per day
3. Number of website visitors per hour
4. Number of Bankruptcies filed per month
5. Number of Network failures per week
6. The pattern of Bombing of London during WWII (more on this in section 1.4.1)

Definition 1.3.22: Poisson Distribution

Let $(\Omega, \mathbb{P})$ be a probability space and $\lambda > 0$. Then if $\mathbb{P}$ satisfies:

$$\mathbb{P}(k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

for $k \in \mathbb{Z}_{\geq 0}$, $\mathbb{P}$ is called a Poisson Distribution. If X is a random variable on some probability space such that

$$\mathbb{P}(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

then X is called a *Poisson Random Variable*, and $X \sim \text{Poiss}(\lambda)$

As an exercise, prove that it is indeed a probability measure. It essentially comes down to the fact that $e^{-\lambda}$ is a constant that can be pulled out, and everything within that's left is the Taylor series of e^λ .

Example 1.3.23: Expected Value of Poisson

If $X \sim \text{Poiss}(\lambda)$, what is its expected value?

Solution Given:

$$\mathbb{E}X = \sum_{k=1}^{\infty} k \mathbb{P}(X = k) = \sum_{k=1}^{\infty} k \frac{\lambda^k}{k!} e^{-\lambda}$$

and manipulating dividing by k and manipulating the sum so that we get back to the Poisson, we can sum it out and get:

$$\mathbb{E}X = \lambda$$

▼

For this reason, λ is usually called the *mean*. It also means that in some exercises you will do, you will be given some information about the average occurrence of an event. More often than not, that implies the Poisson distribution is the appropriate distribution you use to model it.

1.4 Poisson Properties

The Poisson distribution has many interesting properties, and so it's worth taking a moment to understand them more thoroughly.

Stability Property

The first property is a sort of “closure of distribution type” known as a *stability condition*. Let $(\Omega, \mathbb{P})$ be a probability space, and let X_1, X_2 be two i.i.d. Poisson Random Variables on $(\Omega, \mathbb{P})$: $X_1 \sim \text{Poiss}(\lambda_1)$, $X_2 \sim \text{Poiss}(\lambda_2)$. Since they are independent:

$$\mathbb{P}(X_1 = n, X_2 = m) = \mathbb{P}(X_1 = n) \mathbb{P}(X_2 = m) = \frac{\lambda_1^n}{n!} e^{-\lambda_1} \frac{\lambda_2^m}{m!} e^{-\lambda_2}$$

To see that we can construct such a pair of independent random variables, simply take $\Omega = \{0, 1, \dots\} \times \{0, 1, \dots\}$ and re-define X_1 and X_2 in the same way we have done for the product of Bernoulli random variables³.

Lemma 1.4.1: Stability of Poisson

let $X_1 \sim \text{Poiss}(\lambda_1)$, $X_2 \sim \text{Poiss}(\lambda_2)$ be two i.i.d. Poisson random variables. Then the sum $X = X_1 + X_2$ has $\text{Poiss}(\lambda_1 + \lambda_2)$ distribution:

$$X \sim \text{Poiss}(\lambda_1 + \lambda_2)$$

³More generally, this shows we can always take random variables $X_1, X_2, \dots, X_n$ and say they are independent without any problem about whether such random variables exist

Proof:

The following is a common trick for the sum of two random variables^a.

First, note that both X_1 and X_2 only have in $\mathbb{Z}_{\geq 0}$. Thus $X_1 + X_2$ only has values in $\mathbb{Z}_{\geq 0}$ too.

If we have $X_1 + X_2 = n$, then $X_1 = m$, $X_2 = n - m$, for any $0 \leq m \leq n$. Therefore

$$\begin{aligned}
 \mathbb{P}(X = x) &= \sum_{m=0}^n \mathbb{P}(X_1 = m, X_2 = n - m) \\
 &= \sum_{m=0}^n \mathbb{P}(X_1 = m) \mathbb{P}(X_2 = n - m) && \text{independence} \\
 &= \sum_{m=0}^n \frac{\lambda^m}{m!} e^{-\lambda_1} \frac{\lambda^{n-m}}{(n-m)!} e^{-\lambda_2} \\
 &= \frac{1}{n!} \left(\sum_{m=0}^n \frac{n!}{m!(n-m)!} \lambda_1^m \lambda_2^{n-m} \right) e^{-(\lambda_1 + \lambda_2)} \\
 &= \frac{(\lambda_1 + \lambda_2)^n}{n!} e^{-(\lambda_1 + \lambda_2)}
 \end{aligned}$$

where the last equality comes from the binomial formula. Thus $X \sim \text{Pois}(\lambda_1 + \lambda_2)$, as we sought to show

^aThis trick is the discrete form of *convolution*, which reappears for densities in proposition 4.4.4.

As an exercise, show that this generalizes to n i.i.d. Poisson variables (use induction)

1.4.1 Poisson Approximation of the Binomial

We know resume the goal of establishing how every distribution we present is similar in some way to the Bernoulli distribution. As we saw, the binomial distribution is the sum of i.i.d. Bernoulli distributions. The Poisson distribution is related to the binomial distribution in that the limit of a binomial distribution will approach a Poisson given the probability shrinks accordingly:

$$\text{Pois}(\lambda) = \lim_{n \rightarrow \infty} B\left(n, \frac{\lambda}{n}\right)$$

i.e. $B\left(n, \frac{\lambda}{n}\right)$ converges “point-wise” to $\text{Pois}(\lambda)$. To see this, fix any k , and notice that

$$\begin{aligned}
 &= \binom{n}{k} p^k (1-p)^{n-k} \\
 &= \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\
 &= \frac{\lambda^k}{k!} \frac{n}{n} \frac{n-1}{n} \cdots \frac{n-k+1}{n} \cdots \left(1 - \frac{\lambda}{n}\right)^{n-k} \\
 &\rightarrow \frac{\lambda^k}{k!} 1 \cdot 1 \cdots 1 \cdot e^{-\lambda} \\
 &= \frac{\lambda^k}{k!} e^{-\lambda}
 \end{aligned}$$

where it converges to that value since k is fixed, so each term in the middle converges to 1. If we look back at the list of examples given to motivate the Poisson distribution, this limit makes sense: the further “away” you go, the less likely it is that you hit the event. The bombing of London might be particularly insightful: the mathematicians were trying to determine whether or not the Germans were bombing randomly, or targeting specific areas. If there is a central location that is usually bombed, then as you move out radially from that location, and so the amount of area that *can* be bombed there goes up (which parallels our $\frac{\lambda}{n}$), and so the probability will shrink accordingly. If the bombing were random, then we might expect no such concentration, but such concentration was found!

This stability result can be made more precise with a more exact error bound on the difference between the Poisson and the Binomial, and generalized to the product of Bernoulli distributions (and hence, with different p_i 's). The way will be measuring this difference between two random variables is by a new process called *coupling*

Theorem 1.4.2: Approximation of Binomial using Poisson

Let $X_i \sim B(p_i)$ be n Bernoulli random variables on the same probability space with $p_i \in (0, 1)$ for $1 \leq i \leq n$. Define

$$S_n = X_1 + X_2 + \cdots + X_n \quad \lambda = p_1 + p_2 + \cdots + p_n$$

Then for any subset $A \subseteq \mathbb{Z}_{\geq 0}$

$$|\mathbb{P}(S_n \in A) - \text{Poiss}_\lambda(A)| \leq \sum_{i=1}^n p_i^2$$

If $p_1 = p_2 = \cdots = p_n$ and $S_n \sim B(n, p)$, then $\lambda = np$ and the aforementioned equation becomes

$$|\mathbb{P}(S_n \in A) - \text{Poiss}_\lambda(A)| \leq np^2 = \frac{\lambda^2}{n}$$

which if λ is fixed and we take $n \rightarrow \infty$, $B(n, p)$ is approximated by a Poisson.

Proof:

We will start off with a single Bernoulli variable $X_1 \sim B(p)$ and $Y_1 \sim \text{Poiss}(p)$ both defined on the same probability space. Note that the probability space in question does not matter as the random variables that take on these values are not related to the probability space. We will first construct a new random variable that will give us a sense of how X_1 and Y_1 will be “close” when p is small. First, consider the usual sample space for Y_1 :

$$\Omega = \{0, 1, 2, \dots\} \quad \mathbb{P}'(k) = \frac{p^k}{k!} e^{-p}$$

Notice that $\mathbb{P}(0) = e^{-p} \geq 1 - p$ and is only equal if p were 0 (something we are not allowing). What we will do is split e^{-p} into two numbers so that one of them can become the new probability of $\mathbb{P}(0)$ which will be the probability for both X_1 and Y_1 , and the other will be the probability for 1 X_1 , with the second number being the probability of $\mathbb{P}(-1)$ for a new -1 we are adding to Ω to take it into account (say Ω_+). To split e^{-p} , notice that the function $1 - x$ is tangent to

e^{-x} at zero, and so we can slit e^{-p} into two numbers:

$$e^{-p} = (1 - p) + (e^{-p} - 1 + p)$$

Thus, let $\mathbb{P}_p(-1) = 1 - p$, $\mathbb{P}_p(0) = e^{-p} - 1 + p$. All other values of $\mathbb{P}_p$ are defined as for the distribution of Y :

$$\mathbb{P}_p(k) = \frac{p^k}{k!} e^{-p}, \quad k \geq 1$$

Notice now that on our new probability space, we can still define the Bernoulli distribution X and Poisson distribution Y :

$$X(\omega) = \begin{cases} 0 & \omega = -1 \\ 1 & \text{otherwise} \end{cases} \quad Y(\omega) = \begin{cases} 0 & \omega \in \{-1, 0\} \\ \omega & \text{otherwise} \end{cases} \quad (1.1)$$

This construction now let's us compare these two random variables! In particular, we can ask what is $\mathbb{P}_p(X = Y) = \mathbb{P}_p(\{\omega \in \Omega \mid X(\omega) = Y(\omega)\})$ and $\mathbb{P}_p(X \neq Y)$. For the first case, notice the condition is satisfied if and only if $\omega \in \{-1, 1\}$ and so

$$\mathbb{P}(X = Y) = 1 - p + pe^{-p} \geq 1 - p + p(1 - p) = 1 - p^2$$

using the fact that $e^{-p} \geq 1 - p$. Thus, by the properties of a probability measure, the complement of the set will have value:

$$\mathbb{P}(X \neq Y) \leq p^2 \quad (1.2)$$

Notice that when p is small, then the value is very small, showing that the two probabilities are very similar for small p 's.

In the more general case, Let $X_i \sim B(p_i)$ for $1 \leq i \leq n$ and $Y_i \sim \text{Poiss}(p_i)$. We will couple X_i to Y_i on the same distribution in the same way. To achieve this, we will use the Product Bernoulli distribution for the X_i 's and the stability property of the Poisson distribution. For the new sample space, take the same Ω_+ as before, and define

$$\Omega = (\Omega_+)^n$$

If $\omega = (\omega_1, \omega_2, \dots, \omega_n)$, define $\mathbb{P}_+$ on Ω to be

$$\mathbb{P}_+ = \prod_{i=1}^n \mathbb{P}_{p_i}(\omega_i)$$

On this, notice that we can define the X_i in the following way:

$$X_i(\omega) = X(\omega_i) \quad Y_i(\omega) = Y(\omega_i)$$

where X, Y where defined in equation(1.1). Clearly

- $X_i \sim B(p_i)$
- These random variables are independent (For Bernoulli, use the same trick as for the product of Bernoulli's shown earlier. Similar trick for the Poisson random variables)

Let S_n be the sum of Bernoulli's and S'_n be the sum of Poisson's. If $S_n \neq S'_n$, then they are not equal on at least on pair X_i, Y_i , which means that

$$\{S_n \neq S'_n\} \subseteq \bigcup_{i=1}^n \{X_i \neq Y_i\}$$

Thus, by the properties of measures:

$$\mathbb{P}_+(S_n \neq S'_n) \leq \sum_{i=1}^n \mathbb{P}_+(X_i \neq Y_i)$$

By equation (1.2), and independence of the variables, we have that:

$$\mathbb{P}_+(S_n \neq S'_n) \leq \sum_{i=1}^n (p_i)^2$$

To get our desired equality from the definition, notice that $\mathbb{P}(S_n \in A) = \mathbb{P}(S_n \in A, S'_n \in A) + \mathbb{P}(S_n \in A, S'_n \notin A)$, the two sets being disjoint. Then:

$$\begin{aligned} \mathbb{P}_+(S_n \in A) - \mathbb{P}_+(S'_n \in A) &= \mathbb{P}(S_n \in A, S'_n \in A) + \mathbb{P}(S_n \in A, S'_n \notin A) \\ &\quad - \mathbb{P}(S'_n \in A, S_n \in A) - \mathbb{P}(S'_n \in A, S_n \notin A) \\ &= \mathbb{P}(S_n \in A, S'_n \notin A) - \mathbb{P}(S'_n \in A, S_n \notin A) \\ &\leq \mathbb{P}(S_n \in A, S'_n \notin A) + \mathbb{P}(S'_n \in A, S_n \notin A) \\ &= \mathbb{P}(S_n \neq S'_n) \end{aligned}$$

with the same proof working if we take $\mathbb{P}_+(S'_n \in A) - \mathbb{P}_+(S_n \in A)$, hence justifying the absolute values. Combining with our earlier result, this gives us:

$$|\mathbb{P}_+(S_n \in A) - \mathbb{P}_+(S'_n \in A)| \leq \mathbb{P}_+(S_n \neq S'_n) \leq \sum_{i=1}^n (p_i)^2$$

completing the proof

1.5 $\mathbb{E}f(X_1, \dots, X_n)$ and Independence

So far, we showed that the sum of i.i.d. Bernoulli random variables forms a binomial random variable, and the limit of a particular type of Binomial (i.e. $B(n, \frac{\lambda}{n})$) is the Poisson random variable, and that summing two independent Poisson is also a Poisson.

A goal that will pop up once in a while in this book is to understand what happens when we combine random variables in certain ways. In future sections of this book, we will be more general with what functions we are taking between random variables (essentially treating random variables as elements in a function space, and hence the term “variable”). So we will see notation like $f(X_k)$ or $f((X_k)_{k \in I})$ to represent a function on a random variable or variables. I like to think of this as finding relations between probability questions. This idea will be explored in more depth in the next section when we will try to find the expected value of the not just a random variable $\mathbb{E}X$, but of $\mathbb{E}f(X_1, X_2, \dots, X_n)$.

There is a very neat result that will allow us to compute this value easily, and if the random variables are all independent, then this becomes even easier!

1.5.1 Independence Revisited

We now take more time to discuss independence. Recall that if we flip a twice, and assume that there is no relation between the chance of the first and second coin, then the probabilities of the sequence $(1, 1)$ (i.e. heads, heads) is equal to the probability of 1 times the probability of 1. This motivates the following definition:

Definition 1.5.1: Independence of Events

Let $(\Omega, \mathbb{P})$ be a probability space, and let $A, B \subseteq \Omega$ be events. Then the events A and B are said to be *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$$

that, the probability of event A and B happening is equal to the probability of A times the probability of B .

Another way to think of this in terms of conditional

Definition 1.5.2: Conditional Probability

Let $(\Omega, \mathbb{P})$ be a [discrete] probability space, and let $A, B \subseteq \Omega$ be events with $\mathbb{P}(B) > 0$. Then

$$\mathbb{P}(A|B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

My intuition behind this is that we are changing the probability space to the smaller one. To see this, consider:

$$\mathbb{P}(A|\Omega) = \frac{\mathbb{P}(A \cap \Omega)}{\mathbb{P}(\Omega)} = \frac{\mathbb{P}(A)}{1} = \mathbb{P}(A)$$

Proposition 1.5.3: Chain Rule of Probability

Let $(\Omega, \mathbb{P})$ be a probability space, let $n \geq 1$, let $A_1, \dots, A_n \subseteq \Omega$ be events, and write $B_k := A_1 \cap \dots \cap A_k$ for $1 \leq k \leq n$.

1. If $n = 1$ or $\mathbb{P}(B_{n-1}) > 0$, then $\mathbb{P}(B_k) > 0$ for every $k \leq n - 1$, and

$$\mathbb{P}(A_1 \cap \dots \cap A_n) = \mathbb{P}(A_1) \prod_{k=2}^n \mathbb{P}(A_k | B_{k-1})$$

2. If $n \geq 2$ and $\mathbb{P}(B_{n-1}) = 0$, the identity in (1) holds, with both sides equal to 0, whatever values in $[0, 1]$ are assigned to the conditional probabilities $\mathbb{P}(A_k | B_{k-1})$ with $\mathbb{P}(B_{k-1}) = 0$.

Proof. If $j \leq k$, then $B_k \subseteq B_j$, and B_j is the disjoint union of B_k and $B_j \setminus B_k$, so properties (3) and (1) of definition 1.2.1 give $\mathbb{P}(B_j) = \mathbb{P}(B_k) + \mathbb{P}(B_j \setminus B_k) \geq \mathbb{P}(B_k)$.

(1) By this inequality, $\mathbb{P}(B_k) \geq \mathbb{P}(B_{n-1}) > 0$ for $k \leq n-1$, so each conditional probability in the product is defined. Induct on n . For $n = 1$ the product is empty and both sides are $\mathbb{P}(A_1)$. For $n \geq 2$, $B_n = B_{n-1} \cap A_n$ and $\mathbb{P}(B_{n-1}) > 0$, so definition 1.5.2 gives $\mathbb{P}(B_n) = \mathbb{P}(B_{n-1})\mathbb{P}(A_n|B_{n-1})$. The events $A_1, \dots, A_{n-1}$ satisfy the hypothesis of (1), since either $n = 2$, or $n \geq 3$ and $\mathbb{P}(B_{n-2}) > 0$ by the first sentence of (1), so the induction hypothesis gives $\mathbb{P}(B_{n-1}) = \mathbb{P}(A_1) \prod_{k=2}^{n-1} \mathbb{P}(A_k|B_{k-1})$.

(2) Let m be the least index with $\mathbb{P}(B_m) = 0$, which exists and satisfies $m \leq n-1$. For $2 \leq k \leq m$ the conditioning event B_{k-1} has positive probability, so those factors are defined. If $m = 1$ the factor $\mathbb{P}(A_1)$ is 0. If $m \geq 2$, then $\mathbb{P}(B_{m-1}) > 0$ and the factor $\mathbb{P}(A_m|B_{m-1}) = \mathbb{P}(B_m)/\mathbb{P}(B_{m-1})$ is 0. Every factor lies in $[0, 1]$, the defined ones because $\mathbb{P}(B_k) \leq \mathbb{P}(B_{k-1})$ by the first paragraph, so the product is 0. The left side is $\mathbb{P}(B_n) \leq \mathbb{P}(B_m) = 0$. $\square$

This relates to independence by noticing that A and B are independent if $\mathbb{P}(A|B) = \mathbb{P}(A)$, so changing the probability space to looking at B does not change the value of A (i.e. A is not affected).

Independence can be generalized to multiple variables. It is important to remember that to be independent, none of the events must effect any other event. That means we actually need to be stronger than pairwise independent.

Example 1.5.4: Independence of multiple events

Take a tetrahedron coloured red, green, and blue on 3 of the sides, and all 3 colours on the 4th. Let the chances of any side be equal. Calculate the chance of landing on one of the colours. This is pair-wise independent, but not independent.

Thus, we generalize independence in the following way:

Definition 1.5.5: Independence over Multiple Events

Let $(\Omega, \mathbb{P})$ be a probability space and $A_i \subseteq \Omega$ be an events with indexes between $1 \leq i \leq n$. Then the events are called independent if and only if for any finite subset of indices $I \subseteq \{1, 2, \dots, n\}$

$$\mathbb{P}\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} \mathbb{P}(A_i)$$

Just like with the case with two events, (p.39 with conditional event similarity)

Example 1.5.6: exercise

1. with complements of sets
2. example on p.39 with pair-wise independence not implying independence (tetrahedron)

As usual, if we can define something for a probability measure, we can do the same for random variables:

Definition 1.5.7: Independent Random Variables

Let $(\Omega, \mathbb{P})$ be a probability space, and let $X_1, \dots, X_n$ be random variables on Ω . Then these random variables are said to be independent if

$$\mathbb{P}(X_1 = a_1, X_2 = a_2, \dots, X_n = a_n) = \prod_{i=1}^n \mathbb{P}(X_i = a_i)$$

for all $a_1, \dots, a_n \in \mathbb{R}$.

Notice that in this condition we can replace $X_1 = a_1, \dots, X_n = a_n$ with $X_1 \in A_1, \dots, X_n \in A_n$ for arbitrary subsets A_i , by summing the product formula over the values $a_i \in A_i$ of the X_i with proposition 1.3.5 on a discrete space. This also means that $A_i = \mathbb{R}$, and so the value can essentially be “eliminated” from our calculations (since $\mathbb{P}(X \in \mathbb{R}) = 1$), meaning if we have a family of independent random variables, any subfamily is also independent.

Similarly to when we conditional probability, we can define conditional random variables:

Definition 1.5.8: Conditional Random Variables

Let $(\Omega, \mathbb{P})$ be a probability space, let X, Y be two random variables on Ω , and let b satisfy $\mathbb{P}(Y = b) > 0$. Then

$$\mathbb{P}(X = a | Y = b) = \frac{\mathbb{P}(X = a, Y = b)}{\mathbb{P}(Y = b)}$$

This gives another way of writing $\mathbb{P}(X = a, Y = b) = \mathbb{P}(X = a | Y = b) \mathbb{P}(Y = b)$ when $\mathbb{P}(Y = b) > 0$.

To get an intuition independent random variables, let’s explore a generalization of how we defined the Binomial distribution to show how to internalize the idea. We can generalize our previous notion of product of Bernoulli random variables to the following:

Definition 1.5.9: Product Space

Let $(\Omega_i, \mathbb{P}_i)$ be probability spaces for $1 \leq i \leq n$. Define

$$\Omega = \Omega_1 \times \Omega_2 \times \dots \times \Omega_n$$

and

$$\mathbb{P}(\omega) = \mathbb{P}(\omega_1, \omega_2, \dots, \omega_n) = \prod_{i=1}^n \mathbb{P}_i(\omega_i)$$

Then $(\Omega, \mathbb{P})$ is called the *product measure* of the $(\Omega_i, \mathbb{P}_i)$ spaces.

The fact that the product space is defined so that every component (or “coordinate”) is independent has some nice results. For example, if we take some hyper-rectangle

$$A = A_1 \times A_2 \times \dots \times A_n$$

Then

$$\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\omega) = \sum_{\omega \in A} \prod_{i=1}^n \mathbb{P}_i(\omega_i) \stackrel{!}{=} \prod_{i=1}^n \sum_{\omega_i \in A_i} \mathbb{P}_i(\omega_i) = \prod_{i=1}^n \mathbb{P}_i(A_i)$$

showing that product and probability spaces “commute” when the random variables are independent. Due to this, we will define a *random vector* $(X_1, X_2, \dots, X_n)$ to be a n -tuple of random variables, each acting on their particular coordinates.

An important reason to care more about the independence of random variables instead of events is because they behave much more nicely. The following theorem is an excellent illustration of this: If $X_1, X_2, \dots, X_n$ are random variables, and we partition them into different groups forming new random variables $Y_1, \dots, Y_m$, then these random variables are also independent!

In fact, we can be even stronger. If I_k represents the partition of our n random variables for $1 \leq k \leq m$, then define the new random variables Y_k to be

$$Y_k = f_k((X_i)_{i \in I_k})$$

for some arbitrary functions f_k . Then these Y_k 's are still independent! In other words, if we have random variables, and we partition them and define new functions based off of random variable, these are still independent. A simple example would be each $I_k = \{X_k\}$ and $Y_k = X_k^2$. Another would be $I_1 = \{X_1, \dots, X_{n-1}\}$ and $I_2 = \{X_n\}$, and $Y_1 = X_1(X_2)^2(X_3)^3 \cdots (X_{n-1})^{n-1}$, and $Y_2 = \log(X_n)$. Then Y_1 and Y_2 are still independent!

Lemma 1.5.10: Grouping Lemma

Let $X_1, X_2, \dots, X_n$ be independent random variables on a discrete probability space, let $I_1, \dots, I_m$ be pairwise disjoint sets of indices with union $\{1, \dots, n\}$, and let $Y_1, Y_2, \dots, Y_m$ be defined as

$$Y_k = f_k((X_i)_{i \in I_k})$$

for functions f_k . Then the $Y_1, Y_2, \dots, Y_m$ are independent

Proof:

For $1 \leq k \leq m$ write $X^k = (X_i)_{i \in I_k}$. Each X_i takes countably many values, so X^k does too. Sums of products over product sets factor: if $S_1, \dots, S_r$ are countable and $c_l(x) \geq 0$ for $x \in S_l$, with each $\sum_{x \in S_l} c_l(x)$ finite, then

$$\sum_{(x_1, \dots, x_r) \in S_1 \times \cdots \times S_r} c_1(x_1) \cdots c_r(x_r) = \prod_{l=1}^r \sum_{x \in S_l} c_l(x)$$

by induction on r , grouping by the first coordinate with lemma 1.3.3(1) and factoring $c_1(x_1)$ out of each group with [2, The Algebra of Series and the Triangle Inequality](1).

Step 1: the vectors $X^1, \dots, X^m$ are independent. Let $I \subseteq \{1, \dots, n\}$, and let x_i be a value of X_i for each $i \in I$. The event $\{X_i = x_i \text{ for } i \in I\}$ is the disjoint union, over all choices of values x_j of X_j for $j \notin I$, of the events $\{X_1 = x_1, \dots, X_n = x_n\}$. By proposition 1.3.5, definition 1.5.7 and the factorization above,

$$\mathbb{P}(X_i = x_i \text{ for } i \in I) = \prod_{i \in I} \mathbb{P}(X_i = x_i) \cdot \prod_{j \notin I} \sum_{x_j} \mathbb{P}(X_j = x_j) = \prod_{i \in I} \mathbb{P}(X_i = x_i)$$

since each $\sum_{x_j} \mathbb{P}(X_j = x_j) = 1$ by proposition 1.3.5. With $I = I_k$ this gives $\mathbb{P}(X^k = x^k) = \prod_{i \in I_k} \mathbb{P}(X_i = x_i)$, and with $I = \{1, \dots, n\}$ it gives $\mathbb{P}(X^1 = x^1, \dots, X^m = x^m) = \prod_{k=1}^m \mathbb{P}(X^k = x^k)$.

Step 2: the Y_k . Let $A_1, \dots, A_m \subseteq \mathbb{R}$, and let B_k be the set of values x^k of X^k with $f_k(x^k) \in A_k$, so that $\{Y_k \in A_k\} = \{X^k \in B_k\}$. The event $\{Y_1 \in A_1, \dots, Y_m \in A_m\}$ is the disjoint union, over $(x^1, \dots, x^m) \in B_1 \times \dots \times B_m$, of the events $\{X^1 = x^1, \dots, X^m = x^m\}$. By proposition 1.3.5, Step 1 and the factorization,

$$\mathbb{P}(Y_1 \in A_1, \dots, Y_m \in A_m) = \prod_{k=1}^m \sum_{x^k \in B_k} \mathbb{P}(X^k = x^k) = \prod_{k=1}^m \mathbb{P}(Y_k \in A_k)$$

the last step by proposition 1.3.5 again. With $A_k = \{a_k\}$ this is the product formula of definition 1.5.7 for $Y_1, \dots, Y_m$, completing the proof.

Example 1.5.11: Poisson Revisited

Recall that if X_1 and X_2 are independent Poisson Random distributions with λ_1 and λ_2 , then $(X_1 + X_2) \sim \text{Poiss}(\lambda_1 + \lambda_2)$. We also saw in section 1.4.1 that $\lim_{n \rightarrow \infty} B(n, n/\lambda) = \text{Poiss}(\lambda)$. We will show another way of seeing that the sum of two Poisson distributions ought to be Poisson.

Let's say that λ_1 and λ_2 had a rational ration:

$$\frac{\lambda_1}{\lambda_2} = \frac{r_1}{r_2} \quad r_1, r_2 \in \mathbb{Z}$$

If not, then any rational number is arbitrarily close to λ_1/λ_2 , and so modifying one of the lambda's should be no problem. Then re-arranging so the denominator are rational we get:

$$\frac{\lambda_1}{r_1} = \frac{\lambda_2}{r_2}$$

Now, divide the denominator by an n large enough so that

$$\frac{\lambda_1}{nr_1} = \frac{\lambda_2}{nr_2} = p \in (0, 1)$$

Now set $m_1 = nr_1$, $m_2 = nr_2$, and $m = m_1 + m_2$. Since a binomial $B(m, p)$ is the sum of p i.i.d. Bernoulli with probability p , let $X_i \sim B(p)$ and

$$Y_1 = X_1 + X_2 + \dots + X_{m_1} \sim B(m_1, p) \approx \text{Poiss}(m_1 p) = \text{Poiss}\left(m_1 \frac{\lambda_1}{m_1}\right) = \text{Poiss}(\lambda_1)$$

$$Y_2 = X_{m_1+1} + X_{m_1+2} + \dots + X_{m_1+m_2} \sim B(m_2, p) \approx \text{Poiss}(m_2 p) = \text{Poiss}\left(m_2 \frac{\lambda_2}{m_2}\right) = \text{Poiss}(\lambda_2)$$

And since Y_1 and Y_2 are sums of independents random variables,

$$Y_1 + Y_2 \sim B(m, p) \approx \text{Poiss}(mp) = \text{Poiss}(\lambda_1 + \lambda_2)$$

And so we see how connecting the Poisson distribution to The Bernoulli distribution through many many flips of a coin.

1.5.2 Expected Value of Functions of RV's

We now return to the question of finding $\mathbb{E}f(X_1, X_2, \dots, X_n)$ for random variables X_i . We will show that this is simple to compute if the random variables are independent, and the formula for the dependent case is not much more complex! We start off with the case of $f(X, Y) = XY$:

Proposition 1.5.12: Product of Independent Expected Values

Let $(\Omega, \mathbb{P})$ be a discrete probability space, and let X, Y be independent random variables on Ω . If $\mathbb{E}|X| < \infty$ and $\mathbb{E}|Y| < \infty$, then $\mathbb{E}|XY| < \infty$ and

$$\mathbb{E}XY = \mathbb{E}X\mathbb{E}Y$$

Proof:

List the values of X and of Y without repetition as $a_1, a_2, \dots$ and $b_1, b_2, \dots$, and put $p_n = \mathbb{P}(X = a_n)$ and $q_m = \mathbb{P}(Y = b_m)$. The events $\Omega_{nm} = \{X = a_n, Y = b_m\}$, indexed by the countable set of pairs (n, m) , are pairwise disjoint with union Ω , and $\mathbb{P}(\Omega_{nm}) = p_n q_m$ by independence. On Ω_{nm} the term $|X(\omega)Y(\omega)|\mathbb{P}(\omega)$ equals $|a_n||b_m|\mathbb{P}(\omega)$, and the sum of $\mathbb{P}(\omega)$ over Ω_{nm} is $\mathbb{P}(\Omega_{nm})$ by the identity at the start of section 1.3. So lemma 1.3.3(1), applied with these blocks and then to the pairs (n, m) grouped by n , gives, in $[0, +\infty]$,

$$\mathbb{E}|XY| = \sum_{(n,m)} |a_n||b_m| p_n q_m = \sum_n \left(\sum_m |a_n| p_n |b_m| q_m \right)$$

By lemma 1.3.3(1) with the blocks $\{Y = b_m\}$, $\sum_m |b_m| q_m = \mathbb{E}|Y| < \infty$, so the inner sum is $|a_n| p_n \mathbb{E}|Y|$ by [2, The Algebra of Series and the Triangle Inequality](1). Since lemma 1.3.3(1) with the blocks $\{X = a_n\}$ gives $\sum_n |a_n| p_n = \mathbb{E}|X|$, it follows that $\mathbb{E}|XY| = \mathbb{E}|X| \mathbb{E}|Y| < \infty$. Now lemma 1.3.3(2) applies to the same groupings without absolute values, and with proposition 1.3.7 for $\mathbb{E}X = \sum_n a_n p_n$ and $\mathbb{E}Y = \sum_m b_m q_m$ it gives

$$\mathbb{E}XY = \sum_n \left(\sum_m a_n p_n b_m q_m \right) = \sum_n a_n p_n \mathbb{E}Y = \mathbb{E}X \mathbb{E}Y$$

as claimed.

Example 1.5.13: Properties

1. Use the Grouping lemma and induction to show that, for independent random variables $X_1, \dots, X_n$ on a discrete probability space with $\mathbb{E}|X_i| < \infty$ for every i ,

$$\mathbb{E} \left(\prod_{i=1}^n X_i \right) = \prod_{i=1}^n \mathbb{E}(X_i)$$

2. Let $X_1, X_2, \dots, X_n$ be i.i.d. Bernoulli Random variables. Compute

$$\mathbb{E}(X_1 + X_2 + \dots + X_n)^2$$

Notice that in the proof, the fact that we used a “particular” function XY on the random variables X and Y is something that can be generalized. The proof used XY only through its value a_nb_m on the event $\{X = a_n, Y = b_m\}$. For a function f of X and Y the same grouping leads to the double series $\sum_{(n,m)} f(a_n, b_m) \mathbb{P}(X = a_n) \mathbb{P}(Y = b_m)$, and to its iterated form

$$\sum_m \left(\sum_n f(a_n, b_m) \mathbb{P}(X = a_n) \right) \mathbb{P}(Y = b_m)$$

This might actually look quite familiar from a result in calculus:

Theorem 1.5.14: Fubini's Theorem

Let $(\Omega, \mathbb{P})$ be a discrete probability space, and let X, Y be independent random variables with $\mathbb{E}|X| < \infty$ and $\mathbb{E}|Y| < \infty$. Let $a_1, a_2, \dots$ list without repetition the values a of X with $\mathbb{P}(X = a) > 0$, and $b_1, b_2, \dots$ the values b of Y with $\mathbb{P}(Y = b) > 0$. Then if $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ is a function such that $\mathbb{E}|f(X, Y)| < \infty$, then

$$\begin{aligned} \mathbb{E}f(X, Y) &= \sum_j \left(\sum_i f(a_i, b_j) \mathbb{P}(X = a_i) \right) \mathbb{P}(Y = b_j) \\ &= \sum_i \left(\sum_j f(a_i, b_j) \mathbb{P}(Y = b_j) \right) \mathbb{P}(X = a_i) \end{aligned}$$

where every series displayed converges absolutely.

Proof:

Let T be the set of points $\omega \in \Omega$ with $\mathbb{P}(\omega) > 0$. The points of probability 0 contribute 0 to every sum over Ω and to every probability below, so they may be ignored. Since $\mathbb{P}(X = a)$ is the sum of $\mathbb{P}(\omega)$ over the points with $X(\omega) = a$, a value a has $\mathbb{P}(X = a) > 0$ exactly when $a = X(\omega)$ for some $\omega \in T$, so the values of X on T are the a_i . For the same reason applied to Y , the values of Y on T are the b_j .

Step 1: grouping by values, without independence. Let $g: \mathbb{R}^2 \rightarrow \mathbb{R}$ satisfy $\mathbb{E}|g(X, Y)| < \infty$. The sets $T_{ij} = \{\omega \in T : X(\omega) = a_i, Y(\omega) = b_j\}$, indexed by the countable set of pairs (i, j) , are pairwise disjoint with union T , and the sum of $\mathbb{P}(\omega)$ over T_{ij} is $\mathbb{P}(X = a_i, Y = b_j)$. On T_{ij} the term $g(X(\omega), Y(\omega))\mathbb{P}(\omega)$ equals $g(a_i, b_j)\mathbb{P}(\omega)$. By lemma 1.3.3(2), applied with these blocks and then to the pairs (i, j) grouped by i , every series below converges absolutely and

$$\mathbb{E}g(X, Y) = \sum_{(i,j)} g(a_i, b_j) \mathbb{P}(X = a_i, Y = b_j) = \sum_i \sum_j g(a_i, b_j) \mathbb{P}(X = a_i, Y = b_j)$$

Part (1) of the same lemma, applied to $|g|$, gives $\sum_i \sum_j |g(a_i, b_j)| \mathbb{P}(X = a_i, Y = b_j) = \mathbb{E}|g(X, Y)|$.

Step 2: independence. Put $d_{ij} = f(a_i, b_j) \mathbb{P}(X = a_i) \mathbb{P}(Y = b_j)$. By independence, $d_{ij} = f(a_i, b_j) \mathbb{P}(X = a_i, Y = b_j)$, so Step 1, with $g = f$ and with $g = |f|$, gives $\mathbb{E}f(X, Y) = \sum_{(i,j)} d_{ij}$ and $\sum_{(i,j)} |d_{ij}| = \mathbb{E}|f(X, Y)| < \infty$. By lemma 1.3.3(2), applied to the pairs (i, j) grouped by i and then grouped by j , $\mathbb{E}f(X, Y) = \sum_i \sum_j d_{ij} = \sum_j \sum_i d_{ij}$, where every row series $\sum_j d_{ij}$, every column series $\sum_i d_{ij}$, and both outer series converge absolutely. Since $\mathbb{P}(X = a_i) > 0$, the series $\sum_j f(a_i, b_j) \mathbb{P}(Y = b_j)$ is $\mathbb{P}(X = a_i)^{-1}$ times the row series $\sum_j d_{ij}$, so it converges

absolutely, and $\mathbb{P}(X = a_i)$ factors out of the row sum by [2, The Algebra of Series and the Triangle Inequality](1). Since $\mathbb{P}(Y = b_j) > 0$, the series $\sum_i f(a_i, b_j)\mathbb{P}(X = a_i)$ is $\mathbb{P}(Y = b_j)^{-1}$ times the column series $\sum_i d_{ij}$, so it converges absolutely, and $\mathbb{P}(Y = b_j)$ factors out of the column sum. This gives the two expressions in the statement, as claimed.

Essentially, we can take the average over all fixed b_j , and then average over all b_j 's! This is Analogous to the famous Calculus Theorem:

$$\iint_{[0,1]^2} f(x, y) dx dy = \int_0^1 \left(\int_0^1 f(x, y) dy \right) dx$$

Example 1.5.15: Different Perspective

Fubini's Theorem can be "brought back down" to the probability space to get a second perspective. In particular, if X and Y are independent and $\mathbb{E}|f(X, Y)| < \infty$, then, with both sums over the points of positive probability,

$$\mathbb{E}f(X, Y) = \sum_{\omega} \left[\sum_{\omega'} f(X(\omega), Y(\omega')) \mathbb{P}(\omega') \right] \mathbb{P}(\omega)$$

where we sum over Ω twice. For a point ω with $X(\omega) = a_i$, group the points ω' by the value $Y(\omega') = b_j$. By lemma 1.3.3(1) the inner sum of absolute values is $\sum_j |f(a_i, b_j)|\mathbb{P}(Y = b_j)$, which is finite by theorem 1.5.14, so part (2) of the lemma turns the inner sum into $h(a_i) = \sum_j f(a_i, b_j)\mathbb{P}(Y = b_j)$. Grouping the points ω of the outer sum by the value $X(\omega) = a_i$, the two parts of the lemma, with the absolute convergence of $\sum_i h(a_i)\mathbb{P}(X = a_i)$ from theorem 1.5.14, turn the outer sum into $\sum_i h(a_i)\mathbb{P}(X = a_i)$, the second expression of theorem 1.5.14.

What if X and Y are dependent? Then our calculations will break down at the following step:

$$\begin{aligned} \mathbb{E}f(X, Y) &= \sum_{\omega \in \Omega} f(X(\omega), Y(\omega)) \mathbb{P}(\omega) \\ &= \sum_{n, m} f(a_n, b_m) \mathbb{P}(X = a_n, Y = b_m) \quad \text{Step 1 of the proof of theorem 1.5.14} \end{aligned}$$

At this point, we were able to use independence to split the result. In the dependent case, we can introduce a sneaky 1 in the form of $\frac{\mathbb{P}(X=a_n)}{\mathbb{P}(X=a_n)}$, for the values a_n with $\mathbb{P}(X = a_n) > 0$, and get:

$$\begin{aligned} \sum_{n, m} f(a_n, b_m) \mathbb{P}(X = a_n, Y = b_m) &= \sum_{n, m} f(a_n, b_m) \frac{\mathbb{P}(X = a_n, Y = b_m)}{\mathbb{P}(X = a_n)} \mathbb{P}(X = a_n) \\ &= \sum_n \left[\sum_m f(a_n, b_m) \frac{\mathbb{P}(X = a_n, Y = b_m)}{\mathbb{P}(X = a_n)} \right] \mathbb{P}(X = a_n) \end{aligned}$$

where the last line is by lemma 1.3.3(2). Notice that the inside term is the conditional distribution! This result is important enough to get called out as a theorem:

Theorem 1.5.16: Fubini's Theorem II

Let $(\Omega, \mathbb{P})$ be a discrete probability space, and let X, Y be random variables (not necessarily independent) with $\mathbb{E}|X| < \infty$ and $\mathbb{E}|Y| < \infty$. Let $a_1, a_2, \dots$ list without repetition the values a of X with $\mathbb{P}(X = a) > 0$, and $b_1, b_2, \dots$ the values b of Y with $\mathbb{P}(Y = b) > 0$. Then if $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ is a function such that $\mathbb{E}|f(X, Y)| < \infty$, then

$$\mathbb{E}f(X, Y) = \sum_i \left(\sum_j f(a_i, b_j) \mathbb{P}(Y = b_j \mid X = a_i) \right) \mathbb{P}(X = a_i)$$

where every series displayed converges absolutely.

Proof:

Step 1 of the proof of theorem 1.5.14 does not use independence, and with $g = f$ it gives $\mathbb{E}f(X, Y) = \sum_i \sum_j f(a_i, b_j) \mathbb{P}(X = a_i, Y = b_j)$, where for each i the series over j converges absolutely and the series of the sums of absolute values over i converges. Since $\mathbb{P}(X = a_i) > 0$, definition 1.5.8, with the roles of X and Y exchanged, gives $\mathbb{P}(X = a_i, Y = b_j) = \mathbb{P}(Y = b_j \mid X = a_i) \mathbb{P}(X = a_i)$. So the series $\sum_j f(a_i, b_j) \mathbb{P}(Y = b_j \mid X = a_i)$ is $\mathbb{P}(X = a_i)^{-1}$ times the absolutely convergent series $\sum_j f(a_i, b_j) \mathbb{P}(X = a_i, Y = b_j)$, hence converges absolutely, and factoring $\mathbb{P}(X = a_i)$ out of it ([2, The Algebra of Series and the Triangle Inequality](1)) gives the formula. The outer series converges absolutely because its terms are bounded in absolute value by the sums $\sum_j |f(a_i, b_j)| \mathbb{P}(X = a_i, Y = b_j)$ ([2, The Algebra of Series and the Triangle Inequality](3)), whose series over i converges, and [2, The Comparison Test] applies, as claimed.

For fixed b with $\mathbb{P}(Y = b) > 0$, the numbers $\mathbb{P}(X = a \mid Y = b)$, as a runs over the values of X , form a probability distribution: they are nonnegative, and they sum to $\sum_a \mathbb{P}(X = a, Y = b) / \mathbb{P}(Y = b) = 1$ by proposition 1.3.5, since the events $\{X = a, Y = b\}$ are disjoint with union $\{Y = b\}$.

Definition 1.5.17: Conditional Distribution

Let X, Y be random variables, let $b_1, b_2, \dots$ be the values of Y , and let a_n be a value of X with $\mathbb{P}(X = a_n) > 0$. Then $m \mapsto \mathbb{P}(Y = b_m \mid X = a_n)$ is called the *conditional distribution with respect to a_n* .

For a value a of X with $\mathbb{P}(X = a) > 0$ and a function f for which the series converges absolutely, the expected value of this distribution is

$$\mathbb{E}(f(a, Y) \mid X = a) := \sum_m f(a, b_m) \mathbb{P}(Y = b_m \mid X = a)$$

Example 1.5.18: Conditional Distribution

Let's say we had random variable X and Y on some space Ω for which the following table outlines the probability:

$Y \backslash X$	x_1	x_2	x_3	x_4	$p_Y(y) \downarrow$
y_1	4/32	2/32	1/32	1/32	8/32
y_2	3/32	6/32	3/32	3/32	15/32
y_3	9/32	0	0	0	9/32
$p_X(x) \rightarrow$	16/32	8/32	4/32	4/32	32/32

Then any row or column in this table is a conditional distribution if we switch the denominator to the sum of the numerators of the row's or columns (given what we are taking it "with respect to").

Notice that in the image, the table actually represents the probability of $\mathbb{P}((X, Y) = (x_i, y_j))$. And there is a bottom row and right column that holds the values of what is summed of the rows and columns: this is given a name:

Definition 1.5.19: Marginal Distribution

Let X and Y be random variables on Ω . We define $\mathbb{P}_Y(X = x) = \sum_{y_i} \mathbb{P}((X, Y) = (x, y_i))$ to be the *marginal distribution with respect to Y* , and $\mathbb{P}_X(Y = y)$ by exchanging the roles of X and Y . More generally, for random variables $X_1, \dots, X_n$ on Ω , the *marginal distribution* of X_i is

$$x \mapsto \sum \mathbb{P}(X_1 = x_1, \dots, X_{i-1} = x_{i-1}, X_i = x, X_{i+1} = x_{i+1}, \dots, X_n = x_n)$$

the sum running over all values x_j of X_j for $j \neq i$.

Note that the expected value of the marginal distribution is calculated precisely in the way Fubini's Theorem is calculated.

Definition 1.5.20: Joint Distribution

Let $(\Omega, \mathbb{P})$ be a probability space and let X, Y be two random variables on Ω . Define the *joint distribution* (X, Y) to be

$$\mathbb{P}((X, Y) = (a, b)) := \mathbb{P}(X = a, Y = b)$$

which equals $\mathbb{P}(Y = b|X = a)\mathbb{P}(X = a)$ when $\mathbb{P}(X = a) > 0$.

Extending this description of a joint distribution from two random variables to a sequence of dependent ones leads to Markov chains, studied in chapter 5.

Lemma 1.5.21: Poisson Colouring

Let $\lambda > 0$ and $p \in (0, 1)$, and let N and N_1 be random variables on a discrete probability space with $N \sim \text{Poiss}(\lambda)$ and:

$$\mathbb{P}(N_1 = k | N = n) = \binom{n}{k} p^k (1-p)^{n-k}$$

for all integers $n \geq 0$ and $0 \leq k \leq n$, that is, for every $N = n$, get a binomial distribution $B(n, p)$. Let $N_2 = N - N_1$. Then N_1 and N_2 are independent and have distributions $\text{Poiss}(p\lambda)$ and $\text{Poiss}((1-p)\lambda)$ respectively.

Here is a way to make this theorem intuitive: If a Chicken lays a Poisson number of eggs, and then we colour them red or blue depending on a coin toss, then knowing the number of red eggs gives us nothing about the number of blue eggs, and the number of red and blue eggs are also Poisson

Proof:

For integers $k, j \geq 0$ put

$$q_k = \frac{(p\lambda)^k}{k!} e^{-p\lambda} \quad \text{and} \quad r_j = \frac{((1-p)\lambda)^j}{j!} e^{-(1-p)\lambda}$$

the probabilities of $\text{Poiss}(p\lambda)$ and $\text{Poiss}((1-p)\lambda)$ from definition 1.3.22. By the exponential series ([2, The Exponential, Cosine and Sine Functions]), $\sum_k q_k = 1$, $\sum_j r_j = 1$ and $\sum_n \mathbb{P}(N = n) = 1$. By proposition 1.3.5, $\mathbb{P}(N \in \{0, 1, 2, \dots\}) = \sum_n \mathbb{P}(N = n) = 1$, so the event $\{N \notin \{0, 1, 2, \dots\}\}$ has probability 0. Every $\mathbb{P}(N = n)$ is positive, so the conditional probabilities in the statement are defined (definition 1.5.8).

Step 1: the joint law. Let $k, j \geq 0$ be integers. Since $N_2 = N - N_1$, the events $\{N_1 = k, N_2 = j\}$ and $\{N_1 = k, N = k + j\}$ are equal, so

$$\begin{aligned} \mathbb{P}(N_1 = k, N_2 = j) &= \mathbb{P}(N_1 = k | N = k + j) \mathbb{P}(N = k + j) \\ &= \frac{(k+j)!}{k! j!} p^k (1-p)^j \frac{\lambda^{k+j}}{(k+j)!} e^{-\lambda} = q_k r_j \end{aligned}$$

using $\lambda^{k+j} = \lambda^k \lambda^j$ and $e^{-\lambda} = e^{-p\lambda} e^{-(1-p)\lambda}$ ([2, The Functional Equation of the Exponential]).

Step 2: the marginals. Given $N = n$, the variable N_1 lies in $\{0, \dots, n\}$ with conditional probability $\sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} = (p + 1 - p)^n = 1$ (theorem 0.1.3), that is, $\mathbb{P}(N_1 \in \{0, \dots, n\}, N = n) = \mathbb{P}(N = n)$. So $\mathbb{P}(N_1 = k, N = n) = 0$ whenever $k > n$. The event $\{N_1 = k\}$ is the disjoint union of the events $\{N_1 = k, N = n\}$ for $n \geq 0$ and of an event contained in $\{N \notin \{0, 1, 2, \dots\}\}$, which has probability 0. So proposition 1.3.5 gives, for each integer $k \geq 0$,

$$\mathbb{P}(N_1 = k) = \sum_{n \geq 0} \mathbb{P}(N_1 = k, N = n) = \sum_{j \geq 0} \mathbb{P}(N_1 = k, N = k + j) = \sum_{j \geq 0} q_k r_j = q_k$$

Hence $\mathbb{P}(N_1 \in \{0, 1, 2, \dots\}) = \sum_k q_k = 1$, by proposition 1.3.5. The event $\{N_2 = j\}$ is the disjoint union of the events $\{N_1 = k, N_2 = j\}$ for $k \geq 0$ and of an event contained in $\{N_1 \notin \{0, 1, 2, \dots\}\}$, which has probability 0. So proposition 1.3.5 gives, for each integer $j \geq 0$, $\mathbb{P}(N_2 = j) = \sum_{k \geq 0} \mathbb{P}(N_1 = k, N_2 = j) = r_j \sum_k q_k = r_j$, and it also gives $\mathbb{P}(N_2 \in \{0, 1, 2, \dots\}) = \sum_j r_j = 1$. So $N_1 \sim \text{Poiss}(p\lambda)$ and $N_2 \sim \text{Poiss}((1-p)\lambda)$.

Step 3: independence. For integers $k, j \geq 0$, Steps 1 and 2 give $\mathbb{P}(N_1 = k, N_2 = j) = q_k r_j = \mathbb{P}(N_1 = k)\mathbb{P}(N_2 = j)$. If a or b is not a non-negative integer, then $\mathbb{P}(N_1 = a) = 0$ or $\mathbb{P}(N_2 = b) = 0$ by Step 2, and $\mathbb{P}(N_1 = a, N_2 = b)$, which is at most each of these two probabilities, is also 0. So the product formula of definition 1.5.7 holds for all values, and N_1 and N_2 are independent, completing the proof.

1.6 Inequality With Expected Value

In this section, we start exploring one of the basic types of result we try to find in Probability: *approximation*. Approximating is probability the most important statistical concept as it lets you bound difficult problems with simpler variables, and gives important heuristics in solving more complicated problems. Usually, this means we bound using inequalities the values of a problem from above and/or below. In this section, we introduce two famous inequalities that let's us do exactly that. In particular, we will be focusing on bounding from above.

Theorem 1.6.1: Chebyshev's Inequality

Let X be a random variable on a discrete probability space with $X \geq 0$ and $\mathbb{E}X < \infty$, and let $x > 0$. Then:

$$\mathbb{P}(X \geq x) \leq \frac{\mathbb{E}X}{x}$$

Furthermore,

$$\mathbb{P}(X \geq x) \leq \frac{\mathbb{E}(XI(|X| \geq x))}{x}$$

Proof:

Since $X \geq 0$, we have $|X| = X$ and, at every point,

$$x \cdot I(X \geq x) \leq X \cdot I(X \geq x) \leq X$$

because where $X < x$ the first two terms are 0, and where $X \geq x$ the first is $x \leq X$. Multiplying by $\mathbb{P}(\omega)$ and summing over the points ω , the three series are ordered term by term, so their sums are ordered ([2, Limits Preserve Weak Inequalities]), and they converge because the largest one, the series of $\mathbb{E}X$, does ([2, The Comparison Test]). The sum of the first is $x\mathbb{P}(X \geq x)$, by the identity at the start of section 1.3. So

$$x\mathbb{P}(X \geq x) \leq \mathbb{E}(X \cdot I(X \geq x)) \leq \mathbb{E}X$$

and dividing by $x > 0$ gives both inequalities, completing the proof.

Though simple, this inequality gives us one of our first real insights on approximating problems, in particular that the “tail” of a probability given a random variable will never really exceed the expected value! For example, let's say X random variable such that $\{X > 0\} = \{X \geq 1\}$ ⁴. Then using Chebyshev's inequality at $x = 1$ give us

$$\mathbb{P}(X > 0) = \mathbb{P}(X \geq 1) \leq \mathbb{E}X$$

⁴Do not forget this is a shorthand for all values such that $X > 0$ and $X \geq 1$, see example 1.2.5

If $\mathbb{E}X \ll 1$, then that means that $\mathbb{P}(X = 0) = 1 - \mathbb{P}(X \geq 1)$ is very large, meaning most of our values will be 0!

Example 1.6.2: Chebyshev's Inequality

Let σ be a uniformly random permutation of $\{1, \dots, n\}$, each of the $n!$ permutations having probability $1/n!$, and let $F = \sum_{i=1}^n I(\sigma(i) = i)$ be its number of fixed points. Exactly $(n-1)!$ permutations fix a given i , so $\mathbb{P}(\sigma(i) = i) = 1/n$, and linearity of the expected value gives $\mathbb{E}F = 1$. Theorem 1.6.1 with $x = k > 0$ gives $\mathbb{P}(F \geq k) \leq 1/k$ for every n . For instance, at most a third of all permutations have three or more fixed points.

Jensen's inequality compares $f(\mathbb{E}X)$ with $\mathbb{E}f(X)$. Recall that a function is convex if the line segment between any two points of its graph does not lie below the graph, and concave if no such segment lies above the graph.

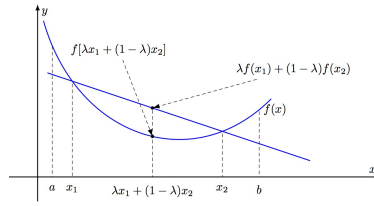


Figure 1.1: Convex Function

Lemma 1.6.3: Jensen's Inequality

Let $I \subseteq \mathbb{R}$ be an interval, let $f: I \rightarrow \mathbb{R}$ be continuous on I and differentiable at every interior point of I , and let X be a random variable on a discrete probability space with values in I , $\mathbb{E}|X| < \infty$ and $\mathbb{E}|f(X)| < \infty$. Suppose that $\mathbb{E}X$ is an interior point of I .

1. If f' is increasing on the interior of I , for instance $f(x) = x^2$ on $\mathbb{R}$, then

$$f(\mathbb{E}X) \leq \mathbb{E}f(X)$$

2. If f' is decreasing on the interior of I , for instance $f(x) = \sqrt{x}$ on $[0, \infty)$, then

$$f(\mathbb{E}X) \geq \mathbb{E}f(X)$$

Proof:

Let $\mu = \mathbb{E}X$.

Step 1: in case (1), the graph lies above the tangent line at μ . Let $x \in I$ with $x \neq \mu$. The open interval between μ and x lies in the interior of I , so [2, The Mean Value Theorem] gives a ξ strictly between μ and x with $f(x) - f(\mu) = f'(\xi)(x - \mu)$. If $x > \mu$ then $f'(\xi) \geq f'(\mu)$ and $x - \mu > 0$, and if $x < \mu$ then $f'(\xi) \leq f'(\mu)$ and $x - \mu < 0$. In both cases $f'(\xi)(x - \mu) \geq f'(\mu)(x - \mu)$, so

$$f(x) \geq f(\mu) + f'(\mu)(x - \mu) \quad \text{for every } x \in I$$

Step 2: case (1). By Step 1 at $x = X(\omega)$, the random variable $f(X) - f(\mu) - f'(\mu)(X - \mu)$ is

nonnegative at every point. Its expected value is defined because those of $f(X)$ and X are, and it is nonnegative, since the series that defines it has nonnegative terms ([2, Limits Preserve Weak Inequalities]). By proposition 1.3.6,

$$0 \leq \mathbb{E}f(X) - f(\mu) - f'(\mu)(\mathbb{E}X - \mu) = \mathbb{E}f(X) - f(\mathbb{E}X)$$

Step 3: case (2). The function $-f$ satisfies the hypotheses of case (1): its derivative $-f'$ is increasing on the interior of I , and $\mathbb{E}|-f(X)| = \mathbb{E}|f(X)| < \infty$. Case (1) gives $-f(\mathbb{E}X) \leq \mathbb{E}(-f(X)) = -\mathbb{E}f(X)$, that is, $f(\mathbb{E}X) \geq \mathbb{E}f(X)$, as claimed.

Example 1.6.4: Jensen's inequality

Suppose $\mathbb{E}X^2 < \infty$. Since $(|X| - 1)^2 \geq 0$, $|X| \leq \frac{1}{2}(X^2 + 1)$ at every point, so comparing the series that define the two expected values term by term ([2, Limits Preserve Weak Inequalities]) gives $\mathbb{E}|X| < \infty$. The function $f(x) = x^2$ on $I = \mathbb{R}$ has the increasing derivative $2x$, so lemma 1.6.3(1) gives

$$(\mathbb{E}X)^2 \leq \mathbb{E}X^2$$

On the other hand, $f(x) = \sqrt{x}$ is continuous on $I = [0, \infty)$ ([2, The n -th Root Function Is Continuous]), with the decreasing derivative $\frac{1}{2\sqrt{x}}$ on $(0, \infty)$ ([2, The Derivative of a Real Power]). Apply lemma 1.6.3(2) to the random variable X^2 , which takes values in I and has $\mathbb{E}\sqrt{X^2} = \mathbb{E}|X| < \infty$. If $\mathbb{E}X^2 > 0$, then $\mathbb{E}X^2$ is an interior point of I , and

$$\sqrt{\mathbb{E}X^2} \geq \mathbb{E}\sqrt{X^2} = \mathbb{E}|X|$$

If $\mathbb{E}X^2 = 0$, then $X = 0$ at every point of positive probability, and both sides are 0. Furthermore, $X \leq |X|$, and so, comparing the two series that define them term by term ([2, Limits Preserve Weak Inequalities]), $\mathbb{E}|X| \geq \mathbb{E}X$. In particular $\mathbb{E}X \leq \mathbb{E}|X| \leq \sqrt{\mathbb{E}X^2}$.

1.7 Computation trick: Linearity of Expectation

In the preceding sections, the importance of the expected value of a random variable should now be evident in how often it is a value we want to know. In this section, we show how it is sometimes very easy to find the expected value of a random variable if the random variable is the sum of indicators:

$$N = \sum_{i=1}^n I_{A_i} \quad A_i \subseteq \Omega$$

The key idea is that this will circumnavigate the change of variable trick we have been doing so far, which can be sometime difficult to come up with. As an example of the power of random variables which are sums of indicators, recall that when we computed the expected value of the Binomial distribution, we took advantage of the fact that we can represent it as the sum of i.i.d. Bernoulli variables. This trick allowed us to incredibly quickly calculate the expected value. Similarly, we can take the expected value of N by taking advantage of the linearity of the expected value and the

known value of the expected value of an indicator function (example 1.2.7):

$$\mathbb{E}N = \mathbb{E}\left(\sum_{i=1}^n I_{A_i}\right) = \sum_{i=1}^n \mathbb{E}(I_{A_i}) = \sum_{i=1}^n \mathbb{P}(A_i)$$

the key insight here is that we avoided the possible ugliness of taking $\mathbb{E}X = \sum_{k=1}^{\infty} k\mathbb{P}(X = k)$ by “evaluating” the expected value only once we reached the indicator functions, which have a very simple expected value!

It is often easier to compute the probabilities $\mathbb{P}(A_i)$ than to find the distribution $\mathbb{P}(N = k)$ and then sum $\mathbb{E}N = \sum_k k\mathbb{P}(N = k)$.

1.7.1 Functions on Finite sets

Let $X = \{1, 2, \dots, n\}$ and let Ω be the set of all functions from X to X . The uniform probability on Ω , which gives each function the same chance of being chosen, is

$$\mathbb{P}(\omega) = \frac{1}{n^n}$$

For $\omega \in \Omega$, let $N(\omega) = \#\omega(X)$ be the cardinality of the range:

$$N = |\{\omega(1), \omega(2), \dots, \omega(n)\}|$$

Then N is a random variable, and $\mathbb{E}N$ is defined because Ω is finite. Computing $\mathbb{E}N = \sum_{k=1}^n k\mathbb{P}(N = k)$ directly requires the distribution of N , which is found at the end of this subsection. It is simpler to write N as a sum of indicators,

$$N(\omega) = \sum_{i=1}^n I(i \in \omega(X))$$

which checks, for each i , whether i is in the range of ω . The functions with $i \notin \omega(X)$ are the functions from X to $X \setminus \{i\}$, of which there are $(n-1)^n$, so

$$\mathbb{P}(i \in \omega(X)) = 1 - \mathbb{P}(i \notin \omega(X)) = 1 - \frac{(n-1)^n}{n^n}$$

for every i . By linearity of the expected value,

$$\mathbb{E}N = \sum_{i=1}^n \mathbb{P}(i \in \omega(X)) = n \left(1 - \left(1 - \frac{1}{n}\right)^n\right)$$

For every real x ,

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x \tag{1.3}$$

For $x = 0$ both sides equal 1. For $x \neq 0$ and $n > |x|$, $(1 + x/n)^n = \exp(n \log(1 + x/n))$ by [2, The Functional Equation of the Exponential] and $\exp(\log y) = y$. With $h = x/n$, $n \log(1 + x/n) = x \cdot \frac{\log(1+h) - \log 1}{h}$, which tends to $x \log'(1) = x$ as $n \rightarrow \infty$ by [2, The Derivative of the Logarithm] and

[2, The Sequential Criterion for Functional Limits]. Continuity of $\exp$ ([2, The Continuity of the Exponential, Cosine and Sine]) gives (1.3). With $x = -1$,

$$\mathbb{E}N = n \left(1 - \left(1 - \frac{1}{n} \right)^n \right) \sim n \left(1 - \frac{1}{e} \right)$$

where $\sim$ means the ratio between the two sides tends to 1 as $n \rightarrow \infty$.

Computing $\mathbb{E}N = \sum_{k=1}^n k \mathbb{P}(N = k)$ instead requires $\mathbb{P}(N = k)$. If $N(\omega) = k$, the range of ω is one of the $\binom{n}{k}$ subsets of X of size k , and ω maps X onto it. So $\mathbb{P}(N = k)$ is $\binom{n}{k}$ times the number of surjective functions from X onto $\{1, 2, \dots, k\}$, divided by n^n . Let Ω_k be the set of all k^n functions from X to $\{1, 2, \dots, k\}$, and for $1 \leq i \leq k$ let

$$A_i = \{\omega \in \Omega_k \mid i \notin \omega(X)\}$$

A function in Ω_k is surjective exactly when it lies in none of the A_i , so by the principle of inclusion-exclusion in its indicator form, expanding the product,

$$I(\omega \text{ is surjective}) = \prod_{i=1}^k (1 - I_{A_i}(\omega)) = \sum_{T \subseteq \{1, \dots, k\}} (-1)^{|T|} \prod_{i \in T} I_{A_i}(\omega) = \sum_{T \subseteq \{1, \dots, k\}} (-1)^{|T|} I_{A_T}(\omega)$$

where $A_T = \bigcap_{i \in T} A_i$ is the set of functions that miss every value in T , that is, the functions from X to $\{1, \dots, k\} \setminus T$, so $\#(A_T) = (k - |T|)^n$. Summing over $\omega \in \Omega_k$ and grouping the sets T by their size ℓ , of which there are $\binom{k}{\ell}$, the number of surjections is $\sum_{\ell=0}^k (-1)^\ell \binom{k}{\ell} (k - \ell)^n$. Hence

$$\mathbb{P}(N = k) = \frac{1}{n^n} \binom{n}{k} \sum_{\ell=0}^k (-1)^\ell \binom{k}{\ell} (k - \ell)^n$$

and so

$$\mathbb{E}N = \sum_{k=1}^n k \left(\frac{1}{n^n} \binom{n}{k} \sum_{\ell=0}^k (-1)^\ell \binom{k}{\ell} (k - \ell)^n \right)$$

This double sum with alternating signs is much harder to evaluate than the sum of indicators above.

1.7.2 Longest Increasing Subsequence

Let's now restrict our attention to all bijective functions from X to X . This probability space already has the name of S_n from group theory, and is usually called the *symmetric group* (since it has a group structure). We will not need any reference to it as a group, but we will use the S_n notation for mathematical consistency. Notice that $|S_n| = n!$, and so we can define the uniform probability to be $\mathbb{P}(\sigma) = \frac{1}{n!}$.

An *increasing subsequence* of σ of length k is a set of positions $i_1 < i_2 < \dots < i_k$ with

$$\sigma(i_1) < \sigma(i_2) < \dots < \sigma(i_k)$$

and $L(\sigma)$ is the largest length of an increasing subsequence of σ .

Example 1.7.1: Chain Examples

If $(\sigma(1), \dots, \sigma(5)) = (1, 5, 2, 4, 3)$, then $L(\sigma) = 3$, and the values 1, 2, 4 and 1, 2, 3 form the increasing subsequences of length 3. If $(\rho(1), \dots, \rho(5)) = (5, 4, 3, 2, 1)$, then every single position is an increasing subsequence and no two positions form one, so $L(\rho) = 1$.

The distribution of L is difficult to find. Hammersley showed that $L/\sqrt{n}$ converges to a constant γ , and Logan and Shepp and Vershik and Kerov showed in 1977 that $\gamma = 2$ ([3, Example 6.5.2]), so $\mathbb{P}(|L/\sqrt{n} - 2| > \epsilon) \rightarrow 0$ for every $\epsilon > 0$.

Proposition 1.7.2: An Upper Bound for the Longest Increasing Subsequence

Let N_k be the number of increasing subsequences of length k of a uniformly random $\sigma \in S_n$. Then

$$\mathbb{E}N_k = \binom{n}{k} \frac{1}{k!} \quad \text{and} \quad \limsup_{n \rightarrow \infty} \frac{\mathbb{E}L}{\sqrt{n}} \leq e$$

Proof:

Step 1: $\mathbb{E}N_k$. Fix positions $T = \{i_1 < \dots < i_k\}$. Group the permutations into classes, two permutations being in the same class when they agree outside T . Each class has $k!$ elements, obtained by placing the same k values on the positions of T in every order, and exactly one of them is increasing on T . So $\mathbb{P}(\sigma(i_1) < \dots < \sigma(i_k)) = 1/k!$. Writing N_k as the sum over the $\binom{n}{k}$ sets T of the indicators that σ is increasing on T , linearity of the expected value gives $\mathbb{E}N_k = \binom{n}{k}/k!$.

Step 2: the tail of L . The exponential series gives $e^k = \sum_{j \geq 0} k^j/j! \geq k^k/k!$ ([2, The Exponential, Cosine and Sine Functions]), so $k! \geq (k/e)^k$. Also $\binom{n}{k} \leq n^k/k!$. Since $L \geq k$ exactly when $N_k \geq 1$, theorem 1.6.1 with $x = 1$ gives

$$\mathbb{P}(L \geq k) \leq \mathbb{E}N_k \leq \frac{n^k}{(k!)^2} \leq \left(\frac{e\sqrt{n}}{k} \right)^{2k}$$

Step 3: the expected value. By proposition 1.3.9, and since $L \leq n$ and $\mathbb{P}(L \geq j) \leq \mathbb{P}(L \geq k)$ for $j \geq k$, every integer $k \geq 1$ gives

$$\mathbb{E}L = \sum_{j=1}^n \mathbb{P}(L \geq j) \leq (k-1) + n\mathbb{P}(L \geq k)$$

Fix $\delta > 0$ and take $k = \lceil e(1+\delta)\sqrt{n} \rceil$. Then $e\sqrt{n}/k \leq 1/(1+\delta)$, so by Step 2 $n\mathbb{P}(L \geq k) \leq n(1+\delta)^{-2k} \leq ne^{-y}$ with $y = 2e \log(1+\delta)\sqrt{n}$. Since $n = (y/(2e \log(1+\delta)))^2$, [2, The Exponential Outgrows Every Power] gives $ne^{-y} \rightarrow 0$. Hence

$$\frac{\mathbb{E}L}{\sqrt{n}} \leq \frac{e(1+\delta)\sqrt{n} + n\mathbb{P}(L \geq k)}{\sqrt{n}} \rightarrow e(1+\delta)$$

so $\limsup_n \mathbb{E}L/\sqrt{n} \leq e(1+\delta)$ for every $\delta > 0$, and therefore $\limsup_n \mathbb{E}L/\sqrt{n} \leq e$, as claimed.

1.7.3 Cliques in Erdős-Rényi graph

Let $G = G(n, p)$ be the Erdős-Rényi random graph on n vertices, in which each of the $\binom{n}{2}$ possible edges is present independently with probability p , where $0 < p < 1$ is fixed. A *clique* is a set of vertices any two of which are joined by an edge. Let N_k be the number of cliques of size k . There are $\binom{n}{k}$ subsets of vertices of size k , and such a subset W is a clique exactly when all $\binom{k}{2}$ edges between its vertices are present, so

$$\mathbb{P}(W \text{ is a clique}) = p^{\binom{k}{2}}$$

Writing N_k as the sum over all W of the indicators that W is a clique, linearity of the expected value gives

$$\mathbb{E}N_k = \binom{n}{k} p^{\binom{k}{2}}$$

Now let $f(k) = \mathbb{E}N_k$ for $1 \leq k \leq n$. Then $f(1) = n$, $f(n) = p^{n(n-1)/2} < 1$ for $n \geq 2$, and

$$\frac{f(k+1)}{f(k)} = \frac{n-k}{k+1} p^k$$

which decreases as k increases, so f increases while this ratio is at least 1 and decreases afterwards. Let $k_0 = k_0(n)$ be the largest $k \leq n$ with $f(k) \geq 1$. Then $k_0 < n$, $f(k) < 1$ for every $k > k_0$, and

$$f(k_0) \geq 1 > f(k_0 + 1)$$

Lemma 1.7.3: Clique Number

Let $0 < p < 1$ be fixed and let $k_0 = k_0(n)$ be as above. Then

$$\lim_{n \rightarrow \infty} \frac{k_0(n) \log(1/p)}{2 \log n} = 1$$

that is, $k_0 \sim \frac{2}{\log(1/p)} \log n$.

Proof:

Put $b = 1/p > 1$ and $\ell_n = \log n / \log b$, so that $p^{x\ell_n} = n^{-x}$ for every real x .

Step 1: bounds on f . For $1 \leq k \leq n$,

$$\binom{n}{k} = \prod_{j=0}^{k-1} \frac{n-j}{k-j}$$

and each factor satisfies $\frac{n}{k} \leq \frac{n-j}{k-j} \leq n$, the first inequality because $k(n-j) - n(k-j) = j(n-k) \geq 0$ and the second because $k-j \geq 1$. Hence $(n/k)^k \leq \binom{n}{k} \leq n^k$, and since $p^{\binom{k}{2}} = (p^{(k-1)/2})^k$,

$$\left(\frac{n}{k} p^{(k-1)/2}\right)^k \leq f(k) \leq \left(n p^{(k-1)/2}\right)^k$$

Step 2: upper bound. If $k > 2\ell_n + 1$, then $(k-1)/2 > \ell_n$, so $p^{(k-1)/2} < p^{\ell_n} = 1/n$ and $f(k) < 1$ by Step 1. Hence $k_0 \leq 2\ell_n + 1$.

Step 3: lower bound. Let $0 < \eta < 1$ and put $k = \lfloor (2 - \eta)\ell_n \rfloor$. With $x = \frac{\eta}{2} \log n$, which tends to ∞ with n ([2, The Logarithm Is Unbounded]),

$$\frac{2\ell_n}{n^{\eta/2}} = \frac{4}{\eta \log b} \cdot \frac{x}{e^x} \rightarrow 0$$

by [2, The Exponential Outgrows Every Power]. So for all large n , $1 \leq k \leq 2\ell_n \leq n^{\eta/2} \leq n$, which gives $n/k \geq n^{1-\eta/2}$. Also $(k-1)/2 < (1-\eta/2)\ell_n$ and $p < 1$, so $p^{(k-1)/2} \geq p^{(1-\eta/2)\ell_n} = n^{-(1-\eta/2)}$. By Step 1, $f(k) \geq (n^{1-\eta/2} n^{-(1-\eta/2)})^k = 1$, so $k_0 \geq k > (2 - \eta)\ell_n - 1$.

Step 4: conclusion. By Steps 2 and 3, for all large n

$$1 - \frac{\eta}{2} - \frac{1}{2\ell_n} < \frac{k_0}{2\ell_n} \leq 1 + \frac{1}{2\ell_n}$$

Since $\ell_n \rightarrow \infty$, both bounds lie within η of 1 for all large n . As η was arbitrary, $k_0/(2\ell_n) \rightarrow 1$, which is the claim because $2\ell_n = 2 \log n / \log(1/p)$.

Corollary 1.7.4: No Cliques Well Above the Threshold

Let $0 < p < 1$, $0 < \epsilon < 1$ and an integer $m \geq 1$ be fixed. Then for all large n

$$\mathbb{P}(N_{k_0+m+1} > 0) \leq \frac{1}{n^{m(1-\epsilon)}}$$

Proof:

Put $\ell_n = \log n / \log(1/p)$. If $k_0 + m + 1 > n$ there is no clique of that size and the claim holds. By lemma 1.7.3, $k_0/(2\ell_n) \rightarrow 1 > 1 - \frac{\epsilon}{2}$, so $k_0 \geq (2 - \epsilon)\ell_n$ for all large n . For such n with $k_0 + m + 1 \leq n$, and for $k_0 \leq j < n$, $p^j \leq p^{(2-\epsilon)\ell_n} = n^{-(2-\epsilon)}$, so

$$\frac{f(j+1)}{f(j)} = \frac{n-j}{j+1} p^j \leq n p^j \leq \frac{1}{n^{1-\epsilon}}$$

Multiplying these ratios for $j = k_0 + 1, \dots, k_0 + m$ and using $f(k_0 + 1) < 1$ gives $f(k_0 + m + 1) \leq f(k_0 + 1) n^{-m(1-\epsilon)} < n^{-m(1-\epsilon)}$. Since N_{k_0+m+1} takes values in $\{0, 1, 2, \dots\}$, $\{N_{k_0+m+1} > 0\} = \{N_{k_0+m+1} \geq 1\}$, and theorem 1.6.1 with $x = 1$ gives $\mathbb{P}(N_{k_0+m+1} > 0) \leq \mathbb{E}N_{k_0+m+1} = f(k_0 + m + 1)$, as claimed.

2

Second Moment: Variation and Correlation

So far, the big probability question we have been asking centered around finding the expected value of a distribution as well as some limits of fixed probability distributions. We now will study some of the probability information around the expected value, in particular, how concentrated are values around the expected value.

Like in the previous chapter, all distributions will be discrete, however many results easily generalize to the non-discrete case.

2.1 Variance and Covariance

Let X be a random variable. For now, the following definition is introduced for technical reasons. We will call $\mathbb{E}(X^k)$ the k th (raw) moment of X . The k th moment is defined if $\mathbb{E}|X|^k < \infty$. Note that if $\mathbb{E}X^k < \infty$ then $\mathbb{E}|X|^{k-i} < \infty$ for all $0 \leq i \leq k$. To see this, Take the case of $k = 2$. Then

$$|X| = |X|I(|X| < 1) + |X|I(|X| > 1) \leq 1 + X^2$$

And if we add absolute values to X^k , the inequality becomes even stronger.

The next concept gives us some idea of how dense values are around the expected value:

Definition 2.1.1: Variance

Let X be a random variable and $\mu = \mathbb{E}X$, $\mathbb{E}|X|^2 < \infty$. Then the *variance* of X is defined as

$$\text{Var}(X) = \mathbb{E}(X - \mu)^2$$

i.e. the variance is the second moment of the random variable $X - \mu$. The value $\sqrt{\text{Var}(X)}$ is called the *standard deviation*

Some common terminology is associated with the aforementioned definition. If X is any random variable, then $X - \mathbb{E}X$ is called *centering*, since $\mathbb{E}(X - \mathbb{E}X) = 0$:

$$\mathbb{E}(X - \mathbb{E}X) = \mathbb{E}(X) - \mathbb{E}X = 0$$

since $\mathbb{E}(\mathbb{E}(X)) = \mathbb{E}(X)$ since $\mathbb{E}(X)$ is a constant. For that reason, we will usually denote it as μ . In general, if $\mathbb{E}X = 0$ a random variable X , then X is called centered.

Suppose the codomain of X is $\mathbb{N}$, so that

$$\mathbb{E}(X - \mu)^2 = \sum_{x=1}^{\infty} (x - \mu)^2 \mathbb{P}(X = x)$$

If most of the probability sits on values close to μ , the terms with x far from μ carry small weights $\mathbb{P}(X = x)$, and the variance is small. Values far from μ that carry substantial probability make it large.

We can also interpret μ as the point where

$$f(c) = \mathbb{E}[|X - c|^2]$$

is smallest. Expanding $(X - c)^2 = (X - \mu)^2 + 2(X - \mu)(\mu - c) + (\mu - c)^2$ and using $\mathbb{E}(X - \mu) = 0$ gives $f(c) = \text{Var}(X) + (\mu - c)^2$, so f is minimized exactly at $c = \mu$, with minimum value $\text{Var}(X)$.

For computational purposes, it is often easier to represent the variance like so:

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}(X - \mu)^2 \\ &= \mathbb{E}(X^2 - 2\mu X + \mu^2) \\ &= \mathbb{E}X^2 - 2\mu^2 + \mu^2 \\ &= \mathbb{E}X^2 - \mu^2 \\ &= \mathbb{E}X^2 - (\mathbb{E}X)^2 \end{aligned}$$

Thus, we get

$$\text{Var}(X) = \mathbb{E}(X - \mu)^2 = \mathbb{E}X^2 - (\mathbb{E}X)^2$$

Applied to $|X|$, which has $\mathbb{E}|X|^2 = \mathbb{E}X^2$, this gives $0 \leq \text{Var}(|X|) = \mathbb{E}X^2 - (\mathbb{E}|X|)^2$, so $\mathbb{E}|X| \leq \sqrt{\mathbb{E}X^2}$.

Example 2.1.2: Variance Examples

1. if $X \sim B(p)$ show that $\text{Var}(X) = p(1 - p)$. What if $X \sim B(n, p)$?

2. if $X \sim \text{Pois}(\lambda)$ show that $\text{Var}(X) = \lambda$. That means that not only is the mean λ , but it varies by λ ! [Hint: Compute $\mathbb{E}(X(X-1))$]
3. If $\mathbb{E}X^2 < \infty$, then $\text{Var}(aX + b) = a^2\text{Var}(X)$ for all $a, b \in \mathbb{R}$

The next value we will look at is a way of seeing how much two random variable make an “observable” effect on each other

Definition 2.1.3: Covariance

Let X, Y be random variables where $\mathbb{E}|XY|, \mathbb{E}|X|, \mathbb{E}|Y| < \infty$ (which holds when $\mathbb{E}X^2, \mathbb{E}Y^2 < \infty$). Then we define the *covariance* between X and Y to be

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}(X)) \cdot (Y - \mathbb{E}(Y))] = \mathbb{E}XY - \mathbb{E}X\mathbb{E}Y$$

The second equality comes from expanding the product and proposition 1.3.6. The finiteness condition holds when $\mathbb{E}X^2, \mathbb{E}Y^2 < \infty$ because

$$|XY| \leq \frac{1}{2}(X^2 + Y^2) \leq X^2 + Y^2$$

which follows from $(|X| - |Y|)^2 \geq 0$. With 1 in place of Y it gives $|X| \leq \frac{1}{2}(X^2 + 1)$, so $\mathbb{E}|X| < \infty$, and with 1 in place of X it gives $\mathbb{E}|Y| < \infty$.

Lemma 2.1.4: Cauchy-Schwarz Inequality

Let X, Y be random variables on the same discrete probability space, with $\mathbb{E}X^2 < \infty$ and $\mathbb{E}Y^2 < \infty$. Then $\mathbb{E}|XY| < \infty$ and

$$\mathbb{E}|XY| \leq (\mathbb{E}X^2)^{1/2}(\mathbb{E}Y^2)^{1/2}$$

Proof:

List the points of the space as $\omega_1, \omega_2, \dots$. For each k put $u_k = |X(\omega_k)|\mathbb{P}(\omega_k)^{1/2}$ and $v_k = |Y(\omega_k)|\mathbb{P}(\omega_k)^{1/2}$. For every N up to the length of the list, theorem 0.3.1 for finite sums, followed by taking square roots, gives

$$\sum_{k=1}^N |X(\omega_k)Y(\omega_k)|\mathbb{P}(\omega_k) = \sum_{k=1}^N u_k v_k \leq \left(\sum_{k=1}^N u_k^2 \right)^{1/2} \left(\sum_{k=1}^N v_k^2 \right)^{1/2} \leq (\mathbb{E}X^2)^{1/2}(\mathbb{E}Y^2)^{1/2}$$

The last step holds because $\sum_{k \leq N} u_k^2 = \sum_{k \leq N} X(\omega_k)^2 \mathbb{P}(\omega_k) \leq \mathbb{E}X^2$ and $\sum_{k \leq N} v_k^2 \leq \mathbb{E}Y^2$, partial sums of series of nonnegative terms being at most their sums ([2, Nonnegative Terms and Bounded Partial Sums]). So the partial sums of the series of nonnegative terms that defines $\mathbb{E}|XY|$ are bounded by $(\mathbb{E}X^2)^{1/2}(\mathbb{E}Y^2)^{1/2}$, and the same proposition gives $\mathbb{E}|XY| < \infty$ together with the inequality, as claimed.

As an exercise, apply the lemma to $X - \mathbb{E}X$ and $Y - \mathbb{E}Y$ to show that $|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X)\text{Var}(Y)}$.

Example 2.1.5: Uncorrelated Random Variables

Let X and Y be independent. Then

$$\text{Cov}(X, Y) = \mathbb{E}XY - \mathbb{E}X\mathbb{E}Y = \mathbb{E}X\mathbb{E}Y - \mathbb{E}X\mathbb{E}Y = 0$$

showing that independent random variables are uncorrelated! Thus being independent implies uncorrelated. However, being uncorrelated does not imply independent. Take for example independent random variables ϵ and Z such that

$$\mathbb{P}(\epsilon = 1) = \mathbb{P}(\epsilon = -1) = \frac{1}{2}$$

$$\mathbb{P}(Z = \pm 2) = \mathbb{P}(Z = \pm 3) = \frac{1}{4}$$

Let

$$X = Z \quad Y = \epsilon Z$$

Then:

$$\mathbb{E}X = 0 \quad \mathbb{E}Y \stackrel{\text{indep.}}{=} \mathbb{E}\epsilon\mathbb{E}Z = 0$$

and

$$\mathbb{E}XY = \mathbb{E}\epsilon Z^2 \stackrel{\text{indep.}}{=} \mathbb{E}\epsilon\mathbb{E}Z^2 = 0$$

Thus

$$\text{Cov}(X, Y) = 0$$

However:

$$0 = \mathbb{P}(Z = 2, \epsilon Z = 3) \neq \mathbb{P}(Z = 2)\mathbb{P}(\epsilon Z = 3) = 1/4 \cdot 1/4 = 1/16$$

Thus, the two random variable are *not* independent! In other words, knowing something about one random variable gives us information about the outcome of the other!

If we have random variables $X_1, X_2, \dots, X_n$ on the same probability space, and f is a linear combination of these functions, then the variance of f is very simple to compute. For simplicity, let's say $f = S_n = X_1 + X_2 + \dots + X_n$ and $\mathbb{E}(X_i)^2 < \infty$. Then to compute $\text{Var}(S_n)$, we first find:

$$\begin{aligned} (S_n - \mathbb{E}S_n)^2 &= \left(\sum_{i=1}^n (X_i - \mathbb{E}X_i) \right)^2 \\ &= \sum_{i,j=1}^n (X_i - \mathbb{E}X_i)(X_j - \mathbb{E}X_j) \end{aligned}$$

Taking the expected value of both sides gets us:

$$\text{Var}(S_n) = \sum_{i,j=1}^n \text{Cov}(X_i, X_j)$$

Since $\text{Cov}(X_i, X_i) = \text{Var}(X_i)$ and $\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$, we get that:

$$\text{Var}(S_n) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

If the random variables are uncorrelated, we get that:

$$\text{Var}(S_n) = \sum_{i=1}^n \text{Var}(X_i)$$

Furthermore, if they are i.i.d., then:

$$\text{Var}(S_n) = n \text{Var}(X_i)$$

Example 2.1.6: Variance of Binomial

Using what we have just proved, find $\text{Var}(B(n, p))$

2.2 Law of Large Numbers

You might have heard that if you flip a coin 100 times, you will get ~ 50 coins to be heads, i.e., half the coins will be heads. The idea here is that as the amount of times you repeat an experiment becomes larger, you expected the changes of the event to happen to regress to the mean. This is known as the law of Large Numbers and will be the focus of this section. In a sense, what the law of large number's is trying to gain from the world is whether there particular scenario we have was quantified in such a way that we can see patterns happen if it is repeated, in particular whether there is a particular "favoured direction" in which the scenario will lean towards (the expected value). Using the law of large numbers will be how we actually determine from the noise of repeated experiments whether there are actually something going on in the randomness! Naturally in the real world, it is very hard (one can even say impossible) to have an identical experiment repeated multiple times. However, with enough control, we can get close enough to get hopefully meaningful results.

We will first generalize Chebyshev's inequality to use Variances. In some textbooks, this is considered *the* Chebyshev's inequality:

Lemma 2.2.1: Chebyshev's Inequality II

Let X be a random variable on a discrete probability space with $\mathbb{E}X^2 < \infty$, and let $\mu = \mathbb{E}X$. Then for any $x > 0$

$$\mathbb{P}(|X - \mu| \geq x) \leq \frac{\text{Var}(X)}{x^2}$$

The key is that the variance is bounded, so the variance *exists*, so to speak, and so we can bound our values:

Proof:

Put $Y = (X - \mu)^2 \geq 0$. Since $Y \leq 2X^2 + 2\mu^2$, $\mathbb{E}Y$ is finite, and $\mathbb{E}Y = \text{Var}(X)$ by definition 2.1.1. Since $|X - \mu|$ and x are nonnegative, $\{|X - \mu| \geq x\} = \{Y \geq x^2\}$, and theorem 1.6.1, applied to Y at $x^2 > 0$, gives

$$\mathbb{P}(|X - \mu| \geq x) = \mathbb{P}(Y \geq x^2) \leq \frac{\mathbb{E}Y}{x^2} = \frac{\text{Var}(X)}{x^2}$$

completing the proof.

Essentially, the probability that the random variable X is far away from the mean gets smaller at a rate of $\mathcal{O}(x^2)$.

One more thing we need before proving the law of large numbers: if $X_1, \dots, X_n$ are random variables, then let

$$\overline{X}_n := \frac{1}{n} \sum_{i=1}^n X_i$$

which will be our random variable that takes into consideration the “average outcome” of many events

Theorem 2.2.2: Law of Large Numbers

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables on a discrete probability space and define $\overline{X}_n$ to be their average. Let $\mu = \mathbb{E}X_1$ and $\sigma^2 = \text{Var}(X_1) < \infty$. Then for any $\epsilon > 0$

$$\mathbb{P}(|\overline{X}_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}$$

Proof:

Since the X_i have the distribution of X_1 , lemma 1.3.3, grouping the points by the value of X_i , gives $\mathbb{E}X_i = \mu$ and $\mathbb{E}(X_i - \mu)^2 = \sigma^2$ for every i . By proposition 1.3.6, $\mathbb{E}\overline{X}_n = \frac{1}{n} \sum_i \mathbb{E}X_i = \mu$. For $i \neq j$, the variables $X_i - \mu$ and $X_j - \mu$ are independent by lemma 1.5.10, applied with the groups $\{i\}$, $\{j\}$ and the remaining indices, the last with the constant function 0. They have expected value 0, so proposition 1.5.12 gives $\mathbb{E}(X_i - \mu)(X_j - \mu) = 0$. Expanding the square and using proposition 1.3.6, each product having finite expectation because $|uv| \leq \frac{1}{2}(u^2 + v^2)$,

$$\text{Var}(\overline{X}_n) = \mathbb{E}\left(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)\right)^2 = \frac{1}{n^2} \sum_{i,j=1}^n \mathbb{E}(X_i - \mu)(X_j - \mu) = \frac{1}{n^2} \sum_{i=1}^n \mathbb{E}(X_i - \mu)^2 = \frac{\sigma^2}{n}$$

Finally, lemma 2.2.1, applied to $\overline{X}_n$, whose second moment is finite by the same bound, gives

$$\mathbb{P}(|\overline{X}_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}$$

as claimed.

Notice that no matter what ϵ is chosen, however small, as $n \rightarrow \infty$, $\frac{\sigma^2}{n\epsilon^2} \rightarrow 0$, meaning the average of the i.i.d. random variables (i.e. the same experiment repeated several times perfectly and independently) will approach the average.

2.3 Bernstein Polynomials

It is a famous result in analysis that all continuous functions can be arbitrarily well approximated by polynomials¹. There is a way to prove this fact using statistics!

¹i.e. the polynomials are dense in the metric space of continuous functions

Definition 2.3.1: Bernstein Polynomials

Let $f : [0, 1] \rightarrow \mathbb{R}$ be a [uniformly] continuous function. Then the *Bernstein Polynomial of order n* associated to the function f is:

$$B_n(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}$$

Note that f is automatically uniformly continuous since it's domain is compact.

Theorem 2.3.2: Weierstrass Approximation Theorem

If $f : [0, 1] \rightarrow \mathbb{R}$ is continuous function, then:

$$\lim_{n \rightarrow \infty} \max_{x \in [0, 1]} |f(x) - B_n(x)| = 0$$

Proof:

Let $p \in [0, 1]$, $X_1, \dots, X_n$ be i.i.d. Bernoulli random variables. $B(p)$, and

$$S_n = X_1 + \dots + X_n \quad \bar{X}_n = \frac{S_n}{n}$$

Since S_n has a binomial distribution, we have:

$$\mathbb{E}(f(\bar{X}_n)) = \mathbb{E}\left(f\left(\frac{S_n}{n}\right)\right) = \sum_{k=1}^n f\left(\frac{k}{n}\right) \binom{n}{k} p^k (1-p)^{n-k}$$

which is equal to the Bernstein polynomial evaluated at $x = p$: $B_n(x)$. In other words, the Bernstein polynomials are the expected value of taking the “average” of random choices which are put through f . Now, consider

$$\begin{aligned} |B_n(p) - f(p)| &= |\mathbb{E}f(\bar{X}_n) - f(p)| \\ &= |\mathbb{E}(f(\bar{X}_n) - f(p))| \\ &\leq \mathbb{E}|f(\bar{X}_n) - f(p)| && \text{proposition 1.2.8} \\ &= \mathbb{E}[|f(\bar{X}_n) - f(p)| I(|\bar{X}_n - p| \leq \epsilon)] \\ &\quad + \mathbb{E}[|f(\bar{X}_n) - f(p)| I(|\bar{X}_n - p| > \epsilon)] \end{aligned}$$

the last line is a common trick in many cases where a function acts differently on two different parts (ex. For x^2 , if $x \geq 1$ then $x^2 \geq x$, but if $0 < x < 1$ then $x^2 < x$). For the first term, since f is differentiable, we can bound it by the modulus of continuity:

$$\Delta(\epsilon) = \max_{|x-y| \leq \epsilon} |f(x) - f(y)|$$

For the second term, we use uniform continuity again to use the fact that there exists a C such that $|f| \leq C$, so we get

$$\mathbb{E}|f(\bar{X}_n) - f(p)| \leq \mathbb{E}[\Delta(\epsilon) + 2C \cdot I(|\bar{X}_n - p| > \epsilon)] = \Delta(\epsilon) + 2C\mathbb{P}(|\bar{X}_n - p| > \epsilon)$$

Finally, we are at a position where we can use Chebyshev's inequality. First, $\mathbb{E}(B_n(p)) = n \frac{\mathbb{E}(B(p))}{n} = p$. Next, since the variance of the Bernoulli random variable is $\text{Var}(B(p)) = p(1-p) \leq 1/4$ (i.e., is bounded by a fair coin), we have $\text{Var}(B_n(p)) = \frac{n}{n^2} \text{Var}(B(p)) = \frac{1}{4n} \leq \frac{1}{4}$. Combining these, we get:

$$|B_n(p) - f(p)| \leq \Delta(\epsilon) + \frac{C}{2n\epsilon^2}$$

Notice that p is not in the upper bound. So this is true for all p , including the one that maximizes the equation:

$$\max_{p \in [0,1]} |B_n(p) - f(p)| \leq \Delta(\epsilon) + \frac{C}{2n\epsilon^2}$$

And taking $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \max_{p \in [0,1]} |B_n(p) - f(p)| \leq \Delta(\epsilon)$$

and since f is uniformly continuous, as $\epsilon \rightarrow 0$, $\Delta(\epsilon) \rightarrow 0$, and since this true for all $\epsilon > 0$, it must be that:

$$\lim_{n \rightarrow \infty} \max_{p \in [0,1]} |B_n(p) - f(p)| = 0$$

as we sought to show

2.4 Cliques on Erdős-Rényi Graph

In section 1.7.3 we found that the expected number of cliques of size k is

$$f(k) = \mathbb{E}N_k = \binom{n}{k} p^{\binom{k}{2}}$$

that $f(k_0) \geq 1 > f(k_0 + 1)$ for a threshold $k_0 \sim \frac{2}{\log(1/p)} \log n$ (lemma 1.7.3), and that for fixed $0 < \epsilon < 1$ and $m \geq 1$, for all large n ,

$$\mathbb{P}(N_{k_0+m+1} > 0) \leq \frac{1}{n^{m(1-\epsilon)}}$$

(corollary 1.7.4). So for large n a clique of size $k_0 + 2$ or more is unlikely. Below the threshold the expected number of cliques is large, and the variance shows that N_k concentrates around it.

Fix a small $\epsilon > 0$ and consider k with

$$\frac{(2-\epsilon) \log(n)}{\log(1/p)} \leq k < k_0 \quad (2.1)$$

Since $p < 1$, the left inequality gives $p^k \leq p^{(2-\epsilon) \log n / \log(1/p)} = n^{-(2-\epsilon)}$, so the ratio of consecutive values of f satisfies

$$\frac{f(k+1)}{f(k)} = \frac{n-k}{k+1} p^k \leq n p^k \leq \frac{1}{n^{1-\epsilon}}$$

Since $f(k_0) \geq 1$, we get $f(k_0 - 1) \geq n^{1-\epsilon}$, $f(k_0 - 2) \geq n^{2(1-\epsilon)}$, and by induction $f(k_0 - m) \geq n^{m(1-\epsilon)}$ as long as $k_0 - m$ satisfies (2.1). So the expected number of cliques of each size in the range (2.1) is large.

2.4.1 Chebyshev's Inequality in Relative Form Suppose $\mathbb{E}X > 0$, and take $x = \delta \mathbb{E}X$. Take some $\delta > 0$ and consider:

$$\mathbb{P}(|X - \mathbb{E}X| \geq \delta \mathbb{E}X) \leq \frac{\text{Var}(X)}{\delta^2 (\mathbb{E}X)^2}$$

usually, $\delta \ll 1$. The key comes if the variance is very small as compared to $\mathbb{E}(X)^2$: $\text{Var}(X) \ll \mathbb{E}(X)^2$. Then since we can re-write $|X - \mathbb{E}X| < \delta \mathbb{E}X$ as:

$$1 - \delta \leq \frac{X}{\mathbb{E}X} \leq 1 + \delta$$

then the probability of the complement of the Chebyshev inequality presented earlier is:

$$\mathbb{P}(1 - \delta \leq \frac{X}{\mathbb{E}X} \leq 1 + \delta) \geq 1 - \frac{\text{Var}(X)}{\delta^2 (\mathbb{E}X)^2}$$

Proposition 2.4.2: Cliques Below the Threshold

Let $0 < p < 1$ and $0 < \epsilon < 1$ be fixed, and let $k = k(n) = \lfloor (2 - \epsilon) \log n / \log(1/p) \rfloor$. Then

$$\lim_{n \rightarrow \infty} \frac{\text{Var}(N_k)}{(\mathbb{E}N_k)^2} = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \mathbb{P}(N_k > 0) = 1$$

Proof:

Put $b = 1/p > 1$ and $\ell_n = \log n / \log b$, so that $k \leq (2 - \epsilon)\ell_n$ and $b^{\ell_n} = n$.

Step 1: the variance as a sum over pairs. For a set W of k vertices let I_W be the indicator that W is a clique, so that $N_k = \sum_W I_W$. By the computation of $\text{Var}(S_n)$ following example 2.1.5,

$$\text{Var}(N_k) = \sum_{W, W'} \text{Cov}(I_W, I_{W'})$$

over ordered pairs of k -sets. If $|W \cap W'| \leq 1$, no edge lies inside both W and W' , so I_W and $I_{W'}$ are functions of disjoint sets of the independent edge indicators. They are then independent by lemma 1.5.10, and $\text{Cov}(I_W, I_{W'}) = 0$ by example 2.1.5. If $|W \cap W'| = \ell \geq 2$, then W and W' are both cliques exactly when the $2\binom{k}{2} - \binom{\ell}{2}$ edges inside W or inside W' are all present, so

$$\text{Cov}(I_W, I_{W'}) = p^{2\binom{k}{2} - \binom{\ell}{2}} - p^{2\binom{k}{2}} \leq p^{2\binom{k}{2}} b^{\binom{\ell}{2}}$$

There are $\binom{n}{k} \binom{k}{\ell} \binom{n-k}{k-\ell}$ ordered pairs with $|W \cap W'| = \ell$: choose W , then the ℓ common vertices, then the other $k - \ell$ vertices of W' outside W . Dividing by $(\mathbb{E}N_k)^2 = \binom{n}{k}^2 p^{2\binom{k}{2}}$ gives

$$\frac{\text{Var}(N_k)}{(\mathbb{E}N_k)^2} \leq \sum_{\ell=2}^k \frac{\binom{k}{\ell} \binom{n-k}{k-\ell}}{\binom{n}{k}} b^{\binom{\ell}{2}}$$

Step 2: each term. Let $2 \leq \ell \leq k < n$. Then $\binom{k}{\ell} \leq k^\ell$, $\binom{n-k}{k-\ell} \leq \binom{n}{k-\ell}$, and

$$\frac{\binom{n}{k-\ell}}{\binom{n}{k}} = \prod_{j=1}^{\ell} \frac{k-\ell+j}{n-k+j} \leq \left(\frac{k}{n-k} \right)^\ell$$

Also $b^{\binom{\ell}{2}} = (b^{(\ell-1)/2})^\ell$, and $b^{(\ell-1)/2} \leq b^{k/2} \leq b^{(1-\epsilon/2)\ell_n} = n^{1-\epsilon/2}$. So the ℓ -th term is at most q_n^ℓ , where

$$q_n = \frac{k^2 n^{1-\epsilon/2}}{n-k}$$

Step 3: $q_n \rightarrow 0$. With $x = \frac{\epsilon}{4} \log n$, which tends to ∞ ([2, The Logarithm Is Unbounded]),

$$\frac{k}{n^{\epsilon/4}} \leq \frac{2\ell_n}{n^{\epsilon/4}} = \frac{8}{\epsilon \log b} \cdot \frac{x}{e^x} \rightarrow 0$$

by [2, The Exponential Outgrows Every Power]. So for all large n , $k \leq n^{\epsilon/4} \leq n/2$, hence $n-k \geq n/2$ and $q_n \leq 2k^2 n^{-\epsilon/2} = 2(k/n^{\epsilon/4})^2$, which tends to 0.

Step 4: conclusion. For all large n , $k \geq 2$ and $q_n \leq \frac{1}{2}$, and Steps 1 to 3 give

$$\frac{\text{Var}(N_k)}{(\mathbb{E}N_k)^2} \leq \sum_{\ell=2}^k q_n^\ell \leq \frac{q_n^2}{1-q_n} \leq 2q_n^2 \rightarrow 0$$

If $N_k = 0$ then $|N_k - \mathbb{E}N_k| = \mathbb{E}N_k$, so lemma 2.2.1 with $x = \mathbb{E}N_k > 0$ gives

$$\mathbb{P}(N_k = 0) \leq \mathbb{P}(|N_k - \mathbb{E}N_k| \geq \mathbb{E}N_k) \leq \frac{\text{Var}(N_k)}{(\mathbb{E}N_k)^2} \rightarrow 0$$

and $\mathbb{P}(N_k > 0) = 1 - \mathbb{P}(N_k = 0) \rightarrow 1$, as claimed.

By proposition 2.4.2 and corollary 1.7.4 with $m = 1$, for fixed $0 < \epsilon < 1$ the largest clique of $G(n, p)$ has size at least $\lfloor (2 - \epsilon) \log n / \log(1/p) \rfloor$ and at most $k_0 + 1$ with probability tending to 1, and $k_0 \sim \frac{2}{\log(1/p)} \log n$ by lemma 1.7.3.

Exponential Inequalities

So far, the best we have been able to bound probabilities is with the use of Chebyshev's inequality. We used it when analyzing the sum of i.i.d. random variables with $\mathbb{E}X^2 < \infty$ to get:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\text{Var}(X)}{n\epsilon^2}$$

The key assumption here is that $\mathbb{E}X^2 < \infty$. If we can assume the decay is faster (which, in fact, it often is!), then we can get a much stronger bound. In particular, if

$$\mathbb{E}e^{\lambda X} < \infty \quad \text{for all } \lambda \in \mathbb{R}$$

then we can get some much stronger bounds; the error of this bound is much better, since e^λ for negative lambda shrinks much faster than $1/x^2$.

3.1 Markov and Hoeffding's Inequality

This bound is not arbitrary: notice that if a random variable X is bounded by a constant, then this condition holds.

Example 3.1.1: Bounded Random Variables

Let $X < c$. Then

$$\mathbb{E}X \leq \mathbb{E}c = c$$

and so

$$\mathbb{E}e^{X\lambda} \leq \mathbb{E}e^{c\lambda} = e^{c\lambda}$$

Furthermore, the exponential is one of the quicker shrinking functions we have at our disposal: it shrinks faster than any polynomial time. Thus, if we find ourselves being able to study random variables that grow rather quickly, we can bound our function even further:

Theorem 3.1.2: Markov's Inequality

Let X be a random variable on a discrete probability space with $\mathbb{E}e^{\lambda X} < \infty$ for all $\lambda \geq 0$. Then for every $t \in \mathbb{R}$ and every $\lambda \geq 0$:

$$\mathbb{P}(X \geq t) \leq e^{-\lambda t} \mathbb{E}e^{\lambda X}$$

or more concisely, since $\mathbb{P}(X \geq t)$ is independent of any λ , we can write:

$$\mathbb{P}(X \geq t) \leq \inf_{\lambda \in \mathbb{R}_{\geq 0}} \frac{\mathbb{E}e^{\lambda X}}{e^{\lambda t}}$$

Proof:

If $\lambda = 0$, the right hand side is equal to 1, so the inequality holds. Thus, assume $\lambda > 0$. Since e^x is a monotone function, $X \geq t$ if and only if $e^{\lambda X} \geq e^{\lambda t}$. Furthermore, e^x is strictly positive, and so we can apply the non-negative version of Chebyshev's inequality (the one without the indicator function):

$$\mathbb{P}(X \geq t) = \mathbb{P}(e^{\lambda X} \geq e^{\lambda t}) \stackrel{\text{C.}}{\leq} \frac{\mathbb{E}e^{\lambda X}}{e^{\lambda t}}$$

completing the proof.

Keep the minimum formula in the back of your mind when working with Markov's inequality; we will often take advantage of this fact. As mentioned earlier, many random variables satisfy the exponential condition. As a key example, let ϵ_i be a random variable such that:

$$\mathbb{P}(\epsilon = -1) = \mathbb{P}(\epsilon = 1) = 1/2$$

that is, it is the Bernoulli random variable, but modified so that $\mathbb{E}\epsilon_i = 0$. This is called the *Rademacher* random variable. Now, consider collection of i.i.d. ϵ_i and the new random variable:

$$X = \sum_{i=1}^n a_i \epsilon_i$$

for some arbitrary $a_1, \dots, a_n \in \mathbb{R}$. This random variable takes on as values all possible $\pm$ combinations of the elements a_i . As a use of such a random variable, imagine there are n events for which if you win, you get a_i amount of money, and if you loose, you give a_n amount of money. In each event, you have a 50% chance of winning. Then this random variable captures the amount of money you will win. Notice that this is a bounded random variable, and so we can apply Markov's inequality and see that the chances that all the signs are positive is in fact rather slim!

Theorem 3.1.3: Hoeffding's Inequality

Let $\epsilon_1, \epsilon_2, \dots, \epsilon_i$ be i.i.d Rademacher random variables and $a_1, a_2, \dots, a_n \in \mathbb{R}$. Then for any $t \geq 0$

$$\mathbb{P}\left(\sum_{i=1}^n \epsilon_i a_i \geq t\right) \leq \exp\left(-\frac{t^2}{2 \sum_{i=1}^n a_i^2}\right)$$

and

$$\mathbb{P}\left(\left|\sum_{i=1}^n \epsilon_i a_i\right| \geq t\right) \leq 2 \exp\left(-\frac{t^2}{2 \sum_{i=1}^n a_i^2}\right)$$

Proof:

First, we can directly apply Markov's inequality:

$$\mathbb{P}\left(\sum_{i=1}^n \epsilon_i a_i \geq t\right) \leq \frac{\mathbb{E}e^{\lambda \sum_{i=1}^n \epsilon_i a_i}}{e^{\lambda t}} = e^{-\lambda t} \mathbb{E} \prod_{i=1}^n e^{\lambda \epsilon_i a_i}$$

Next, applying independence of the ϵ_i , by induction on proposition 1.5.12 ($\mathbb{E}XY = \mathbb{E}X\mathbb{E}Y$), we have:

$$e^{-\lambda t} \mathbb{E} \prod_{i=1}^n e^{\lambda \epsilon_i a_i} = e^{-\lambda t} \prod_{i=1}^n \mathbb{E} e^{\lambda \epsilon_i a_i}$$

At this point, we can directly compute the expected value of each $\mathbb{E}e^{\lambda \epsilon_i}$, since we can then use some knowledge from complex analysis to equate this value to a known function:

$$\mathbb{E}e^{\lambda \epsilon_i} = \frac{1}{2}e^{\lambda a_i} + \frac{1}{2}e^{-\lambda a_i} = \cosh(\lambda a_i)$$

Since we have that every expected value is related to $\cosh(x)$, we can use some Taylor series to show that $\cosh(x) \leq e^{x^2/2}$:

$$\cosh(x) = \frac{e^x + e^{-x}}{2} = \sum_{k=0}^{\infty} \frac{1}{(2k)!} x^{2k} \leq \sum_{k=0}^{\infty} \frac{1}{2^k k!} x^{2k} = e^{x^2/2}$$

Thus, for every ϵ_i , we get that $\mathbb{E}e^{\lambda \epsilon_i a_i} \leq e^{\frac{\lambda^2 a_i^2}{2}}$. Substituting back into our equation, we get:

$$\mathbb{P}\left(\sum_{i=1}^n \epsilon_i a_i \geq t\right) \leq \exp\left(-\lambda t + \frac{\lambda^2}{2} \sum_{i=1}^n a_i^2\right)$$

This inequality is true over all λ , and so we can minimize it over our choice of λ :

$$\mathbb{P}\left(\sum_{i=1}^n \epsilon_i a_i \geq t\right) \leq \min_{\lambda \in \mathbb{R}_{\geq 0}} \left\{ \exp\left(-\lambda t + \frac{\lambda^2}{2} \sum_{i=1}^n a_i^2\right) \right\}$$

To find this, we take advantage of some calculus knowledge. In particular, we start by finding the critical points of the value in the exponent (since e^x is monotone, it is equivalent to minimize the exponent). Thus:

$$\frac{d}{d\lambda} \left(-\lambda t + \frac{\lambda^2}{2} \sum_{i=1}^n a_i^2 \right) = -t + \lambda \sum_{i=1}^n a_i^2 \quad \Leftrightarrow \quad \lambda = \frac{t}{\sum_{i=1}^n a_i^2}$$

and so, plugging this value back in, we get:

$$\begin{aligned} \exp \left(-t \frac{t}{\sum_{i=1}^n a_i^2} + \frac{\left(\frac{t}{\sum_{i=1}^n a_i^2} \right)^2}{2} \sum_{i=1}^n a_i^2 \right) &= \exp \left(-\frac{t^2}{\sum_{i=1}^n a_i^2} + \frac{t^2}{2 \left(\sum_{i=1}^n a_i^2 \right)^2} \sum_{i=1}^n a_i^2 \right) \\ &= \exp \left(-\frac{t^2}{2 \sum_{i=1}^n a_i^2} \right) \end{aligned}$$

Thus:

$$\mathbb{P} \left(\sum_{i=1}^n \epsilon_i a_i \geq t \right) \leq \exp \left(-\frac{t^2}{2 \sum_{i=1}^n a_i^2} \right)$$

proving the first inequality. For the second inequality, we take advantage of the symmetry of ϵ_i . In particular, the random variable $(-\epsilon_i)$ has the same distribution:

$$\mathbb{P}(-\epsilon_i = 1) = \mathbb{P}(-\epsilon_i = -1) = 1/2$$

and so:

$$\mathbb{P} \left(\sum_{i=1}^n -\epsilon_i a_i \geq t \right) = \mathbb{P} \left(\sum_{i=1}^n \epsilon_i a_i \leq -t \right) \leq \exp \left(-\frac{t^2}{2 \sum_{i=1}^n a_i^2} \right)$$

Since:

$$\left\{ \left| \sum_{i=1}^n a_i \epsilon_i \right| \geq t \right\} = \left\{ \sum_{i=1}^n a_i \epsilon_i \geq t \right\} \cup \left\{ \sum_{i=1}^n a_i \epsilon_i \leq -t \right\}$$

Then since $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$ (i.e., the union bound), we have the desired result.

As an immediate use-case of this inequality, let's improve our bound from the law of large numbers when we have i.i.d. Bernoulli random variables:

Example 3.1.4: Law of Large Numbers with fair coins

In theorem 3.1.3, if we let $a_i = \frac{1}{n}$ for all a_i , we get:

$$\mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n \epsilon_i \right| \geq t \right) \leq 2 \exp \left(-\frac{n^2 t^2}{2 \sum_{i=1}^n 1} \right) = 2e^{-\frac{nt^2}{2}}$$

We can relate this to $\bar{X}_n = \frac{1}{n} \sum X_i$ where the X_i are i.i.d. Bernoulli random variables by taking $X_i = (\epsilon_i + 1)/2$. Since each ϵ_i are i.i.d, each X_i is i.i.d, and further more $X_i \sim B(1/2)$. Isolating $\epsilon_i = 2X_i - 1$ and substituting back in, we get:

$$\mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n 2X_i - 1 \right| \geq t \right) = \mathbb{P} \left(|\bar{X}_n - 1/2| \geq \frac{t}{2} \right) \leq 2e^{-\frac{nt^2}{2}}$$

Substituting $\epsilon = t/2$, we get:

$$\mathbb{P} (|\bar{X}_n - 1/2| \geq \epsilon) \leq 2e^{-2n\epsilon^2}$$

If we were to do Chebyshev's inequality, we get:

$$\mathbb{P} (|\bar{X}_n - 1/2| \geq \epsilon) \leq \frac{1}{4n\epsilon^2}$$

Notice how much stronger our new inequality is! Let's take $n = 10000$ and $\epsilon = 0.02$. Then the two bounds we get are:

$$0.00067 \quad \text{vs.} \quad 0.0625$$

meaning the Hoeffding bound is over 100 times smaller than what we get with Chebyshev!

We might further ask if we can estimate any better than this. We will ask this question again when we cover the Central Limit Theorem, at which point we will show that n and ϵ on the right hand side is almost as good as it gets.

Another use of the inequality comes from studying higher moments of the Rademacher Random variables:

Example 3.1.5: Higher moments of Rademacher RV

Let $a_1, a_2, \dots, a_n \in \mathbb{R}$ such that:

$$a_1^2 + a_2^2 + \dots + a_n^2 = 1$$

that is, $(a_1, \dots, a_n)$ is a unit vector of $\mathbb{R}^n$. Let $\epsilon_1, \dots, \epsilon_n$ be i.i.d. Rademacher random variables, and take the random variable:

$$X := a_1\epsilon_1 + \dots + a_n\epsilon_n$$

Take a moment to see that all of its odd moments are equal to zero:

$$\mathbb{E}X^{2k+1} = \mathbb{E}(a_1\epsilon_1 + \dots + a_n\epsilon_n)^{2k+1} = 0$$

Its second moment is:

$$\mathbb{E}X^2 = \mathbb{E}(a_1\epsilon_1 + \dots + a_n\epsilon_n)^2 = a_1^2 + \dots + a_n^2 = 1$$

We will use Hoeffding's inequality to bound the other even moments, using the tail formula for the expected value of example 1.3.10(1), which rests on proposition 1.3.9:

$$\mathbb{E}X^{2k} = \int_0^\infty 2kt^{2k-1}\mathbb{P}(|X| \geq t)dt \stackrel{!}{\leq} \int_0^\infty 4kt^{2k-1}e^{-t^2/2}dt$$

where we used Hoeffding's inequality in $\stackrel{!}{\leq}$ by replacing $\mathbb{P}(|X| \geq t) \leq 2e^{-t^2/2}$. If we set $t = \sqrt{2u}$, we get:

$$\mathbb{E}X^{2k} \leq k2^{k+1} \int_0^\infty u^{k-1}e^{-u}du = k2^{k+1}\Gamma(k) = k2^{k+1}(k-1)! = 2^{k+1}k!$$

where we recognized the integral is the gamma function, and we can evaluate the Gamma function to be $(k-1)!$ (we will in fact take lots of time to explore this function in the next chapter, so no need to worry about not recognizing it at this stage).

Given Rademacher random variables $\epsilon_1, \dots, \epsilon_n$ and real numbers $a_1, \dots, a_n$, let $X = a_1\epsilon_1 + \dots + a_n\epsilon_n$ and

$$Y = X^2 - 1 = (a_1\epsilon_1 + \dots + a_n\epsilon_n)^2 - 1$$

Lemma 3.1.6: Exponential Moments of Y

Let $\epsilon_1, \dots, \epsilon_n$ be i.i.d. Rademacher random variables, let $a_1, \dots, a_n \in \mathbb{R}$ such that $a_1^2 + \dots + a_n^2 = 1$, and let Y be defined as above. Then for any $0 \leq \lambda \leq \frac{1}{4}$,

$$\mathbb{E}e^{\lambda Y} \leq e^{16\lambda^2} \text{ and } \mathbb{E}e^{-\lambda Y} \leq e^{16\lambda^2}$$

Proof:

Write $X = a_1\epsilon_1 + \dots + a_n\epsilon_n$, so $Y = X^2 - 1$, and $\epsilon = (\epsilon_1, \dots, \epsilon_n)$. By independence, $\mathbb{P}(\epsilon = s) = 2^{-n}$ for every $s \in \{-1, 1\}^n$ (definition 1.2.9). For a function $G: \{-1, 1\}^n \rightarrow \mathbb{R}$, the random variable $G(\epsilon)$ is the finite sum $\sum_s G(s) I(\epsilon = s)$, and the expected value of the indicator of an event is its probability, so proposition 1.3.6 gives

$$\mathbb{E}G(\epsilon) = \sum_{s \in \{-1, 1\}^n} G(s) \mathbb{P}(\epsilon = s) = 2^{-n} \sum_{s \in \{-1, 1\}^n} G(s)$$

Step 1: even moments. For integers $m_1, \dots, m_n \geq 0$, the display above and the expansion of a product of sums give

$$\mathbb{E}\epsilon_1^{m_1} \dots \epsilon_n^{m_n} = 2^{-n} \sum_{s \in \{-1, 1\}^n} \prod_{i=1}^n s_i^{m_i} = \prod_{i=1}^n \frac{1 + (-1)^{m_i}}{2}$$

which is 1 if every m_i is even and 0 otherwise. Let $k \geq 1$. Expanding X^{2k} by theorem 0.1.5 and applying proposition 1.3.6, only the terms with every exponent even survive. Writing these exponents as $2j_1, \dots, 2j_n$, and using $(2j)! = 2^j j! \prod_{l=1}^j (2l-1) \geq 2^j j!$,

$$\mathbb{E}X^{2k} = \sum_{j_1 + \dots + j_n = k} \frac{(2k)!}{(2j_1)! \dots (2j_n)!} a_1^{2j_1} \dots a_n^{2j_n} \leq \frac{(2k)!}{2^k k!} \sum_{j_1 + \dots + j_n = k} \binom{k}{j_1, \dots, j_n} (a_1^2)^{j_1} \dots (a_n^2)^{j_n}$$

The last sum is $(a_1^2 + \dots + a_n^2)^k = 1$, by theorem 0.1.5. For $k = 1$ every term has a single j_i equal to 1, so the inequality is an equality, and $\mathbb{E}X^2 = 1$. In general $\frac{(2k)!}{2^k k!} = \prod_{l=1}^k (2l-1) \leq 2^k k!$, so $\mathbb{E}X^{2k} \leq 2^k k!$.

Step 2: a bound for the exponential. For every real x , with $x_+ = \max(x, 0)$,

$$e^x \leq 1 + x + \frac{x^2}{2} + \sum_{k=3}^{\infty} \frac{(x_+)^k}{k!}$$

For $x \geq 0$ this is the exponential series ([2, The Exponential, Cosine and Sine Functions]), with equality. For $x < 0$ the sum vanishes, and [2, Lagrange Form of the Remainder], applied to the exponential at 0 with the Taylor polynomial of degree 2, gives $e^x = 1 + x + \frac{x^2}{2} + \frac{e^\xi}{6} x^3$ for some ξ between x and 0, and the last term is negative.

Step 3: the two bounds. Let $0 \leq \lambda \leq \frac{1}{4}$ and $\sigma \in \{1, -1\}$. Apply Step 2 with $x = \sigma\lambda Y$ at each of the 2^n values of ϵ , and average with the display before Step 1. The average of finitely many convergent series is the series of the averages ([2, The Algebra of Series and the Triangle Inequality](1)), so

$$\mathbb{E}e^{\sigma\lambda Y} \leq 1 + \sigma\lambda \mathbb{E}Y + \frac{\lambda^2}{2} \mathbb{E}Y^2 + \sum_{k=3}^{\infty} \frac{\lambda^k}{k!} \mathbb{E}((\sigma Y)_+)^k$$

By Step 1, $\mathbb{E}Y = \mathbb{E}X^2 - 1 = 0$ and $\mathbb{E}Y^2 = \mathbb{E}X^4 - 2\mathbb{E}X^2 + 1 = \mathbb{E}X^4 - 1 \leq 2^2 \cdot 2!$. For $\sigma = 1$, $0 \leq Y_+ \leq X^2$ gives $\mathbb{E}(Y_+)^k \leq \mathbb{E}X^{2k} \leq 2^k k!$, and for $\sigma = -1$, $0 \leq (-Y)_+ = (1 - X^2)_+ \leq 1$ gives $\mathbb{E}((-Y)_+)^k \leq 1 \leq 2^k k!$, both by the display before Step 1, since each inequality holds at every one of the 2^n values of ϵ . Since $0 \leq 2\lambda \leq \frac{1}{2}$, [2, The Geometric Series] gives

$$\mathbb{E}e^{\sigma\lambda Y} \leq 1 + \sum_{k=2}^{\infty} \frac{\lambda^k}{k!} 2^k k! = 1 + \sum_{k=2}^{\infty} (2\lambda)^k = 1 + \frac{4\lambda^2}{1-2\lambda} \leq 1 + 8\lambda^2$$

Every term of the exponential series at $8\lambda^2 \geq 0$ is nonnegative, so $1+8\lambda^2 \leq e^{8\lambda^2}$, and $e^{8\lambda^2} \leq e^{16\lambda^2}$. Taking $\sigma = 1$ and $\sigma = -1$ gives the two inequalities, as claimed.

3.2 Johnson-Lindenstrauss lemma

In this section, we are going to take a detour to show an application of the previous result to prove a really important result that comes up often in the analysis of big data and functional analysis.

Let $N \geq 1$, and set pick m points in $\mathbb{R}^N$:

$$V = \{v_1, \dots, v_m\} \subseteq \mathbb{R}^N$$

Usually, we think of N being able to be arbitrarily large and m being large. What the Johnson-Lindenstrauss lemma shows is that there exists a *linear map* $\mathbb{R}^N$ into $\mathbb{R}^n$ (where n is possibly of much lower dimension) that preserves the distances between all points in the set V up to a small relative error! This is called a *low-distortion embedding*. It was discovered by Johnson and Lindenstrauss while studying functional analysis, but it has been very useful in computational algorithms as a pre-processing tool to reduce the number of dimensions of high-dimensional data (which is becoming important). We formalise this idea in the following theorem:

Theorem 3.2.1: Johnson-Lindenstrauss Lemma

Let $\{v_1, \dots, v_m\} \subseteq \mathbb{R}^N$ be m distinct points in $\mathbb{R}^N$. Then for any $\epsilon \in (0, 1)$ and any integer n with

$$n > \frac{128}{\epsilon^2} \log(m)$$

there exists a linear map $f : \mathbb{R}^N \rightarrow \mathbb{R}^n$ such that

$$\sqrt{1-\epsilon} \leq \frac{\|f(v_k) - f(v_l)\|}{\|v_k - v_l\|} \leq \sqrt{1+\epsilon}$$

for all $1 \leq k < l \leq m$, with $\|\cdot\|$ being the euclidean norm.

Notice that the dimension n is only dependent on m and ϵ , and not N . This means that this estimate can even work if N is small, however in most cases we are more interesting in having a large N and embedding into dimensions. Furthermore, the dependence on m is logarithmic, meaning that in general, increasing the number of vector only slightly changes the bound on n . Finally, the number 128 is not optimized. It can be shown that it can be diminished to 8, but such a bound is harder to prove.

For this proof, we will define a linear map with random variable as entries (i.e. a random matrix)

$$\mathcal{E} = \begin{pmatrix} \epsilon_{11} & \epsilon_{12} & \cdots & \epsilon_{1N} \\ \epsilon_{21} & \epsilon_{22} & \cdots & \epsilon_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \epsilon_{n1} & \epsilon_{n2} & \cdots & \epsilon_{nN} \end{pmatrix}$$

where ϵ_{ij} are i.i.d. Rademacher random variables. We will show that there exists an n satisfying the inequality such that:

$$f(x) = \frac{1}{\sqrt{n}} \mathcal{E}x$$

will satisfy the low distortion property and exist with positive probability. That is, there exists at least one matrix with ± 1 entries that satisfy the bounding equation given a properly chosen dimension n .

Before giving a proof, we will reformulate the bounds. Let $1 \leq k < l \leq m$ and define

$$a_{kl} = \frac{v_k - v_l}{\|v_k - v_l\|}$$

that is, each a_{kl} represents the difference between any two vectors v_k and v_l which is normalized to have length 1. Then if we square the main bound and replace f with the linear function we define above, we get:

$$1 - \epsilon \leq \frac{1}{n} \|\mathcal{E}a_{kl}\|^2 \leq 1 + \epsilon$$

Since each a_{kl} was rescaled so that $\|a_{kl}\| = 1$, then each a_{kl} belong to the unit sphere $a_{kl} \in S^{N-1} \subseteq \mathbb{R}^N$. We will further manipulate this so that the values we want will cluster (just like in the law of large number's). To get a feeling of how the Large numbers apply, consider an arbitrary vector

$$a = (a_1, \dots, a_N) \in S^{N-1}$$

and denote the i th coordinate of $\mathcal{E}a$ by

$$X_i(a) := (\mathcal{E}a)_i = \epsilon_{i1}a_1 + \cdots + \epsilon_{iN}a_N$$

and with this denote:

$$Y_i(a) := X_i^2(a) - 1 = (\epsilon_{i1}a_1 + \cdots + \epsilon_{iN}a_N)^2 - 1$$

With the new X_i notation, we can re-write our bounds again as:

$$1 - \epsilon \leq \frac{1}{n} \sum_{i=1}^n X_i^2(a) \leq 1 + \epsilon$$

and if we re-write as Y_i , we get:

$$\left| \frac{1}{n} \sum_{i=1}^n Y_i(a) \right| \leq \epsilon$$

At this point, this inequality is starting to look like the law of large numbers for $Y_1(a), \dots, Y_n(a)$ since $\mathbb{E}Y_i(a) = 0$ as we have shown in last section. Even better, we have managed to bound these inequalities in lemma 3.1.6 for $0 \leq \lambda \leq \frac{1}{4}$:

$$\mathbb{E}e^{\lambda Y_i(a)} \leq e^{16\lambda^2} \quad \text{and} \quad \mathbb{E}e^{-\lambda Y_i(a)} \leq e^{16\lambda^2}$$

Furthermore, notice that by the grouping lemma, the random variables $Y_1(a), \dots, Y_n(a)$ are independent since the rows of the matrix $\mathcal{E}$ are all independent by construction (as a high-level way of seeing this, if you choose values for one row of the random matrix, it tells you nothing about what values you will choose for the other row of the matrix)

Using this inequality and independence, we get to prove the following lemma:

Lemma 3.2.2: Bounding $Y_i(a)$

Let $Y_1(a), \dots, Y_n(a)$ be the random variables we defined above from the $n \times N$ matrix $\mathcal{E}$ of i.i.d. Rademacher entries, let $\epsilon \in (0, 1)$ and let $a = (a_1, \dots, a_N) \in S^{N-1}$. Then:

$$\mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n Y_i(a) \right| \geq \epsilon \right) \leq 2e^{-\frac{n\epsilon^2}{64}}$$

Proof:

By independence of the nN entries, $\mathbb{P}(\mathcal{E} = s) = 2^{-nN}$ for every $n \times N$ matrix s with entries ± 1 (definition 1.2.9). As in the proof of lemma 3.1.6, the expected value of a function of $\mathcal{E}$ is therefore the average of its values over these 2^{nN} sign matrices. Let $\lambda \in \mathbb{R}$. The random variable $e^{\lambda \sum_i Y_i(a)} = \prod_{i=1}^n e^{\lambda Y_i(a)}$ is a product whose i -th factor depends only on the i -th row of $\mathcal{E}$. A sign matrix is a list of n rows, each a sign vector of length N , so by the distributive law the average over sign matrices of such a product is the product over i of the averages of the i -th factor over the 2^N possible i -th rows, and each of these averages is $\mathbb{E}e^{\lambda Y_i(a)}$:

$$\mathbb{E}e^{\lambda \sum_{i=1}^n Y_i(a)} = \prod_{i=1}^n \mathbb{E}e^{\lambda Y_i(a)}$$

Each $Y_i(a)$ is the variable Y of lemma 3.1.6 built from the i.i.d. Rademacher entries $\epsilon_{i1}, \dots, \epsilon_{iN}$ of the i -th row and from coefficients with $a_1^2 + \dots + a_N^2 = 1$. Let $\sigma \in \{1, -1\}$ and $\lambda = \epsilon/32$, so that $0 < \lambda \leq \frac{1}{4}$. Theorem 3.1.2, applied to $\sigma \sum_i Y_i(a)$ at $n\epsilon$, then the display with $\sigma\lambda$ in place of λ , then lemma 3.1.6, give

$$\mathbb{P} \left(\frac{\sigma}{n} \sum_{i=1}^n Y_i(a) \geq \epsilon \right) \leq e^{-n\lambda\epsilon} \prod_{i=1}^n \mathbb{E}e^{\sigma\lambda Y_i(a)} \leq e^{-n\lambda\epsilon + 16n\lambda^2} = e^{-n\epsilon^2/64}$$

The event in the statement is the union of these events for $\sigma = 1$ and $\sigma = -1$. An outcome in the union lies in at least one of them, and the probability of an event is the sum of the probabilities of its outcomes, so the probability of the union is at most $2e^{-n\epsilon^2/64}$, as claimed.

Proof of theorem 3.2.1:

Let $\mathcal{E}$ be the random $n \times N$ matrix above, whose entries are i.i.d. Rademacher random variables on the finite space of sign matrices, and for each value of $\mathcal{E}$ let $f(x) = \frac{1}{\sqrt{n}}\mathcal{E}x$, a linear map $\mathbb{R}^N \rightarrow \mathbb{R}^n$. For $1 \leq k < l \leq m$ the points v_k and v_l are distinct, so a_{kl} is a unit vector, and

$f(v_k) - f(v_l) = \|v_k - v_l\| f(a_{kl})$ by linearity. Since $Y_i(a) = X_i(a)^2 - 1$ and $\sum_{i=1}^n X_i(a)^2 = \|\mathcal{E}a\|^2$,

$$\frac{1}{n} \sum_{i=1}^n Y_i(a_{kl}) = \frac{1}{n} \|\mathcal{E}a_{kl}\|^2 - 1 = \|f(a_{kl})\|^2 - 1 = \frac{\|f(v_k) - f(v_l)\|^2}{\|v_k - v_l\|^2} - 1$$

Let B_{kl} be the event that $|\frac{1}{n} \sum_{i=1}^n Y_i(a_{kl})| \geq \epsilon$. Lemma 3.2.2, applied with $a = a_{kl} \in S^{N-1}$, gives $\mathbb{P}(B_{kl}) \leq 2e^{-n\epsilon^2/64}$. The probability of an event is the sum of the probabilities of its outcomes, and an outcome in the union of the $\binom{m}{2}$ events B_{kl} lies in at least one of them, so

$$\mathbb{P}\left(\bigcup_{1 \leq k < l \leq m} B_{kl}\right) \leq \sum_{1 \leq k < l \leq m} \mathbb{P}(B_{kl}) \leq \binom{m}{2} \cdot 2e^{-n\epsilon^2/64} \leq m^2 e^{-n\epsilon^2/64}$$

The hypothesis $n > \frac{128}{\epsilon^2} \log(m)$ says that $\frac{n\epsilon^2}{64} > 2 \log(m)$, that is, $m^2 e^{-n\epsilon^2/64} < 1$. So the complement of the union has positive probability, and in particular contains an outcome: a sign matrix $\mathcal{E}$ with $|\frac{1}{n} \sum_{i=1}^n Y_i(a_{kl})| < \epsilon$ for all $1 \leq k < l \leq m$. For the corresponding linear map f , the display above gives

$$1 - \epsilon < \frac{\|f(v_k) - f(v_l)\|^2}{\|v_k - v_l\|^2} < 1 + \epsilon$$

for all $1 \leq k < l \leq m$, and taking square roots gives the claim.

As a final observation, notice that in the end we had $m^2 e^{-n\epsilon^2/64} < 1$, and so there is a balance between m and n to make the value less than 1. However, since the n is in the exponential decay, by increasing the dimension of n , we can make sure that the random matrix we picked will yield a low-distortion embedding with not just positive probability, but with high probability (i.e. close to 1).

3.3 Hoeffding-Chernoff Inequality

In the proof of the Hoeffding inequality, we relied on the symmetry of the Rademacher random variables. Therefore, the same calculation does not apply for, say, a biased coin flip $B(p)$ ($p \neq 1/2$). We want to generalize our result, so we need to replace the i.i.d. restriction with a more general one. Since many random variables are bounded, we will replace the i.i.d. of random variables to the sum of bound independent random variables. Since the random variables are bounded, they can all be re-scaled to be within $[0, 1]$.

Let $X_1, \dots, X_n$ be independent random variables taking values in $[0, 1]$ with common expectation $p = \mathbb{E}X_i \in [0, 1]$. Notice that they are not identically distributed, and so they may have different distributions. In particular, there will be a bias now towards one part of the distribution. This is captured in the following concept:

Definition 3.3.1: Kullback-Leibler divergence

Let $p \in (0, 1)$ and $q \in [0, 1]$. Then define:

$$D(q||p) := q \log \left(\frac{q}{p} \right) + (1 - q) \log \left(\frac{1 - q}{1 - p} \right)$$

This function is called the *Kullback-Leibler divergence* or *relative entropy*, where if $q \in \{0, 1\}$, it is understood that $0 \cdot \log(0) = 0$

The last condition guarantees that $D(q||p)$ is continuous in q .

Proposition 3.3.2: Nonnegativity of the Kullback-Leibler Divergence

Let $p \in (0, 1)$ and $q \in [0, 1]$. Then $D(q||p) \geq 0$, with equality if and only if $q = p$.

Proof:

Step 1: $\log x \leq x - 1$ for every $x > 0$, with equality only at $x = 1$. Put $h(x) = x - 1 - \log x$ on $(0, \infty)$. By [2, The Derivative of the Logarithm], h is differentiable with $h'(x) = 1 - 1/x$, which is negative for $x < 1$ and positive for $x > 1$. Let $x \neq 1$. By [2, The Mean Value Theorem], on the closed interval with endpoints x and 1, there is a point c strictly between them with $h(x) - h(1) = h'(c)(x - 1)$. If $x < 1$ then $c < 1$, and if $x > 1$ then $c > 1$, so in both cases $h'(c)$ and $x - 1$ have the same sign and $h(x) > h(1) = 0$.

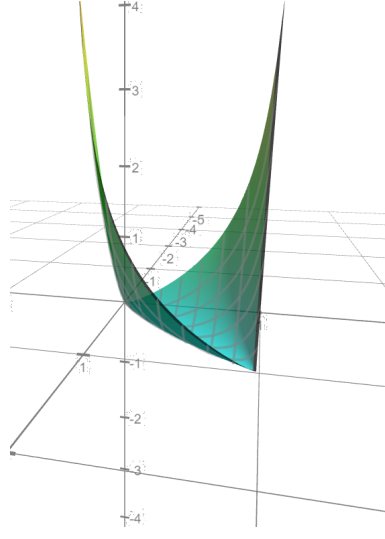
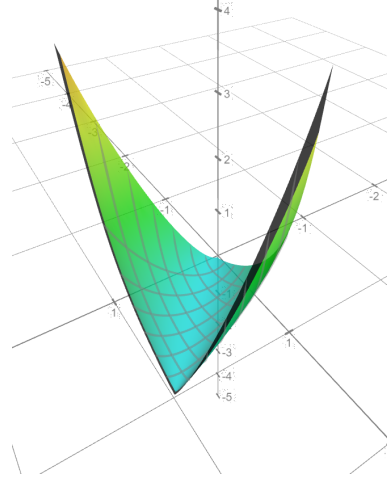
Step 2: $0 < q < 1$. Both p/q and $(1 - p)/(1 - q)$ are positive. Multiplying the two instances of Step 1 by $q > 0$ and by $1 - q > 0$ and adding,

$$-D(q||p) = q \log \frac{p}{q} + (1 - q) \log \frac{1 - p}{1 - q} \leq q \left(\frac{p}{q} - 1 \right) + (1 - q) \left(\frac{1 - p}{1 - q} - 1 \right) = (p - q) + (q - p) = 0$$

with equality if and only if both instances are equalities, that is, $p/q = 1$ and $(1 - p)/(1 - q) = 1$, which happens exactly when $q = p$.

Step 3: $q \in \{0, 1\}$. With the convention $0 \cdot \log(0) = 0$ of definition 3.3.1, $D(0||p) = \log \frac{1}{1 - p} = -\log(1 - p)$ and $D(1||p) = \log \frac{1}{p} = -\log p$. Both are positive, because Step 1 gives $\log x \leq x - 1 < 0$ for $0 < x < 1$, and p and $1 - p$ lie in $(0, 1)$. Since $q \neq p$ in this case, the claimed equivalence holds, completing the proof.

So D has one property of a metric: it is nonnegative, and it vanishes exactly when its two arguments agree. It is not a metric, however, because it is not symmetric. For instance $D(\frac{1}{2}||\frac{1}{4}) = \frac{1}{2} \log \frac{4}{3} = \frac{1}{4} \log \frac{16}{9}$ and $D(\frac{1}{4}||\frac{1}{2}) = \frac{1}{4} \log \frac{1}{2} + \frac{3}{4} \log \frac{3}{2} = \frac{1}{4} \log \frac{27}{16}$, and these differ because $\frac{16}{9} \neq \frac{27}{16}$ and the logarithm is injective. The way I like to think about it is imagine a metric is capturing the energy required to go from point a to point b on a flat surface. However, if your surface has hills and dips, then the energy required to go from point a to point b might be different from the energy required for going from point b to point a (due to relative entropy). $D(p||q)$ captures a particular such surface:

Figure 3.1: $D(p||q)$ one angleFigure 3.2: $D(p||q)$ another angle

Using this, we can generalize Hoeffding's inequality:

Theorem 3.3.3: Hoeffding-Chernoff Inequality

Let $X_1, \dots, X_n$ be bounded independent random variables on a discrete probability space, where $X_i \in [0, 1]$ and $p = \mathbb{E}X_i \in (0, 1)$ for each i . Then:

$$\mathbb{P}(\bar{X}_n \geq p + t) \leq e^{-nD(p+t||p)}$$

for any $0 \leq t \leq 1 - p$ and

$$\mathbb{P}(\bar{X}_n \leq p - t) \leq e^{-nD(1-p+t||1-p)}$$

for any $0 \leq t \leq p$

Just like Hoeffding's inequalities, the n is in the exponent, so these are again much better bounds than Chebyshev's inequalities. Furthermore, notice that since the average is always less than 1, the probability that $p + t > 1$ is 0, so the restriction that $t \leq 1 - p$ is in fact superfluous (and similarly for the $t \leq p$ constraint).

Furthermore, the fact $D(p+t||p)$ in the upper tail bound is not the same as $D(1-p+t||1-p)$ in the exponent in the lower tail bound, so these bounds are *not* symmetric. This should make sense, since heuristically, when $p < 1/2$, there is “less room” between 0 and p than above p between p and 1.

Proof:

First, apply Markov's inequality: for any $\lambda \geq 0$,

$$\begin{aligned}\mathbb{P}(\bar{X}_n \geq p + t) &= \mathbb{P}\left(\sum_{i=1}^n X_i \geq n(p + t)\right) \\ &\leq \min_{\lambda} e^{-\lambda n(p+t)} \mathbb{E} e^{\lambda \sum_{i=1}^n X_i} \\ &= \min_{\lambda} e^{-\lambda n(p+t)} \prod_{i=1}^n \mathbb{E} e^{\lambda X_i}\end{aligned}$$

the last step by proposition 1.5.12, applied $n - 1$ times to the variables $e^{\lambda(X_1 + \dots + X_k)}$ and $e^{\lambda X_{k+1}}$, which are independent by lemma 1.5.10. At this point, we bound the exponential by a chord, which removes x from the exponent. First, $e^v \geq 1 + v$ for every real v : the function $h(v) = e^v - 1 - v$ has $h'(v) = e^v - 1$ by [2, The Exponential Is Its Own Derivative], which is ≤ 0 for $v \leq 0$ and ≥ 0 for $v \geq 0$ because exp is increasing with $e^0 = 1$ ([2, The Exponential Is a Strictly Increasing Bijection onto $(0, \infty)$]). By [2, Monotonicity From the Sign of the Derivative], h is decreasing on $(-\infty, 0]$ and increasing on $[0, \infty)$, so $h(v) \geq h(0) = 0$. Now let $x \in [0, 1]$ and $\lambda \geq 0$. Applying this with $v = \lambda(1 - x)$ and with $v = -\lambda x$, and combining with the nonnegative weights x and $1 - x$,

$$xe^{\lambda(1-x)} + (1-x)e^{-\lambda x} \geq x(1 + \lambda(1-x)) + (1-x)(1 - \lambda x) = 1$$

and multiplying by $e^{\lambda x} > 0$,

$$e^{\lambda x} \leq xe^{\lambda} + (1-x)e^0 = 1 - x + xe^{\lambda}$$

Since each $X_i \in [0, 1]$, this holds with $x = X_i(\omega)$ at every point ω . Comparing the series that define the expected values term by term ([2, Limits Preserve Weak Inequalities]) and using proposition 1.3.6,

$$\mathbb{E} e^{\lambda X_i} \leq 1 - \mathbb{E} X_i + \mathbb{E} X_i e^{\lambda} = 1 - p + pe^{\lambda}$$

thus:

$$\min_{\lambda} e^{-\lambda n(p+t)} \prod_{i=1}^n \mathbb{E} e^{\lambda X_i} \leq \min_{\lambda} e^{-\lambda n(p+t)} (1 - p + pe^{\lambda})^n$$

To minimize λ , notice that since e^x and log is monotone, minimizing the log of the expression is equivalent:

$$\min_{\lambda} (-\lambda n(p+t) + n \log(1 - p + pe^{\lambda}))$$

which we do by finding the critical point as we have done in similar proofs:

$$-n(p+t) + \frac{npe^{\lambda}}{1 - p + pe^{\lambda}} = 0$$

and solving for λ , we get:

$$e^{\lambda} = \frac{(1-p)(p+t)}{p(1-p-t)}$$

This expression is greater than or equal to 1 since $p + t \geq p$ and $1 - p \geq 1 - p - t$, which implies $\lambda \geq 0$, as needed for Markov's inequality! Thus, we can plug this back into our expression and get:

$$\begin{aligned} e^{-\lambda n(p+t)}(1-p+pe^\lambda)^n &= \left[\left(\frac{(1-p)(p+t)}{p(1-p-t)} \right)^{p+t} \left(1-p + \frac{(1-p)(p+t)}{p(1-p-t)} \right) \right]^n \\ &= \left[\left(\frac{p}{p+t} \right)^{p+t} \left(\frac{1-p}{1-p-t} \right)^{1-p-t} \right]^n \\ &= \exp \left[-n \left((p+t) \log \frac{p+t}{p} + (1-p-t) \log \frac{1-p-t}{1-p} \right) \right] \\ &= e^{-nD(p+t||p)} \end{aligned}$$

and squishing these inequalities, we get:

$$\mathbb{P}(\bar{X}_n \geq p+t) \leq e^{-nD(p+t||p)}$$

proving the first inequality.

For the lower-tail bound, consider $Z_i = 1 - X_i \in [0, 1]$. Then $\mathbb{E}Z_i = 1 - p$. Applying what we have just proved, we get:

$$\mathbb{P}(\bar{Z}_n \geq 1 - p + t) \leq e^{-nD(1-p+t||1-p)}$$

since

$$\bar{Z}_n = 1 - \bar{X}_n \geq 1 - p + t \quad \Leftrightarrow \quad \bar{X}_n \leq p - t$$

completing the lower tail bound, and hence completing the proof

The exponents $D(p+t||p)$ and $D(1-p+t||1-p)$ depend on p . A lower bound on the divergence that depends only on the gap $|q-p|$ turns both tail bounds into the form of Hoeffding's inequality.

Lemma 3.3.4: Pinsker's Inequality for Bernoulli Parameters

For $p \in (0, 1)$ and $q \in [0, 1]$,

$$D(q||p) \geq 2(q-p)^2$$

Proof:

Fix p and set $g(q) = D(q||p) - 2(q-p)^2$ for $q \in [0, 1]$. With the convention $0 \log 0 = 0$ of definition 3.3.1, g is continuous on $[0, 1]$, since $q \log q \rightarrow 0$ as $q \rightarrow 0$. On $(0, 1)$,

$$g'(q) = \log \frac{q}{p} - \log \frac{1-q}{1-p} - 4(q-p) \quad \text{and} \quad g''(q) = \frac{1}{q} + \frac{1}{1-q} - 4 = \frac{1}{q(1-q)} - 4 \geq 0$$

because $q(1-q) \leq 1/4$. So g' is nondecreasing on $(0, 1)$, and $g'(p) = 0$ gives $g' \leq 0$ on $(0, p]$ and $g' \geq 0$ on $[p, 1)$. By the mean value theorem g is nonincreasing on $(0, p]$ and nondecreasing on $[p, 1)$, so $g(q) \geq g(p) = 0$ on $(0, 1)$, and by continuity also at $q = 0$ and $q = 1$, as we sought to show.

Applying lemma 3.3.4 with $q = p + t$ in the upper tail, and with $q = 1 - p + t$ and $1 - p$ in place of p in the lower tail, weakens theorem 3.3.3 to

$$\mathbb{P}(\bar{X}_n \geq p + t) \leq e^{-2nt^2} \quad \text{and} \quad \mathbb{P}(\bar{X}_n \leq p - t) \leq e^{-2nt^2}$$

which depend on p only through the admissible range of t .

Example 3.3.5: Generalization error of Classification Algorithm

A classification algorithm takes in a lot of data, and tries to map it to the appropriate classification. For example, given pictures of cats and dogs to a classification algorithm, the algorithm must correctly output “cat” or “dog”. For every “image object”, we will have a tuple (X, Y) with X representing the data (let’s say, the pixels of the image) and Y representing the label (in this example, it can either cat or dog). The algorithm will then be a function f that should take in X and give Y : $f(X) = Y$. The way these algorithms are created these days is by training them with a lot of sample data (X_i, Y_i) (for which Y_i is the known value) which the algorithm uses to find patterns to be able to pick up on non-sample data.

The algorithm will try to find a function f that minimizes the proportion of incorrectly guessed output’s given the training data:

$$E_n(f) := \frac{1}{n} \sum_{i=1}^n I(Y_i \neq f(X_i)) \rightarrow \text{minimize}$$

Or more generally, instead of using an indicator, a different function called a *loss function* can be used:

$$E_n(f) := \frac{1}{n} \sum_{i=1}^n L(Y_i, f(X_i)) \rightarrow \text{minimize}$$

This average is called the *empirical error* of the classifier f . Different algorithms will give different ways to minimize this error over different classes of functions $\mathcal{F}$ to find a good candidate function $f \in \mathcal{F}$. Now, let’s say the algorithm found a function that has a low error estimate on the training data. How do we know it will do a good job outside the training data? If we give it new input not in the training data, will it correctly classify the object?

One way to try and answer this question is to assume that the examples are coming from an [unknown] probability distribution $\mathbb{P}$ over a population of possible examples Ω , and that the training data consists of i.i.d. observations (let’s say the finite collection (X_i, Y_i) for $1 \leq i \leq n$) from the distribution $\mathbb{P}$. Then the ability to classify future examples (called the *generalisation ability* is measured by the *generalization error*, which we will define as the expected value of the loss function:

$$E(f) := \mathbb{E}(L(Y, f(X)))$$

We can ask ourselves if a small value in $E_n(f)$ means there is a small value for $E(f)$? If f is fixed, then as n gets larger, the values will concentrate; in fact this is precisely a use-case of the law of large numbers. However, f is not fixed, it depends on the training data, and so we cannot apply the law of large numbers. To see intuitively why the Law of Large Numbers cannot be implied, imagine that the algorithm brute-force memorizes the labels given an X_i (like memorizing the answers to a solution without understanding them). Then even if the amount of solution’s memorized increased, it will not necessarily generalize well. In statistics, this is called *over-fitting*

3.4 Azuma's Inequality

In the Hoeffding-Chernoff inequality we dealt with sums of bounded independent variables. Azuma's inequality bounds the deviations of a general function

$$Z = f(X_1, X_2, \dots, X_n)$$

of independent random variables, not only of their sum. Here $X_1, \dots, X_n$ are random variables on a discrete probability space, and each X_i may be a random vector: it takes values in a countable set S_i of numbers, of vectors, or of other objects. Independence means, as in definition 1.2.9,

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n \mathbb{P}(X_i = x_i)$$

for all $x_i \in S_i$. The function $f: S_1 \times \dots \times S_n \rightarrow \mathbb{R}$ satisfies the following *stability condition*: for some constants $a_1, \dots, a_n \geq 0$,

$$|f(x_1, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_n) - f(x_1, \dots, x_{i-1}, x'_i, x_{i+1}, \dots, x_n)| \leq a_i$$

for all $1 \leq i \leq n$, all $(x_1, \dots, x_n) \in S_1 \times \dots \times S_n$ and all $x'_i \in S_i$. That is, if we change one coordinate and fix all other coordinates, the value of the function changes by at most a_i . Changing the coordinates of a point x into those of a fixed point y one at a time gives $|f(x) - f(y)| \leq a_1 + \dots + a_n$, so f is bounded, and $\mathbb{E}f$ and $\mathbb{E}e^{\lambda f}$ are finite for every real λ .

Lemma 3.4.1: Special Bounding Condition

Let X be a random variable on a discrete probability space that satisfies $|X| \leq 1$ and $\mathbb{E}X = 0$. Then for any $\lambda \geq 0$

$$\mathbb{E}e^{\lambda X} \leq e^{\lambda^2/2}$$

Proof:

Step 1: $e^v \geq 1 + v$ for every real v . This is shown in the proof of theorem 3.3.3.

Step 2: a chord bound. Let $u \in [-1, 1]$, so that the weights $\frac{1+u}{2}$ and $\frac{1-u}{2}$ are nonnegative with sum 1. Step 1 with $v = \lambda(1-u)$ and with $v = -\lambda(1+u)$ gives

$$\frac{1+u}{2}e^{\lambda(1-u)} + \frac{1-u}{2}e^{-\lambda(1+u)} \geq \frac{1+u}{2}(1 + \lambda(1-u)) + \frac{1-u}{2}(1 - \lambda(1+u)) = 1$$

and multiplying by $e^{\lambda u} > 0$,

$$e^{\lambda u} \leq \frac{1+u}{2}e^{\lambda} + \frac{1-u}{2}e^{-\lambda}$$

Step 3: conclusion. Since $|X| \leq 1$, Step 2 holds with $u = X(\omega)$ at every point ω , and $0 < e^{\lambda X} \leq e^{\lambda}$, so $\mathbb{E}e^{\lambda X}$ is finite. Multiplying by $\mathbb{P}(\omega)$ and summing over the points of the space, the two series compare term by term ([2, Limits Preserve Weak Inequalities]), and proposition 1.3.6 and $\mathbb{E}X = 0$ give

$$\mathbb{E}e^{\lambda X} \leq \frac{1 + \mathbb{E}X}{2}e^{\lambda} + \frac{1 - \mathbb{E}X}{2}e^{-\lambda} = \frac{e^{\lambda} + e^{-\lambda}}{2} = \cosh(\lambda)$$

and $\cosh(\lambda) \leq e^{\lambda^2/2}$ by the series comparison in the proof of theorem 3.1.3, as claimed.

Theorem 3.4.2: Azuma's Inequality

Let $X_1, \dots, X_n$ be independent random variables or random vectors on a discrete probability space, with X_i taking values in a countable set S_i , and let $f: S_1 \times \dots \times S_n \rightarrow \mathbb{R}$ satisfy the stability condition with constants $a_1, \dots, a_n$, not all 0. Write $f = f(X_1, \dots, X_n)$. Then for any $t \geq 0$:

$$\mathbb{P}(f - \mathbb{E}f \geq t) \leq \exp\left(\frac{-t^2}{2 \sum_{i=1}^n a_i^2}\right) \quad \text{and} \quad \mathbb{P}(f - \mathbb{E}f \leq -t) \leq \exp\left(\frac{-t^2}{2 \sum_{i=1}^n a_i^2}\right)$$

and therefore

$$\mathbb{P}(|f - \mathbb{E}f| \geq t) \leq 2 \exp\left(\frac{-t^2}{2 \sum_{i=1}^n a_i^2}\right)$$

Proof:

Write $p_i(x_i) = \mathbb{P}(X_i = x_i)$ for $x_i \in S_i$, and $S = S_1 \times \dots \times S_n$, a countable set. For a bounded $g: S \rightarrow \mathbb{R}$, grouping the points ω of the probability space by the value $x = (X_1(\omega), \dots, X_n(\omega))$ (lemma 1.3.3(2)) and using independence gives

$$\mathbb{E}g(X_1, \dots, X_n) = \sum_{x \in S} g(x) p_1(x_1) \cdots p_n(x_n)$$

So all expected values below are sums over S with these weights, and the weights over any set of coordinates have sum 1, by the product-of-sums identity in the proof of lemma 1.5.10.

Step 1: partial averages. For $0 \leq k \leq n$ let $F_k(x_1, \dots, x_k)$ be the average of f over the last $n - k$ coordinates with the first k held fixed:

$$F_k(x_1, \dots, x_k) = \sum_{(y_{k+1}, \dots, y_n)} f(x_1, \dots, x_k, y_{k+1}, \dots, y_n) p_{k+1}(y_{k+1}) \cdots p_n(y_n)$$

the sum running over $S_{k+1} \times \dots \times S_n$. It converges absolutely because f is bounded, and $F_n = f$ and $F_0 = \mathbb{E}f$. Grouping the sum that defines F_{k-1} by the value of y_k (lemma 1.3.3(2)) gives

$$F_{k-1}(x_1, \dots, x_{k-1}) = \sum_{y_k \in S_k} F_k(x_1, \dots, x_{k-1}, y_k) p_k(y_k)$$

Step 2: the differences. For $1 \leq k \leq n$ put $D_k(x_1, \dots, x_k) = F_k(x_1, \dots, x_k) - F_{k-1}(x_1, \dots, x_{k-1})$. The sum telescopes:

$$D_1 + D_2 + \dots + D_n = F_n - F_0 = f - \mathbb{E}f$$

This is called the *martingale-difference representation* of $f - \mathbb{E}f$. By Step 1 and $\sum_{y_k} p_k(y_k) = 1$, averaging D_k over its last coordinate gives 0:

$$\sum_{y_k \in S_k} D_k(x_1, \dots, x_{k-1}, y_k) p_k(y_k) = F_{k-1}(x_1, \dots, x_{k-1}) - F_{k-1}(x_1, \dots, x_{k-1}) = 0$$

Also $|D_k| \leq a_k$. By Step 1,

$$D_k(x_1, \dots, x_k) = \sum_{y_k \in S_k} (F_k(x_1, \dots, x_k) - F_k(x_1, \dots, x_{k-1}, y_k)) p_k(y_k)$$

and each difference in brackets is the average, with weights $p_{k+1}(y_{k+1}) \cdots p_n(y_n)$ of total 1, of the differences $f(x_1, \dots, x_k, y_{k+1}, \dots, y_n) - f(x_1, \dots, x_{k-1}, y_k, y_{k+1}, \dots, y_n)$. These change only the k -th coordinate, so the stability condition bounds each by a_k in absolute value, and the triangle inequality bounds each bracket, and then D_k , by a_k .

Step 3: the exponential moment. Fix $\lambda \geq 0$ and put $A = \sum_{i=1}^n a_i^2$. For $0 \leq k \leq n$ let $M_k(x_1, \dots, x_k) = \exp(\lambda \sum_{j=1}^k D_j(x_1, \dots, x_j))$, so $M_0 = 1$. We show by induction on k that

$$\sum_{(x_1, \dots, x_k)} M_k(x_1, \dots, x_k) p_1(x_1) \cdots p_k(x_k) \leq \exp\left(\frac{\lambda^2}{2} \sum_{j=1}^k a_j^2\right)$$

For $k = 0$ both sides are 1. For $k \geq 1$, group the sum of nonnegative terms by $(x_1, \dots, x_{k-1})$ (lemma 1.3.3(1)). Since $M_k = M_{k-1} e^{\lambda D_k}$, the inner sum over x_k is

$$M_{k-1}(x_1, \dots, x_{k-1}) \sum_{x_k \in S_k} e^{\lambda D_k(x_1, \dots, x_k)} p_k(x_k)$$

If $a_k = 0$ then $D_k = 0$ and this last sum is 1. If $a_k > 0$, then for fixed $x_1, \dots, x_{k-1}$ the function $x_k \mapsto D_k(x_1, \dots, x_k)/a_k$, a random variable on the discrete probability space (S_k, p_k) , has absolute value at most 1 and expected value 0 by Step 2, so lemma 3.4.1 with λa_k in place of λ gives

$$\sum_{x_k \in S_k} e^{\lambda D_k(x_1, \dots, x_k)} p_k(x_k) \leq e^{\lambda^2 a_k^2 / 2}$$

In both cases the sum is at most $e^{\lambda^2 a_k^2 / 2}$, and the induction hypothesis completes the step. For $k = n$, $M_n = e^{\lambda(f - \mathbb{E}f)}$ by Step 2, so $\mathbb{E}e^{\lambda(f - \mathbb{E}f)} \leq e^{\lambda^2 A / 2}$.

Step 4: the upper tail. By theorem 3.1.2 and Step 3, for every $\lambda \geq 0$

$$\mathbb{P}(f - \mathbb{E}f \geq t) \leq e^{-\lambda t} \mathbb{E}e^{\lambda(f - \mathbb{E}f)} \leq \exp\left(-\lambda t + \frac{\lambda^2}{2} A\right)$$

and $\lambda = t/A$ gives $\exp(-t^2/A + t^2/(2A)) = \exp(-t^2/(2A))$.

Step 5: the lower tail. The function $-f$ satisfies the stability condition with the same constants, and $\mathbb{E}(-f) = -\mathbb{E}f$, so Step 4 applied to $-f$ gives $\mathbb{P}(f - \mathbb{E}f \leq -t) = \mathbb{P}((-f) - \mathbb{E}(-f) \geq t) \leq \exp(-t^2/(2A))$. Since $\{|f - \mathbb{E}f| \geq t\}$ is the union of $\{f - \mathbb{E}f \geq t\}$ and $\{f - \mathbb{E}f \leq -t\}$, adding the two bounds gives the last inequality, as claimed.

3.5 Applications of Azuma's Inequality

3.5.1 Chromatic Number of Erdős Rényi graph

Let $G(n, p)$ be the Erdős Rényi graph on n vertices. Now, define $\chi(G(n, p))$ to be the smallest number of colours needed to colour the vertices so that no two adjacent vertices share the same colour. The function $\chi(G(n, p))$ is called the *chromatic number* of $G(n, p)$. To represent this problem, we must keep track of the edges. Let

$$e_{i,j} = I(\text{there is an edge between vertex } v_i \text{ and } v_j)$$

Then by definition, each $e_{i,j} \sim B(p)$ and are all independent of each other. Next, let $X_i = (e_{1,i}, e_{2,i}, \dots, e_{i-1,i})$ for $2 \leq i \leq n$ be random vectors consisting of all possible edges of the random graph. The vectors $X_2, \dots, X_n$ consist of disjoint sets of the independent $e_{i,j}$, so the product formula $\mathbb{P}(X_2 = x_2, \dots, X_n = x_n) = \prod_{i=2}^n \mathbb{P}(X_i = x_i)$ holds by Step 1 of the proof of lemma 1.5.10, and $X_2, \dots, X_n$ are independent random vectors. Therefore, taking as input all possible edges and giving a chromatic number is equivalent as taking as input the random variables $X_2, \dots, X_n$ and giving an output, that is for appropriate f :

$$\chi(G(n, p)) = f(X_2, \dots, X_n)$$

We want to apply Azuma, and so we must verify that f satisfies the stability condition. Let's say that we have changed some X_i to X'_i with different values (i.e. we modify some edges between v_i all vertices with a lower index). Since we can always assign new colours, the chromatic number cannot increase by more than 1, and so

$$f(X_2, \dots, X_i, \dots, X_n) - f(X_2, \dots, X'_i, \dots, X_n) \leq 1$$

and by the same reasoning:

$$f(X_2, \dots, X'_i, \dots, X_n) - f(X_2, \dots, X_i, \dots, X_n) \leq 1$$

and so

$$|f(X_2, \dots, X'_i, \dots, X_n) - f(X_2, \dots, X_i, \dots, X_n)| \leq 1$$

This is the stability condition with $a_i = 1$ for each of the $n - 1$ coordinates, so theorem 3.4.2 gives

$$\mathbb{P}(|\chi(G(n, p)) - \mathbb{E}\chi(G(n, p))| \geq t) \leq 2e^{\frac{-t^2}{2(n-1)}}$$

Taking $t = \sqrt{2n \log n}$ and passing to the complement gives

$$\mathbb{P}(|\chi(G(n, p)) - \mathbb{E}\chi(G(n, p))| \leq \sqrt{2n \log n}) \geq 1 - \frac{2}{n}$$

For large n , with high probability the chromatic number is close to its expected value, and Azuma's inequality gives this without knowing what the expected value is. Bollobás [1] determined its size:

$$\mathbb{E}\chi(G(n, p)) \sim \frac{n}{2 \log n} \log \frac{1}{1-p}$$

(Bollobás proves the asymptotic for $\chi(G(n, p))$ with probability tending to 1, and the concentration above transfers it to the expected value). The deviation $\sqrt{2n \log n}$ is therefore of smaller order than the expected value, so the chromatic number concentrates around its mean as n grows, a law of large numbers for a random variable that is far from a sum of independent terms.

3.5.2 Balls and boxes

Let's say you have m boxes and n balls. You throw the balls at random, with each box having equal chance of having a ball land in it (i.e. $1/m$). Each ball throw is independent of each other. Let N be the random variable representing the number of non-empty boxes. To use Azuma's, we want to characterize N in terms of a function of random variables. To that end, let X_i ($1 \leq i \leq n$) represent the number of the box for which the i th ball lands (i.e. $X_i \in \{1, \dots, m\}$). Then

$$N = |\{X_1, \dots, X_n\}| = \#\{X_1, \dots, X_n\}$$

is the number of distinct boxes hit, meaning we have represented N as a function of $X_1, \dots, X_n$. If we replace X_i with X'_i , the size of the set changes by at most 1, so the stability condition holds with $a_i = 1$, and theorem 3.4.2 gives

$$\mathbb{P}(|N - \mathbb{E}N| \geq t) \leq 2e^{-\frac{t^2}{2n}}$$

Taking $t = \sqrt{2n \log n}$ and passing to the complement gives

$$\mathbb{P}(|N - \mathbb{E}N| \leq \sqrt{2n \log n}) \geq 1 - \frac{2}{n}$$

The computation of $\mathbb{E}N$ in section 1.7.1, with m boxes in place of the n possible values, gives

$$\mathbb{E}N = m \left(1 - \left(1 - \frac{1}{m} \right)^n \right)$$

If $m = \alpha n$ for a fixed $\alpha > 0$, then $(1 - \frac{1}{\alpha n})^n \rightarrow e^{-1/\alpha}$ by (1.3), so

$$\mathbb{E}N \sim n\alpha(1 - e^{-1/\alpha}) \quad (3.1)$$

The deviation $\sqrt{2n \log n}$ is of smaller order than n , and $1 - 2/n \rightarrow 1$, so with probability close to 1 the number of non-empty boxes differs from its expected value by a vanishing fraction of that value.

3.5.3 Max-cut Sparse Graph

This example will be like the last one except we will have a more complicated function and re-interpret the boxes and balls in terms of a random graph.

Let G be a random graph with vertices $V = \{v_1, \dots, v_n\}$ and edges E . We want to find a way to cut the graph in two in such a way that a maximal number of edges connect the two components: this is called the *max-cut problem*. This comes in often in computer science when trying to figure out the layout of electronic circuitry, or for a way to reformulate certain combinatorial problems. For example, consider the set of vertices to be people and edges represent if the two people dislike each other. Then we would want to as much as possible separate the group of people so as many “enemies” are on opposite sides.

To find a random variable that captures this idea, we first construct the random graph in a somewhat different way than we have done before. Start off by choosing some natural number $d > 0$ (the variable d is supposed to hint that it will represent in some way the “degree” of the vertices), and let $m = dn$ (where as usual we will think of n as being very large). The number m represents the number of edges we want to put in our graph. We now distribute them randomly and independently into the $\binom{n}{2}$ possible places they can go into our random graph (if an edge goes to the same place twice, we only “keep” one of them). Notice how this is similar to the balls and boxes scenario from last section: a possible location for an edge is a box, and the edge itself is the ball. So we can define the same random variables $X_1, \dots, X_m$ from the last section that can take on values $1 \leq X_i \leq \binom{n}{2}$. Since $m < \binom{n}{2}$, this model produces a *sparse graph* since there are relatively few edges compared to all possible edges, and there is a fixed number of edges per vertices (namely d). Just like before, we define

$$e_{i,j} = I(\text{there is an edge between } v_i \text{ and } v_j)$$

since edges are not repeated between vertices, this random variable captures all edges between two vertices.

We will now define the function *Max-Cut* as follows: let each vertex be assigned a value from $\{-1, 1\}$ randomly. We consider all vertices with the same value to be part of the same group. Let

$$\sigma = (\sigma_1, \dots, \sigma_n) \in \{-1, 1\}^n$$

Then a σ represents a possible cut of the graph into 2 groups. Let $E(\sigma)$ represent the number of edges connecting the two groups:

$$E(\sigma) = \# \{i < j \mid e_{i,j} = 1, \sigma_i \neq \sigma_j\}$$

This random variable is a bit hard to work with, and so here is another way to represent the same value: notice that $\sigma_i \neq \sigma_j$ if and only if $\sigma_i \sigma_j = -1$ (otherwise $\sigma_i \sigma_j = 1$). Thus, we can write:

$$I(\sigma_i \neq \sigma_j) = \frac{1 - \sigma_i \sigma_j}{2}$$

and so:

$$E(\sigma) = \frac{1}{2} \sum_{i < j} e_{i,j} (1 - \sigma_i \sigma_j)$$

Finally, we define M to represent the maximal number of $E(\sigma)$:

$$M = \max_{\sigma} E(\sigma)$$

The function M satisfies the stability condition as a function of $X_1, \dots, X_m$. Changing one X_i to X'_i moves one edge to a different position, which removes at most one edge and adds at most one edge, so each $E(\sigma)$ changes by at most 1, and so does $M = \max_{\sigma} E(\sigma)$. With $a_i = 1$ for the $m = dn$ coordinates, theorem 3.4.2 gives

$$\mathbb{P}(|M - \mathbb{E}M| \geq t) \leq 2e^{-\frac{t^2}{2m}} = 2e^{-\frac{t^2}{2dn}}$$

and with $t = \sqrt{2dn \log(dn)}$, taking the complement,

$$\mathbb{P}(|M - \mathbb{E}M| \leq \sqrt{2dn \log(dn)}) \geq 1 - \frac{2}{dn}$$

Whether the typical max-cut is close to its expected value depends on whether the deviation $\sqrt{2dn \log(dn)}$ is of smaller order than $\mathbb{E}M$. It is, because every graph has a cut containing at least half of its edges. For a fixed graph, average $E(\sigma)$ over all 2^n vectors σ . For $i \neq j$, changing the sign of σ_i is a bijection of $\{-1, 1\}^n$ that changes the sign of $\sigma_i \sigma_j$, so $\sum_{\sigma} \sigma_i \sigma_j = 0$. Since the maximum is at least the average,

$$M = \max_{\sigma} E(\sigma) \geq \frac{1}{2^n} \sum_{\sigma} E(\sigma) = \frac{1}{2} \sum_{i < j} e_{i,j} \left(1 - \frac{1}{2^n} \sum_{\sigma} \sigma_i \sigma_j\right) = \frac{1}{2} \sum_{i < j} e_{i,j}$$

which is half the number of edges. Taking the expected value over the graph, on a finite space (proposition 1.2.8),

$$\mathbb{E}M \geq \frac{1}{2} \sum_{i < j} \mathbb{P}(\text{edge between } v_i, v_j) = \frac{1}{2} \binom{n}{2} \left(1 - \left(1 - \frac{1}{\binom{n}{2}}\right)^{dn}\right)$$

because the position $\{v_i, v_j\}$ stays empty exactly when each of the dn independent edges misses it, as in the computation of $\mathbb{E}N$ in section 1.7.1. To simplify this lower bound, apply $e^v \geq 1+v$ (Step 1 of the proof of lemma 3.4.1) with $v = -1/\binom{n}{2}$, raise to the power dn , and use $dn/\binom{n}{2} = 2d/(n-1) \geq 2d/n$:

$$\left(1 - \frac{1}{\binom{n}{2}}\right)^{dn} \leq e^{-dn/\binom{n}{2}} \leq e^{-\frac{2d}{n}}$$

So $1 - (1 - 1/\binom{n}{2})^{dn} \geq 1 - e^{-2d/n}$. For $x > 0$, [2, Lagrange Form of the Remainder], applied to e^{-x} at 0 with the Taylor polynomial of degree 2, gives a $\xi \in (0, x)$ with $e^{-x} = 1 - x + \frac{x^2}{2} - \frac{e^{-\xi}x^3}{6} \leq 1 - x + \frac{x^2}{2}$, so $1 - e^{-x} \geq x - \frac{x^2}{2}$. With $x = 2d/n$,

$$\mathbb{E}M \geq \frac{1}{2} \binom{n}{2} \left(\frac{2d}{n} - \frac{2d^2}{n^2} \right) = \frac{(n-1)d}{2} \left(1 - \frac{d}{n} \right)$$

which is of order n , while the deviation $\sqrt{2dn \log(dn)}$ is of smaller order. So with probability close to 1 the max-cut differs from its expected value by a vanishing fraction of that value.

3.5.4 Hamming Cube

The vertices of the 3-dimensional cube can be represented by the points of $\{0, 1\}^3$, and more generally $\{0, 1\}^n$ represents the vertices of the n -dimensional cube. For any $x, y \in \{0, 1\}^n$, the number of coordinates in which they differ is called the *Hamming distance*:

$$\rho(x, y) = \sum_{i=1}^n I(x_i \neq y_i)$$

It is the restriction of the taxi-cab metric of $\mathbb{R}^n$ to $\{0, 1\}^n$, and it counts the edges of the cube on a shortest path from x to y . Distances in the cube range up to n . Azuma's inequality shows that if $A \subseteq \{0, 1\}^n$ contains more than a fixed proportion ϵ of all points, then all but a proportion δ of the points lie within distance of order $\sqrt{n}$ of A .

The ball of radius t around A is the set of points at distance at most t from A :

$$B_t(A) = \{x \in \{0, 1\}^n \mid \rho(x, y) \leq t \text{ for some } y \in A\}$$

For any $\epsilon, \delta \in (0, 1)$, define

$$t_{\epsilon, \delta} = \sqrt{2 \log \frac{1}{\epsilon}} + \sqrt{2 \log \frac{1}{\delta}}$$

Lemma 3.5.1: Hamming Lemma

For any $\epsilon, \delta \in (0, 1)$, if $A \subseteq \{0, 1\}^n$ and $\#(A) > \epsilon 2^n$, then

$$\#B_{t_{\epsilon, \delta} \sqrt{n}}(A) \geq (1 - \delta) 2^n$$

That is, the $t_{\epsilon, \delta} \sqrt{n}$ -neighbourhood of A contains at least a proportion $1 - \delta$ of all points of $\{0, 1\}^n$.

Proof:

Let $X = (X_1, X_2, \dots, X_n)$, where $X_1, \dots, X_n$ are independent with $\mathbb{P}(X_i = 0) = \mathbb{P}(X_i = 1) = \frac{1}{2}$. Then each point of $\{0, 1\}^n$ has probability 2^{-n} , so $\mathbb{P}(X \in S) = \#(S)/2^n$ for every $S \subseteq \{0, 1\}^n$. The set A is nonempty because $\#(A) > \epsilon 2^n > 0$. Let $Z = g(X)$ be the distance from X to A , where $g(x) = \min_{y \in A} \rho(x, y)$.

Step 1: stability. Let $x, x' \in \{0, 1\}^n$ differ only in coordinate i . For every $y \in A$, $\rho(x, y) \leq \rho(x', y) + 1$, so $g(x) \leq \rho(x', y) + 1$, and taking y with $\rho(x', y) = g(x')$ gives $g(x) \leq g(x') + 1$. Exchanging x and x' gives $|g(x) - g(x')| \leq 1$. So g satisfies the stability condition with $a_i = 1$ for all i , and $\sum_i a_i^2 = n$. By theorem 3.4.2 with $t = s\sqrt{n}$, for every $s \geq 0$,

$$\mathbb{P}(Z - \mathbb{E}Z \leq -s\sqrt{n}) \leq e^{-s^2/2} \quad \text{and} \quad \mathbb{P}(Z - \mathbb{E}Z \geq s\sqrt{n}) \leq e^{-s^2/2}$$

Step 2: $\mathbb{E}Z \leq t_\epsilon \sqrt{n}$, where $t_\epsilon = \sqrt{2 \log(1/\epsilon)}$. Note $e^{-t_\epsilon^2/2} = \epsilon$. Suppose $\mathbb{E}Z > t_\epsilon \sqrt{n}$. On the event $\{X \in A\}$, $Z = 0$, so $Z - \mathbb{E}Z = -\mathbb{E}Z < -t_\epsilon \sqrt{n}$. Hence $\{X \in A\} \subseteq \{Z - \mathbb{E}Z \leq -t_\epsilon \sqrt{n}\}$, and Step 1 gives

$$\frac{\#(A)}{2^n} = \mathbb{P}(X \in A) \leq \mathbb{P}(Z - \mathbb{E}Z \leq -t_\epsilon \sqrt{n}) \leq \epsilon$$

contradicting $\#(A) > \epsilon 2^n$. So $\mathbb{E}Z \leq t_\epsilon \sqrt{n}$.

Step 3: conclusion. Put $t_\delta = \sqrt{2 \log(1/\delta)}$, so that $e^{-t_\delta^2/2} = \delta$ and $t_{\epsilon, \delta} = t_\epsilon + t_\delta$. By Step 2, $Z \geq t_{\epsilon, \delta} \sqrt{n}$ implies $Z - \mathbb{E}Z \geq t_\delta \sqrt{n}$, so by Step 1

$$\mathbb{P}(Z \geq t_{\epsilon, \delta} \sqrt{n}) \leq \mathbb{P}(Z - \mathbb{E}Z \geq t_\delta \sqrt{n}) \leq \delta$$

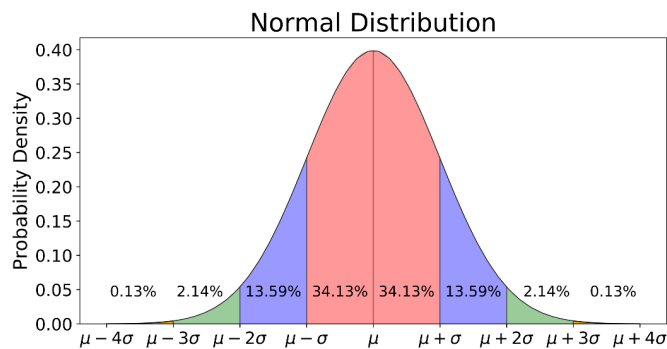
Taking the complement, $\mathbb{P}(Z < t_{\epsilon, \delta} \sqrt{n}) \geq 1 - \delta$. The event $\{Z \leq t_{\epsilon, \delta} \sqrt{n}\}$ contains it and is exactly $\{X \in B_{t_{\epsilon, \delta} \sqrt{n}}(A)\}$, so

$$\frac{\#B_{t_{\epsilon, \delta} \sqrt{n}}(A)}{2^n} = \mathbb{P}(Z \leq t_{\epsilon, \delta} \sqrt{n}) \geq 1 - \delta$$

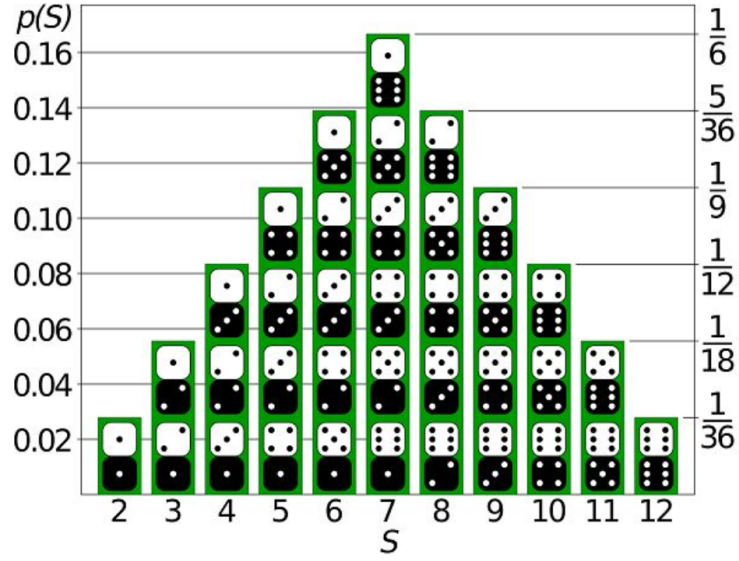
as claimed.

Gaussian Distribution

Many might have heard of the normal distribution: many metrics (such as I.Q., weight of people, height, SAT scores, blood pressure, error's in measurements produced by a machine) turn out to fall into a normal distribution, with the chances of being within 1 standard deviation away to be 68%, being 2 standard deviations away to be 95%, and 3 standard deviation to be 99%:



All of the given examples were required a large pool of data to show, but we can do it with a pair of dice: the values are normally distributed:



Since we can relate it to dice, we can relate it to a fair coin (take 12 i.i.d. $B(1/2)$ random variables). In fact, we almost encountered this distribution in example 3.1.4 (we will replace t with x because t will be used for a different purpose in a moment): recall that

$$\bar{X}_n = \frac{S_n}{n} = \frac{X_1, X_2, \dots, X_n}{n}$$

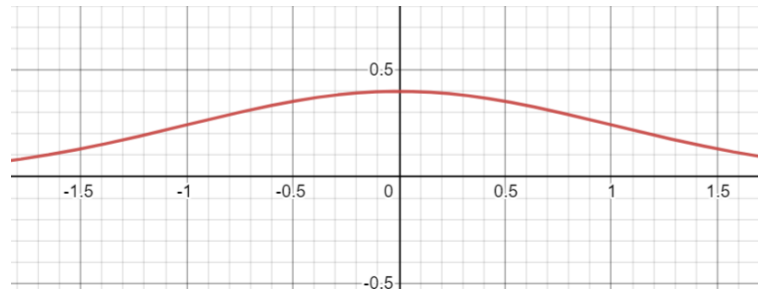
with $X_1, X_2, \dots, X_n$ being i.i.d. Bernoulli random variable $B(1/2)$. Given $x \geq 0$, we proved in example 3.1.4 that:

$$\mathbb{P}(|\bar{X}_n - 1/2| \geq x) \leq 2e^{-2nx^2}$$

doing a change of variables $x = \frac{t}{2\sqrt{n}}$ and defining $Z_n := 2\sqrt{n}(\bar{X}_n - 1/2)$, we get:

$$\mathbb{P}(|Z_n| \geq t) \leq 2e^{-\frac{t^2}{2}} \quad (4.1)$$

If we graph this function, it will be really close to the normal distribution we discussed earlier. In fact, if we replace the constant 2 with $\sqrt{1}\sqrt{2\pi}$, then the function $\frac{1}{\sqrt{2\pi}}e^{-\frac{t^2}{2}}$ will be:



Taking the integral of this function on $[-a, a]$ for $a \in \{1, 2, 3\}$, and letting $f(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{t^2}{2}}$, we get:

$$\int_{-1}^1 f(x) dx \approx 0.6826 \quad \int_{-2}^2 f(x) dx \approx 0.9545 \quad \int_{-3}^3 f(x) dx \approx 0.9973$$

showing that we got back the normal distribution! Even better, notice that

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

(we will prove this soon), and if $A, B \subseteq \mathbb{R}$ are disjoint, then:

$$\int_{A \sqcup B} f(x) dx = \int_A f(x) dx + \int_B f(x) dx$$

meaning the function $\mathbb{P}(E) = \int_E f(x) dx$ satisfies the definition of a probability! How then do we get $f(x)$, in particular the constant $\frac{1}{\sqrt{2\pi}}$ from equation 4.1? That will be the contents of the *Central Limit Theorem*, a key result of this section. What the C.L.T. will tell us is given the sum of independent random variable actually converge towards this distribution!

In particular, we will get that:

$$\lim_{t \rightarrow \infty} \mathbb{P}(Z_n \leq t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-\frac{x^2}{2}} dx$$

The name central limit theorem might not seem clear. Under a different perspective, it might be more clear: recall that $\mu = \mathbb{E}X_1 = \frac{1}{2}$, $\sigma^2 = \text{Var}(X_1) = \frac{1}{4}$, so we can re-write Z_n as :

$$Z_n = \frac{\sqrt{n}}{\sigma}(\bar{X}_n - \mu) = \frac{S_n - n\mu}{\sqrt{n\sigma^2}}$$

From this perspective, the aforementioned limit becomes:

$$\lim_{t \rightarrow \infty} \mathbb{P}\left(\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \leq t\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-\frac{x^2}{2}} dx$$

which looks a lot more like the law of large numbers modified in some way! This hopefully gives some motivation for the coming study of the Gaussian distribution

4.1 Gaussian Distribution on $\mathbb{R}$

We will first put a box around the Gaussian distribution for ease of reference:

Definition 4.1.1: Gaussian Distribution and Measure

Let $p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$. Then this function is called the *standard Gaussian distribution*. Define $\gamma : P(\mathbb{R}) \rightarrow \mathbb{R}$ to be:

$$\gamma(A) = \int_A p(x) dx$$

Then γ is called the *standard Gaussian measure on $\mathbb{R}$* . The continuous probability space with Gaussian probability measure is usually denoted $(\mathbb{R}, \gamma)$.

A quick word before we continue: the domain of γ was said to be $P(\mathbb{R})$, however this is not strictly speaking true. As would be covered in a course on measure theory, this set is too large (there exists sets that will contradict γ being a probability¹). Therefore, we would usually limit ourselves to “measurable set” and give a new symbol for the domain (for example $\mathcal{M}(\mathbb{R})$) to show the restriction from $P(\mathbb{R})$ to the set of measurable sets². In this book, we will not worry about that; instead we will only work with countable subsets (since the integral over a countable subset is always 0), and sets of the form:

$$(a, b), (a, b], [a, b), [a, b]$$

where $a, b \in \overline{\mathbb{R}}$ (i.e. $\mathbb{R} \cup \{\pm\infty\}$), and all finite unions of such intervals³. Notice that since the integral does not care if we drop a finite number of points, then it does not really matter if we deal with (a, b) , $(a, b]$, $[a, b)$, or $[a, b]$ (if $-\infty < a, b < \infty$, since we do not define the integral on infinity by considering infinity to be a point, but as the asymptotic behavior of $\lim_{b \rightarrow \infty} \int_a^b f(x) dx$)

Notice that we have just defined our first *continuous distribution*! Unlike discrete distributions, we define the probability of events over the events of the form described above, and not for each individual outcome. That means we will have to go over how to define change of variables, expected value, random variable, and so forth. We will do all of this in a moment, but let us first verify that γ does indeed satisfy the conditions of a probability. In particular, we should show that $\gamma(\mathbb{R}) = 1$ (since $\gamma(A \sqcup B) = \gamma(A) + \gamma(B)$ follows from the additivity of the integral over disjoint sets). By theorem 0.4.1, with $\varphi(x) = x/\sqrt{2}$ from $\mathbb{R}$ onto $\mathbb{R}$ and the function e^{-y^2} , which is integrable over $\mathbb{R}$ by proposition 0.4.3,

$$\sqrt{\pi} = \int_{\mathbb{R}} e^{-y^2} dy = \int_{\mathbb{R}} e^{-x^2/2} \frac{1}{\sqrt{2}} dx \quad \text{so} \quad \gamma(\mathbb{R}) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-x^2/2} dx = \frac{\sqrt{2}\sqrt{\pi}}{\sqrt{2\pi}} = 1$$

showing that γ is indeed a probability space! This means that we can use the $\mathbb{P}$ to represent it as well, however we will often stick to γ to make it more explicit that we are dealing with a continuous probability space. As an exercise, all the properties covered in section 1.2. One important comment to make is proving countable additivity of disjoint sets. This is not always possible with if we use the Riemann integral definition (prove that χ_q for $q \in \mathbb{Q}$ is Riemann integrable, but that $\chi_{\mathbb{Q}}$ is not Riemann integrable, in particular it is continuous nowhere).

Let us now ponder over how random variables would look like over this distribution. The simplest such random variable would be the identity function $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = x$. For any event $A \subseteq \mathbb{R}$ that:

$$\mathbb{P}(g \in A) = \gamma(\{x \mid g(x) \in A\}) = \gamma(A)$$

A little more generally, for any event A , we can define χ_A to be a random variable (i.e. the indicator function on the event, i.e. finite union and intersections of intervals, A). Another example would be $g(x) = -x$ (you should prove this as an exercise). However, to define random variable's more generally is not as easy as it was in the discrete case. In particular, if we let $X : \mathbb{R} \rightarrow \mathbb{R}$ a meaningful

¹Google Vitali sets to see the construction of such a set

²In fact, there is not real “natural” restriction to some subsets $\mathcal{M}(\mathbb{R})$, meaning we have to make some choice of what restriction we make. This leads to measure-theorists needing to keep track of what choice of $\mathcal{M}(\mathbb{R})$ was made, and so instead of writing $(\mathbb{R}, \gamma)$ we would write $(\mathbb{R}, \mathcal{M}, \gamma)$ to keep track of the choice

³If we wanted to use more machinery from measure theory, we would employ the Lebesgue integral and allow countable unions, however this is not necessary for this book

random variable, then for any event $A \subseteq \mathbb{R}$, we should be able to calculate $\mathbb{P}(X \in A)$, so we must do:

$$\begin{aligned}\mathbb{P}(X \in A) &= \gamma(\{x \mid X(x) \in A\}) \\ &= \gamma(X^{-1}(A)) \\ &= \int_{X^{-1}(A)} p(x) dx\end{aligned}$$

which means we must be able to integrate over $X^{-1}(A)$. However, just like not every set is “measurable”, not every function will produce sets $X^{-1}(A)$ that are “measurable” (in our case, that will be finite unions and intersection of intervals). Function’s that will always produce such sets are called “measurable functions”. The natural environment to define such a function is, again, measure theory, but we want to avoid the technicalities of measure theory to make the study of probability in the continuous case accessible, so instead we will limit ourselves to a smaller class of measurable functions that will suffice for this book. It must be noted that this is not too big of a restriction when it comes to the types of random variables that are used in probability, so if you are not aware of the technicalities of measurable functions you can just think of this next definition as trying to avoid the “pathologies” that would come up when dealing with measure theory that are not interesting from a probability point of view:

Definition 4.1.2: Nice Function

We will say a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *nice* if:

1. The function is piece-wise continuous
2. The function is bounded on compact sets
3. The pre-image of any interval A (i.e. $f^{-1}(A)$) is a Riemann-integrable set

In most cases, we will usually deal with functions where $f^{-1}(A)$ is the finite union of intervals.

Example 4.1.3: Nice Functions

1. Let:

$$X(x) = I(x \geq 0) - I(x < 0)$$

Then the image of X is $\{1, -1\}$. Notice that:

$$\mathbb{P}(X = 1) = \gamma(\{x \mid x \geq 0\}) = \int_0^\infty p(x) dx = 1/2$$

$$\mathbb{P}(X = -1) = \gamma(\{x \mid x < 0\}) = \int_{-\infty}^0 p(x) dx = 1/2$$

which shows that X is in fact a *Rademacher random variable* within a continuous probability space $(\mathbb{R}, \gamma)$.

2. Let

$$X(x) = |x|$$

The image of X is $[0, \infty)$, and so as a start let’s consider events of the form $[a, b]$:

$$\mathbb{P}(X \in [a, b]) = \gamma(\{x \mid X(x) \in [a, b]\}) = [-b, -a] \cup [a, b]$$

Since the Gaussian density $p(x)$ is symmetric, we get that:

$$\mathbb{P}(X \in [a, b]) = 2\gamma([a, b]) = 2 \int_a^b p(x)dx$$

given this new insight on how this random variable gets “converted” into some function of γ (namely 2γ) and that it’s never negative, define:

$$p_X(x) := 2p(x)I(x \geq 0)$$

Then for any interval $[a, b]$ we can write:

$$\mathbb{P}(a \leq x \leq b) = \int_a^b p_X(x)dx$$

and in general, if A is the finite disjoint union of intervals, then:

$$\mathbb{P}(X \in A) = \int_A p_X(x)dx$$

Notice that $\mathbb{P}$ in the previous example is now defined in terms of a new density (the function inside the integral)! Additivity clearly works, and for $\mathbb{P}(\mathbb{R}) = 1$, we still get:

$$\int_{\mathbb{R}} p_X(x)dx = 1$$

meaning $p_X(x)$ *also* defines a probability measure! In fact, we can construct many new probability measures through the use of such random variables:

Definition 4.1.4: Continuous Distributions From Densities

Let $(\mathbb{R}, \gamma)$ be the standard Gaussian probability space and let X be a random variable. Then if there exists a function p_X such that

$$\mathbb{P}(X \in A) = \int_A p_X(x)dx$$

for every interval A , then p_X is called the *density* of X . Taking $A = \mathbb{R}$ shows $\int_{\mathbb{R}} p_X(x)dx = 1$

Note that not all random variables and distributions can be expressed in this way: there does not always exist a function $p_X(x)$ such that $\mathbb{P}(X \in A) = \int_A p_X(x)dx$. In this class, all continuous random variables will have this property. For those taking measure theory, you will encounter some continuous random variables that do not admit distributions. More generally, distributions that admit a density are called *absolutely continuous*.

Example 4.1.5: Continuous Distributions From Densities

1. Define the density

$$p_X(x) = \lambda e^{-\lambda x} I(x \geq 0)$$

then the resulting probability is called the *exponential distribution with parameter λ*

2. Define the density

$$p_X(x) = \frac{1}{b-a} I(a \leq x \leq b)$$

then the resulting probability is called the *uniform distribution on the interval* $[a, b]$

An equivalent to the tail-sum formula for the continuous case is the following:

Definition 4.1.6: Cumulative Distribution Function (c.d.f.)

Let $(\mathbb{R}, \mathbb{P})$ be a probability space. Then $F(x) = \mathbb{P}(X \leq x)$ is called the *cumulative distribution function* (or c.d.f for short).

The nice thing about the c.d.f. is that it gives you an easy way to calculate the probability of an event over an interval, since:

$$\mathbb{P}(a \leq x \leq b) = \mathbb{P}(X \leq b) - \mathbb{P}(X \leq a) = F(b) - F(a)$$

We get some more properties if F has nicer properties. For examples, if F is continuous, then:

$$\mathbb{P}(X = a) \leq \mathbb{P}(a - \epsilon \leq x \leq a + \epsilon) = F(a + \epsilon) - F(a - \epsilon)$$

Letting $\epsilon \rightarrow 0$, we see that $\mathbb{P}(X = a) = 0$! It might seem like an obvious statement that this is the case since we are integrating over a single point, however this consideration becomes more important in more general environments, but it is interesting to see it here as a property of continuity of F .

Furthermore, if F is differentiable, then the derivative $F'(x) = p_X(x)$ is piece-wise continuous (by assumption of nice function), so by the Fundamental Theorem of Calculus:

$$\mathbb{P}(a \leq X \leq b) = F(b) - F(a) = \int_a^b p_X(x) dx$$

Hence, $p_X(x)$ is the density of the distribution of X . Using all this information, we can deduce what a density ought to be given a random variable:

Example 4.1.7: Deducing Density From X

1. Let $X = g^2$ where $g(x) = x$ is the standard Gaussian Random variable on $(\mathbb{R}, \gamma)$. Then for $x > 0$ (since $x < 0$ is not in the range, and we will cover the $x = 0$ case soon):

$$F(x) = \mathbb{P}(X \leq x) = \mathbb{P}(g^2 \leq x)$$

re-arranging so that we may apply the fundamental theorem of calculus and take advantage of the fact that g is the standard Gaussian:

$$\mathbb{P}(-\sqrt{x} \leq g \leq \sqrt{x}) = \int_{-\sqrt{x}}^{\sqrt{x}} p(t) dt$$

Taking the derivative with respect to x and taking advantage of the symmetry of the Gaussian, we get:

$$F'(x) = \frac{1}{2\sqrt{x}} p(\sqrt{x}) - p(-\sqrt{x}) \frac{-1}{2\sqrt{x}} = p(\sqrt{x}) \frac{1}{\sqrt{x}} = \frac{1}{\sqrt{2\pi x}} e^{-x/2}$$

Since X takes on non-negative values, we define the density to be:

$$p_X(x) = \begin{cases} 0 & x \leq 0 \\ \frac{1}{\sqrt{2\pi x}} e^{-x/2} & x > 0 \end{cases}$$

Thus, $p_X(x)$ is the density of distribution for X . This distribution is the Gamma distribution $\Gamma(\frac{1}{2}, \frac{1}{2})$ of section 4.4.

- Let $\mu \in \mathbb{R}$, $\sigma \in \mathbb{R}_{>0}$, and $X = \mu + \sigma g$ where $g(x) = x$ is the standard Gaussian distribution. Then, doing the same calculations as in the last example, we get:

$$p_{\mu, \sigma}(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

This distribution is usually denoted $N(\mu, \sigma)$ and is called the *Gaussian distribution with mean μ and variance σ^2* (the name will become clear shortly). For convenience, when $\mu = 0$, we usually denote $p_\sigma(x) := p_{0, \sigma}(x)$

The next concept to update is that of the expected value:

Definition 4.1.8: Expected Value Over Gaussian

Let $(\mathbb{R}, \gamma)$ be the probability space. The expected value of $X : \mathbb{R} \rightarrow \mathbb{R}$ on $(\mathbb{R}, \gamma)$ is:

$$\mathbb{E}X = \int_{\mathbb{R}} X(x)p(x)dx$$

Assuming $\mathbb{E}|X| < \infty$.

Example 4.1.9: Expected Values Over Gaussian

- Take $g(x) = x$ to be the random variable. Then:

$$\mathbb{E}g = \int_{\mathbb{R}} xp(x)dx = 0$$

Since $p(x)$ is symmetric, and combined with x makes $xp(x)$ an odd function.

- Let $g(x) = x^2$. Then:

$$\begin{aligned} \mathbb{E}g^2 &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} x^2 e^{-x^2/2} dx \\ &= -\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x d(e^{-x^2/2}) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/2} dx \\ &= 1 \end{aligned}$$

by integration by parts. Using this piece of information, we see that $\text{Var}(g) = 1$

3. We can combine these two facts to find the expected value and variance of $X = \mu + \sigma g \cong N(\mu, \sigma^2)$. In particular,

$$\mathbb{E}X = \mu \quad \text{Var}(X) = \mathbb{E}(\sigma^2 g^2) = \sigma^2$$

justifying the names of the parameters of $N(\mu, \sigma^2)$. Note that the usual Gaussian would thus be $N(0, 1)$ since by the result of the last section: $\text{Var}(g) = \mathbb{E}(g^2) - (\mathbb{E}g)^2 = 1 - 0 = 1$.

We next bring in the notion of change of variables with respect to expected value for continuous functions. As a reminder we saw when a distribution is continuous that:

$$\mathbb{P}(X \in A) = \int_{X^{-1}(A)} p(x) dx = \int_A p_X(t) dt$$

for some density $p_X(t)$.

Lemma 4.1.10: Change Of Variable For Integrals

Let X be a random variable on $(\mathbb{R}, \gamma)$, that is, a nice function (definition 4.1.2), such that $\mathbb{E}|X| < \infty$, and suppose that X has a density p_X (definition 4.1.4) that is nonnegative and Riemann integrable on every closed bounded interval. Then $\int_{\mathbb{R}} |t| p_X(t) dt < \infty$ and:

$$\mathbb{E}X = \int_{\mathbb{R}} X(x) p(x) dx = \int_{\mathbb{R}} t p_X(t) dt$$

Proof:

Let $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ be continuous with $\mathbb{E}|\varphi(X)| < \infty$. The claim, which for $\varphi(u) = u$ is the lemma, is that $\int_{\mathbb{R}} |\varphi(t)| p_X(t) dt < \infty$ and $\mathbb{E}\varphi(X) = \int_{\mathbb{R}} \varphi(t) p_X(t) dt$.

Here an integral over $\mathbb{R}$ is the sum of the improper integrals over $(-\infty, 0]$ and over $[0, \infty)$, in the sense of [2, Improper Integral], and [2, Convergence Tests for Improper Integrals] is used on $(-\infty, 0]$ in the reflected form from the proof of theorem 0.3.1. For a set $S \subseteq \mathbb{R}$ whose indicator I_S is Riemann integrable on every closed bounded interval, $\int_S h(x) dx$ means $\int_{\mathbb{R}} h(x) I_S(x) dx$. Every integral below of a function of x converges. On closed bounded intervals its integrand is Riemann integrable, by properties (1) to (3) of definition 4.1.2, [2, Finitely Many Discontinuities Are Harmless] and [2, Products, Absolute Values and Pointwise Extrema], because $\varphi \circ X$ is continuous wherever X is, and is bounded on bounded intervals, X being bounded there and φ being bounded on closed bounded intervals by [2, The Extreme Value Theorem]. The integrand is bounded in absolute value by a constant multiple of p or by $|\varphi(X)|p$, whose integrals over $\mathbb{R}$ converge, so the convergence tests apply. These integrals are additive over finite disjoint unions of such sets, and monotone in the integrand, by [2, Linearity of the Integral] and [2, Order Bounds and Additivity over a Subdivision Point] on bounded intervals followed by passage to the limit. By property (3) of definition 4.1.2, $X^{-1}(A)$ is such a set for every interval A , and definition 4.1.1 together with the density of X gives

$$\int_{X^{-1}(A)} p(x) dx = \gamma(X^{-1}(A)) = \mathbb{P}(X \in A) = \int_A p_X(t) dt$$

Step 1: truncation. Let ψ be φ or $|\varphi|$, a continuous function. Fix $M > 0$ and $\eta > 0$. By [2, Continuity on a Closed Bounded Interval Is Uniform] there is an integer $K \geq 1$ such that $|\psi(u) - \psi(v)| < \eta$ for $u, v \in [-M, M]$ with $|u - v| \leq 2M/K$. Put $h = 2M/K$ and $t_k = -M + kh$

for $0 \leq k \leq K$, and let $S_k = X^{-1}([t_{k-1}, t_k])$ and $q_k = \mathbb{P}(X \in [t_{k-1}, t_k])$. The sets S_k are disjoint, with union $S = X^{-1}([-M, M])$. For $x \in S_k$ we have $|\psi(X(x)) - \psi(t_{k-1})| < \eta$, and for $t \in [t_{k-1}, t_k]$ we have $|\psi(t) - \psi(t_{k-1})| < \eta$. Since p and p_X are nonnegative, monotonicity and the display above place both $\int_{S_k} \psi(X(x))p(x) dx$ and $\int_{t_{k-1}}^{t_k} \psi(t)p_X(t) dt$ in the interval $[(\psi(t_{k-1}) - \eta)q_k, (\psi(t_{k-1}) + \eta)q_k]$, so they differ by at most $2\eta q_k$. Adding over k , with additivity on both sides, and using $\sum_k q_k = \mathbb{P}(X \in [-M, M]) \leq 1$,

$$\left| \int_S \psi(X(x))p(x) dx - \int_{-M}^M \psi(t)p_X(t) dt \right| \leq 2\eta \sum_{k=1}^K q_k \leq 2\eta$$

Since η is arbitrary, for every $M > 0$

$$\int_{X^{-1}([-M, M])} \psi(X(x))p(x) dx = \int_{-M}^M \psi(t)p_X(t) dt$$

Step 2: $\int_{\mathbb{R}} |\varphi(t)|p_X(t) dt$ converges. Step 1 with $\psi = |\varphi|$, and monotonicity, since the integrand $|\varphi(X)|p$ is nonnegative, give

$$\int_{-M}^M |\varphi(t)|p_X(t) dt = \int_{X^{-1}([-M, M])} |\varphi(X(x))|p(x) dx \leq \int_{\mathbb{R}} |\varphi(X(x))|p(x) dx = \mathbb{E}|\varphi(X)|$$

for every $M > 0$. The integrand $|\varphi(t)|p_X(t)$ is nonnegative, so its integrals over $[0, M]$ and over $[-M, 0]$ are at most $\mathbb{E}|\varphi(X)|$, and part (1) of the convergence tests, on each half-line, shows that both halves of $\int_{\mathbb{R}} |\varphi(t)|p_X(t) dt$ converge. Part (3) then shows that $\int_{\mathbb{R}} \varphi(t)p_X(t) dt$ converges, and the tails $\int_{\{|t| \geq M\}} |\varphi(t)|p_X(t) dt$ tend to 0 as $M \rightarrow \infty$.

Step 3: the tails of $\mathbb{E}\varphi(X)$. By definition 4.1.8, applied to $\varphi \circ X$, $\mathbb{E}\varphi(X) = \int_{\mathbb{R}} \varphi(X(x))p(x) dx$, which converges because $\mathbb{E}|\varphi(X)| < \infty$. With $S = X^{-1}([-M, M])$ as in Step 1, the complement of S lies in $\{|X| \geq M\}$, so additivity and monotonicity give

$$\left| \mathbb{E}\varphi(X) - \int_S \varphi(X(x))p(x) dx \right| \leq \int_{\{|X| \geq M\}} |\varphi(X(x))|p(x) dx$$

Let $\epsilon > 0$, and choose $R > 0$ with $\int_{\{|x| > R\}} |\varphi(X(x))|p(x) dx < \epsilon$, which is possible because $\int_{\mathbb{R}} |\varphi(X(x))|p(x) dx$ converges. By property (2) of definition 4.1.2 there is a B with $|X| \leq B$ on $[-R, R]$. For $M > B$ the set $\{|X| \geq M\}$ does not meet $[-R, R]$, so the right-hand side above is at most $\int_{\{|x| > R\}} |\varphi(X(x))|p(x) dx < \epsilon$. Hence $\int_S \varphi(X(x))p(x) dx \rightarrow \mathbb{E}\varphi(X)$ as $M \rightarrow \infty$.

Step 4: conclusion. By Step 1 with $\psi = \varphi$, $\int_S \varphi(X(x))p(x) dx = \int_{-M}^M \varphi(t)p_X(t) dt$ for every $M > 0$. As $M \rightarrow \infty$ the left-hand side tends to $\mathbb{E}\varphi(X)$ by Step 3, and the right-hand side tends to $\int_{\mathbb{R}} \varphi(t)p_X(t) dt$ by Step 2, since the two differ by at most $\int_{\{|t| \geq M\}} |\varphi(t)|p_X(t) dt$. So $\mathbb{E}\varphi(X) = \int_{\mathbb{R}} \varphi(t)p_X(t) dt$, and the case $\varphi(u) = u$ is the lemma, as claimed.

Like in we saw in the discrete case in section 1.5, we can quite easily represent $\mathbb{E}f(x)$. The same holds true for the continuous case:

Lemma 4.1.11: Change Of Variables For Integrals II

Let X be a random variable on $(\mathbb{R}, \gamma)$, that is, a nice function, with a density p_X that is nonnegative and Riemann integrable on every closed bounded interval, and let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous with $\mathbb{E}|f(X)| < \infty$. Then $\int_{\mathbb{R}} |f(t)| p_X(t) dt < \infty$ and:

$$\mathbb{E}f(X) = \int_{\mathbb{R}} f(t) p_X(t) dt$$

Proof:

The proof of lemma 4.1.10 shows both conclusions for every continuous φ with $\mathbb{E}|\varphi(X)| < \infty$, without using the hypothesis $\mathbb{E}|X| < \infty$ of that lemma. With $\varphi = f$ this is the statement, as claimed.

A closing word before continuing to the n -dimensional case: we can also generalize Chebyshev's and Markov's inequality to the continuous case. If X is a random variable with associated density $p_X(x)$, then as an exercise show that:

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}X}{a}$$

the proof essentially mirrors the one for the discrete case.

4.2 Gaussian on $\mathbb{R}^n$

We will now define a probability on $\mathbb{R}^n$. First, recall (or define) the length of a vector $x \in \mathbb{R}^n$ by:

$$|x| = (x_1^2 + \cdots + x_n^2)^{1/2}$$

Since $|x| \in \mathbb{R}$, this will let us define the following

Definition 4.2.1: Gaussian Distribution On $\mathbb{R}^n$

Take $\mathbb{R}^n$ be the standard euclidean space. Then define $\mathbb{P}$ where:

$$\begin{aligned} \mathbb{P}(X \in A) &= \gamma_n(A) \\ &= \int_A p_n(\mathbf{x}) d\mathbf{x} \\ &= \int_A p_n(x_1, \dots, x_n) dx_1 \dots dx_n \\ &= \int_A \frac{1}{(2\pi)^{n/2}} e^{-|x|^2/2} dx_1 dx_2 \cdots dx_n \end{aligned}$$

where $p_n(x)$ is called the *standard Gaussian distribution on $\mathbb{R}^n$*

Just like in the 1-dimensional Gaussian case, most probability measures we will define will be with respect to the Gaussian, and will be defined with respect to a *density*. We will get back to this after

upgrading many previous concepts like independence, random vectors, expected value, and change of variables.

Remember that the Gaussian on $\mathbb{R}$ was $p(x) = \frac{1}{(2\pi)^{1/2}} e^{-x^2/2}$. By definition of $p_n(x)$, notice that:

$$p_n(x) = p(x_1)p(x_2) \cdots p(x_n)$$

So $p_n(x)$ is the product of 1-dimensional Gaussian! We have defined this as a probability measure, however, we should quickly verify that it does indeed satisfy that $\mathbb{P}(\mathbb{R}^n) = 1$, and indeed by Fubini's theorem ([4, Theorem 2.37], which this book uses without proof) and $\gamma(\mathbb{R}) = 1$:

$$\gamma_n(\mathbb{R}^n) = \int_{\mathbb{R}^n} p(x_1) \cdots p(x_n) dx_1 \cdots dx_n = \prod_{i=1}^n \int_{\mathbb{R}} p(x_i) dx_i = 1$$

Notice how this plays an analogous role to independence in the discrete case! In particular, If we let $\gamma_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $\gamma_i(\mathbf{x}) = x_i$ be the coordinate function (also known as projection map), then the collection $\{\gamma_i\}_{i=1}^n$ can be seen as *independent Gaussian random variables*: For any nice set $A_i \subseteq \mathbb{R}$, by Fubini's theorem ([4, Theorem 2.37]):

$$\begin{aligned} \mathbb{P}(g_1 \in A_1, \dots, g_n \in A_n) &= \gamma_n(A_1 \times A_2 \times \cdots \times A_n) \\ &= \int_{A_1 \times A_2 \times \cdots \times A_n} p_n(\mathbf{x}) d\mathbf{x} \\ &= \prod_{i=1}^n \int_{A_i} p(x) dx \\ &= \prod_{i=1}^n \gamma(A_i) \end{aligned}$$

We want to conclude that $\prod_{i=1}^n \gamma(A_i) = \prod_{i=1}^n \mathbb{P}(\gamma_i \in A_i)$ however in order to do that we must make sure that $\mathbb{P}(\gamma_i \in A_i) = \gamma(A_i)$. This comes from by fixing A_i in the first equality in the previous string of equalities and letting all other $A_j = \mathbb{R}$, $i \neq j$. Then:

$$\mathbb{P}(g_i \in A_i) = \gamma(A_i)$$

Thus, we can conclude that

$$\mathbb{P}(g_1 \in A_1, \dots, g_n \in A_n) = \prod_{i=1}^n \mathbb{P}(\gamma_i \in A_i)$$

And so we would say that $g_1, \dots, g_n$ are independent random variables with Gaussian distribution $N(0, 1)$.

Just like in the discrete case, we can define random vectors in the continuous case. As a simple first example, let $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$, with every component function $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ being $g_i(\mathbf{x}) = x_i$. Then any nice set $A \subseteq \mathbb{R}^n$ we can write:

$$\mathbb{P}(g \in A) = \gamma(A) = \int_A p_n(\mathbf{x}) d\mathbf{x}$$

We would say that the random vector g has density $p_n(x)$. Since the density is $p_n(x)$, we call such random vectors *standard Gaussian random vectors*.

Given any function $X : \mathbb{R}^n \rightarrow \mathbb{R}$, we will call it a random variable if it is a nice function (or if you know Lebesgue integration, it is an integrable function with respect to the Lebesgue measure). Thus, we may assume that $X^{-1}(A)$ will be a nice (or rectifiable) set for any interval A . Given a definition of a random variable, we can define the expected value:

Definition 4.2.2: Expected Value For $\mathbb{R}^n$ Gaussian

Let $(\mathbb{R}^n, \gamma_n)$ be the probability space and X a random variable. Then it's *expected value* is:

$$\mathbb{E}X = \int_{\mathbb{R}^n} X(\mathbf{x}) p_n(\mathbf{x}) d\mathbf{x}$$

and is considered well-defined if $\mathbb{E}|X| < \infty$

Like in the 1-dimensional case, we would like it if X were defined as a distribution with respect to a density $p_X(t)$ ($p_X(t) \geq 0$, $\int_{\mathbb{R}} p_X(t) dt = 1$), so that we can compute (for some interval A):

$$\mathbb{P}(X \in A) = \int_{X^{-1}(A)} p(\mathbf{x}) d\mathbf{x} = \int_A p_X(t) dt$$

This is certainly possible: as before, we can define $F(x) = \mathbb{P}(X \leq x)$ (since the range of X is $\mathbb{R}$, $\leq$ is well-defined), and find $p_X(t)$. We need to make sure that the change of variables will indeed return the same result, however this is a simple upgrade of the similar lemma from before:

Lemma 4.2.3: Change Of Variables For $\mathbb{R}^n$ Integral

Let X be a random variable on $(\mathbb{R}^n, \gamma_n)$ with a density p_X that is nonnegative and Riemann integrable on every closed bounded interval, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous with $\mathbb{E}|f(X)| < \infty$. Then

$$\mathbb{E}f(X) = \int_{\mathbb{R}} f(t) p_X(t) dt$$

Proof Key ideas:

The proof is not given here, because it needs a theory of integration on $\mathbb{R}^n$ that this book uses without developing. The argument of lemma 4.1.10 applies once integrals over subsets of $\mathbb{R}^n$ are additive over finite disjoint unions of the sets $X^{-1}(A)$, monotone in the integrand, and convergent as improper integrals over $\mathbb{R}^n$, since its Steps 1 to 4 use nothing else besides the density identity and the boundedness of X on bounded sets. In the language of measure theory, [4, Proposition 10.1] gives $\mathbb{E}f(X) = \int_{\mathbb{R}} f dP_X$ for the distribution P_X of X , and [4, Proposition 3.9], part (a), gives $\int_{\mathbb{R}} f dP_X = \int_{\mathbb{R}} f(t) p_X(t) dt$ when P_X has density p_X with respect to Lebesgue measure. The density identity of this book holds for intervals, and agreement on intervals determines P_X on all Borel sets, since by [4, Theorem 1.16] a Borel measure on $\mathbb{R}$ that is finite on bounded Borel sets is determined by its values on the intervals $(a, b]$.

A quick side note: we have seen the change of variables in many settings at this point: discrete, $\mathbb{R}$ and $\mathbb{R}^n$. The proof seemed to work in all settings regardless of the probability space. There is in fact a generalization of this proof that would be covered in a course on measure theory.

Before continuing to explore other distributions, we introduce an important result for computing them:

Theorem 4.2.4: Gaussian Stability

Let $a = (a_1, \dots, a_n) \in \mathbb{R}^n$. Define the random variable

$$X = (a, g) = a_1 g_1 + \dots + a_n g_n$$

Then $X \sim N(0, |a|^2)$ with variance $|a|^2$.

Remember from example 4.1.9 that $a_i g_i \sim N(0, a_i^2)$, and so we are saying that the sum of independent Gaussian random variable is another Gaussian random variable, justifying the name “Gaussian stability”

Proof:

If $a = 0$, then $X \sim N(0, 0) = 0$ (the zero function), and so this case is trivial. For $a \neq 0$, We compute the C.D.F.:

$$\mathbb{P}(X \leq t) = \mathbb{P}((a, g) \leq t) = \int_H p_n(x) dx$$

where $t \in \mathbb{R}$ and $H = \{x \mid \sum_i a_i g_i(x) \leq t\}$ (i.e., we translated the restriction of $\mathbb{P}$ into a bound of $\int$). The set H is messy to understand, so we will simplify it using a clever change of variable. Let $q_1 = \frac{a}{|a|}$ be the normalized vector a , and let $q_2, \dots, q_n$ be arbitrary vector that together with q_1 form an orthonormal basis for $\mathbb{R}^n$. Let Q be the matrix with rows $q_1, \dots, q_n$. Since Q is orthogonal, $QQ^T = I$, so $\det(Q)^2 = \det(Q) \det(Q^T) = \det(QQ^T) = 1$ and $|\det(Q)| = 1$, and $|Qx| = |x|$. This makes Q a perfect candidate for a change of variables: by theorem 0.4.5, applied to the linear bijection $x \mapsto Qx$ of $\mathbb{R}^n$, whose Jacobian matrix is Q , an integral over H in x becomes an integral over QH in $y = Qx$, with the factor $|\det(Q)| = 1$. Letting $y = Qx$, we get that the bound is now:

$$y_1 = \sum_i \frac{a_i}{|a|} x_i = \frac{1}{|a|} \sum a_i x_i$$

and the density becomes:

$$p_n(Qx) = \frac{1}{(2\pi)^{n/2}} e^{-\frac{|Qx|^2}{2}} = \frac{1}{(2\pi)^{n/2}} e^{-\frac{|y|^2}{2}} = p_n(y)$$

Finally, putting this all together, we get:

$$\begin{aligned} \int_H p_n(x) dx &= \int_{QH} p_n(y) dy \\ &= \int_{|a|y_1 \leq t} p(y_1) \cdots p(y_n) dy_1 \cdots dy_n \\ &= \int_{|a|y_1 \leq t} p(y_2) \cdots p(y_n) dy_2 \cdots dy_n p(y_1) dy_1 \quad \text{Fubini, [4, Theorem 2.37]} \\ &\stackrel{!}{=} \int_{|a|y_1 \leq t} p(y_1) dy_1 \\ &= \mathbb{P}(|a|g_1 \leq t) \end{aligned}$$

where the $\stackrel{!}{=}$ inequality comes from the constraint $|a|y_1 \leq t$ only applying to the first variable, and so all the other iterated integrals range freely over $\mathbb{R}$, and so their integral is 1. Thus, we

have reduced the C.D.F. to

$$\mathbb{P}(X \leq t) = \mathbb{P}(|a|g_1 \leq t)$$

the last C.D.F. defines the distribution with density $|a|g_1$, which in example 4.1.9 we showed was $N(0, |a|^2)$, which is what we sought to show.

Example 4.2.5: Stability And Change Of Variables

Combining the Gaussian stability with the change of variables for $\mathbb{R}^n$ (lemma 4.2.3), for a continuous f with $\mathbb{E}|f(a, g)| < \infty$ and $a \neq 0$, we get that:

$$\begin{aligned} \mathbb{E}(f(a, g)) &= \mathbb{E}(f(a_1g_1 + \cdots + a_ng_n)) \\ &= \int_{\mathbb{R}} f(t)p_{|a|}(t)dt \\ &= \frac{1}{\sqrt{2\pi}|a|} \int_{\mathbb{R}} f(t)e^{\frac{-t^2}{2|a|^2}} dt \end{aligned}$$

This formula will come back in the proof of the C.L.T. since $f^{-1}(A)$ will be an interval and, if we set $X = (a, g)$, then $(f \circ X)^{-1}(A)$ will be a region bounded by parallel hyperplanes.

Finally, note that $\mathbb{E}f(X)$ does *not* depend on a chosen distribution for X or probability space on which X is defined. Because of this, some authors write:

$$\mathbb{E}(f(a, g)) = \mathbb{E}f(|a|g_1)$$

since $|a|g_1$ also has distribution $N(0, |a|^2)$.

4.2.1 General Gaussian distribution on $\mathbb{R}^n$

We will now consider more general Gaussian random variables, in particular Gaussian random vectors (or Gaussian vectors for short). Let A be an n by n matrix and $g = (g_1, \dots, g_n)^T$ be a Gaussian vector. Let:

$$X = (X_1, \dots, X_n)^T = Ag = A(g_1, \dots, g_n)^T$$

We will show that the distribution of X will be dependent on:

$$C = AA^t$$

To get there, we start by defining some terms:

Definition 4.2.6: Covariance Matrix

Let X be a random vector. Then the *covariance matrix* is:

$$\text{Cov}(X) = [\text{Cov}(X_i, X_j)]_{i,j \leq n}$$

assuming all the entries are well-defined.

The covariance matrix is linked to the usual covariance of a random variable ($\text{Cov}(X) = \mathbb{E}[(X - \mathbb{E}(X)) \cdot (X - \mathbb{E}(X))^T]$) if we take $\boldsymbol{\mu} = \mathbb{E}(X) = (\mathbb{E}(X_1), \dots, \mathbb{E}(X_n))^T$. Then:

$$\text{Cov}(X) = \mathbb{E}[(X - \boldsymbol{\mu})(X - \boldsymbol{\mu})^T]$$

As we have seen in example 4.1.9, $\mathbb{E}X_{ij} = 0$, so $\mathbb{E}X_i = 0$ given $X = Ag$. And so, in this special case, we get that:

$$\text{Cov}(Ag) = \mathbb{E}Ag(Ag)^T = \mathbb{E}Ag g^T A = A(\mathbb{E}g g^T)A^T \stackrel{!}{=} AA^T = C$$

where $\mathbb{E}g g^T = [\mathbb{E}g_i g_j]_{i,j \leq n} = I$ using example 4.1.9. With this, we can explore the cases where $\det C \neq 0$ and $\det C = 0$

Lemma 4.2.7: Distribution Of $\det(C)$

Let $X = Ag$, $C = AA^T$, and $\det(C) \neq 0$. Then the distribution of X has density:

$$p_C(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} \sqrt{\det(C)}} e^{-\frac{1}{2} \mathbf{x}^T C^{-1} \mathbf{x}}$$

Meaning:

$$\mathbb{P}(X \in \Delta) = \int_{\Delta} p_C(\mathbf{x}) d\mathbf{x}$$

for nice set Δ

Proof:

Let Δ be any nice set of $\mathbb{R}^n$ (for ex. a rectangle). Since $X = Ag$, and $\det(C) \neq 0$, then A is invertible, and so we can simply write:

$$\mathbb{P}(Ag \in \Delta) = \mathbb{P}(g \in A^{-1}\Delta) = \frac{1}{(2\pi)^{n/2}} \int_{A^{-1}\Delta} e^{-\frac{|x|^2}{2}} d\mathbf{x}$$

Since A is invertible, we can do the change of variables $y = Ax$, so $x = A^{-1}y$: by theorem 0.4.5, applied on $\mathbb{R}^n$ to the linear bijection $y \mapsto A^{-1}y$, whose Jacobian matrix is A^{-1} with $|\det(A^{-1})| = 1/|\det(A)|$, and to the integrand times the indicator of $A^{-1}\Delta$, we get:

$$\mathbb{P}(Ag \in \Delta) = \int_{\Delta} \frac{1}{(2\pi)^{n/2}} e^{-\frac{|A^{-1}y|^2}{2}} \frac{1}{|\det(A)|} dy$$

at this point, we just manipulate notation. Notice that:

$$\det(C) = \det(AA^T) = \det(A) \det(A^T) = \det(A)^2 = |\det(A)|^2$$

so $\sqrt{\det(C)} = |\det(A)|$. Next

$$|A^{-1}y|^2 = (A^{-1}y)^T (A^{-1}y) = y^T (A^{-1})^T A^{-1} y = y^T (AA^T)^{-1} y = y^T C^{-1} y$$

and so:

$$\mathbb{P}(Ag \in \Delta) = \int_{\Delta} \frac{1}{(2\pi)^{n/2}} \frac{1}{\sqrt{\det(C)}} e^{-\frac{y^T C^{-1} y}{2}} dy = \int_{\Delta} p_C(\mathbf{y}) d\mathbf{y}$$

which is $N(0, C)$, completing the proof.

Example 4.2.8: Orthogonal Maps

If we take Q be orthogonal matrix (i.e. $QQ^T = I$) and take $X = Qg$, then lemma 4.2.7 gives us:

$$\mathbb{P}(X \in \Delta) = \int_{\Delta} p_I(y) dy = \int_{\Delta} \frac{1}{(2\pi)^{n/2}} e^{-\frac{|x|^2}{2}} dx$$

i.e. $p_I(x) = p_n(x)$.

If $\det(C) = 0$ or $\det(A) = 0$, then this would be our first example of a random variable with no density (in $\mathbb{R}^n$)! We can still describe this random variable in terms of C with a bit more work. We first review some linear algebra facts:

1. Show that $\text{Cov}(X)$ is symmetric and non-negative
2. If $C = AA^T$ and $C = QDQ^T$ is the eigen-decomposition of C given appropriate orthogonal matrices Q and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then $A = QD^{1/2}R$ for appropriate orthogonal matrix R and $D^{1/2} = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n})$ (hint: use that $B = Q^T A$ satisfies $BB^T = D$ to represent $B = D^{1/2}R$)

4.3 Central Limit Theorem

Before stating the theorem, we need to be able to define independent random variables with arbitrary distributions on the same probability space. Let $X_1, \dots, X_n$ be random variables on $(\Omega, \mathbb{P})$ with respective *independent* distribution $\mathbb{P}_1, \dots, \mathbb{P}_n$, meaning

$$\mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n) = \prod \mathbb{P}(X_i \in A_i) = \prod \mathbb{P}_i(A_i)$$

We have shown such random variables exist when we took the product space of random variables (recall definition 1.5.9). This definition worked well since all the variables were discrete, however we may now work with both continuous and discrete random variables. To update our idea of a product space to this more general scenario, let $X_1, \dots, X_n$ be random variables (discrete or continuous). If X_i is discrete, let Ω_i be the range of X_i . If X_i is continuous, let $\Omega_i = \mathbb{R}$. Let

$$\Omega = \Omega_1 \times \dots \times \Omega_n$$

with $\mathbb{P}$ defined as:

$$\mathbb{P}(A_1 \times \dots \times A_n) = \prod \mathbb{P}_i(A_i)$$

where A_i can be any subset for discrete random variable, and A_i is the union of intervals in the continuous case. As before, we can define coordinate functions $X_i : \Omega \rightarrow \mathbb{R}$ to be:

$$X_i(\omega) = X_i(\omega_1, \dots, \omega_n) = \omega_i$$

and we see that:

$$\mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n) = \mathbb{P}(\omega_1 \in A_1, \dots, \omega_n \in A_n) = \mathbb{P}(A_1 \times \dots \times A_n) = \prod_i \mathbb{P}_i(A_i)$$

showing us that the coordinate functions are still independent random variables with distributions $\mathbb{P}_1, \dots, \mathbb{P}_n$.

Actually, there is a slight problem in how we defined $\mathbb{P}$: we only showed the result for rectangles. To extend to general sets A , we would take the integral:

$$\mathbb{P}(A) = \int_A 1 d\mathbb{P}_1(\omega_1) \cdots d\mathbb{P}_n(\omega_n)$$

where we are automatically going to always apply Fubini's theorem. If we are integrating over a discrete space, $\int_{\Omega_i} \cdots d\mathbb{P}_i(\omega_i)$, then we would be taking the *sum* $\sum \cdots d\mathbb{P}_i(\omega_i)$. When X is continuous, we take the integral.

With this new knowledge we are ready to prove the Central Limit Theorem (commonly abbreviated to C.L.T.). This proof of the C.L.T. is called *Lindeberg's method*. It shows how the Gaussian distribution emerges in the limit through its stability property.

For the C.L.T., the random variables can either be i.i.d. or independent. If $X_1, X_2, \dots$ are independent with distributions $\mathbb{P}_1, \mathbb{P}_2, \dots$, then denote

$$\mathbb{E}X_i = \mu_i, \quad \text{Var}(X_i) = \sigma_i^2$$

with $0 < \sigma_i^2 < \infty$, and assume that

$$\frac{\mathbb{E}|X_i - \mu_i|^3}{\sigma_i^3} \leq K$$

for all i and some constant K . This is a uniform third-moment condition. Now define:

$$Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{X_i - \mu_i}{\sigma_i}$$

which is the “normalized sum”. If all the variables are i.i.d., $\mu = \mu_i$, $\sigma = \sigma_i$, then the sum becomes

$$Z_n = \frac{S_n - n\mu}{\sqrt{n}\sigma} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$$

and the third-moment condition reduces to

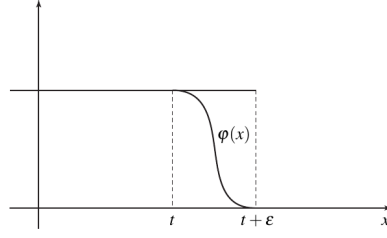
$$\mathbb{E}|X_1 - \mu|^3 < \infty$$

Lindeberg's method compares $\mathbb{E}\varphi(Z_n)$ with $\int_{\mathbb{R}} \varphi(x)p(x) dx$ for smooth functions φ , where $p(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ is the standard Gaussian density, by replacing the summands of Z_n with Gaussian variables one at a time.

Lemma 4.3.1: Smooth Approximation of an Indicator

For $u \in [0, 1]$ let $S(u) = 35u^4 - 84u^5 + 70u^6 - 20u^7$. For $t \in \mathbb{R}$ and $\epsilon > 0$ define $\varphi_{t,\epsilon}(x) = 1$ for $x \leq t$, $\varphi_{t,\epsilon}(x) = 1 - S((x-t)/\epsilon)$ for $t < x < t + \epsilon$, and $\varphi_{t,\epsilon}(x) = 0$ for $x \geq t + \epsilon$. Then $\varphi_{t,\epsilon}$ is three times differentiable on $\mathbb{R}$, its first three derivatives are continuous and bounded, $|\varphi_{t,\epsilon}'''| \leq 210/\epsilon^3$, and

$$I(x \leq t) \leq \varphi_{t,\epsilon}(x) \leq I(x \leq t + \epsilon) \quad \text{for every } x \in \mathbb{R}$$

**Proof:**

Differentiating the polynomial S gives

$$S'(u) = 140u^3(1-u)^3, \quad S''(u) = 420u^2(1-u)^2(1-2u)$$

and

$$S'''(u) = 840u(1-u)(1-5u(1-u))$$

so S' , S'' and S''' vanish at $u = 0$ and at $u = 1$, while $S(0) = 0$ and $S(1) = 35 - 84 + 70 - 20 = 1$. Since $S' \geq 0$ on $[0, 1]$, S is increasing there by [2, Monotonicity From the Sign of the Derivative], so $0 \leq S \leq 1$ on $[0, 1]$.

Put $P(x) = 1 - S((x-t)/\epsilon)$ for $x \in \mathbb{R}$, a polynomial. By the chain rule $P^{(j)}(x) = -\epsilon^{-j} S^{(j)}((x-t)/\epsilon)$, so $P(t) = 1$, $P(t+\epsilon) = 0$, and $P^{(j)}(t) = P^{(j)}(t+\epsilon) = 0$ for $j = 1, 2, 3$. Now let A and B be functions differentiable at a point c with $A(c) = B(c)$ and $A'(c) = B'(c)$, and let h agree with A to the left of c and with B to the right of c , with $h(c) = A(c)$. The one-sided difference quotients of h at c are those of A and of B , which tend to $A'(c) = B'(c)$, so h is differentiable at c with $h'(c) = A'(c)$ by [2, A Limit Exists Exactly When the One-Sided Limits Agree]. The function $\varphi_{t,\epsilon}$ is given by the pieces 1, P , 0 on $(-\infty, t]$, $[t, t+\epsilon]$, $[t+\epsilon, \infty)$, and the values and first derivatives of neighbouring pieces agree at t and at $t+\epsilon$. Hence $\varphi'_{t,\epsilon}$ exists and is given by the pieces 0, P' , 0. The values and first derivatives of these pieces again agree at t and at $t+\epsilon$, because P' and P'' vanish there, so $\varphi''_{t,\epsilon}$ exists and is given by the pieces 0, P'' , 0. Since P'' and P''' vanish at t and at $t+\epsilon$, $\varphi'''_{t,\epsilon}$ exists and is given by the pieces 0, P''' , 0, which agree at t and at $t+\epsilon$, so $\varphi'''_{t,\epsilon}$ is continuous. On $[0, 1]$, $u(1-u) \leq \frac{1}{4}$ gives $0 \leq 5u(1-u) \leq \frac{5}{4}$, so $|1-5u(1-u)| \leq 1$ and $|S'''(u)| \leq 840 \cdot \frac{1}{4} = 210$. Hence $|\varphi'''_{t,\epsilon}| \leq 210/\epsilon^3$, and $\varphi'_{t,\epsilon}$ and $\varphi''_{t,\epsilon}$ are bounded because they are continuous on $[t, t+\epsilon]$ and 0 elsewhere ([2, The Extreme Value Theorem]). Finally, for $x \leq t$ the three terms of the claimed inequality are 1, 1 and 1, for $t < x < t+\epsilon$ they are 0, a number in $[0, 1]$ and 1, and for $x \geq t+\epsilon$ they are 0, 0 and a number in $\{0, 1\}$, as claimed.

Lemma 4.3.2: Lindeberg Replacement

Let $W_1, \dots, W_n$ be independent random variables with $\mathbb{E}W_i = 0$, $\mathbb{E}W_i^2 = s_i^2$ and $\mathbb{E}|W_i|^3 < \infty$, where $s_1^2 + \dots + s_n^2 = 1$. Let $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ be three times differentiable, with φ , φ' , φ'' continuous and bounded and $|\varphi'''| \leq c$. Then

$$\left| \mathbb{E}\varphi(W_1 + \dots + W_n) - \int_{\mathbb{R}} \varphi(x)p(x) dx \right| \leq \frac{c}{6} \sum_{i=1}^n \left(\mathbb{E}|W_i|^3 + \frac{4}{\sqrt{2\pi}} s_i^3 \right)$$

Proof:

The value of $\mathbb{E}\varphi(W_1 + \cdots + W_n)$ depends only on the joint distribution of $(W_1, \dots, W_n)$ ([4, Proposition 10.1]), and by definition 1.2.11 this joint distribution is the product $\mu_{W_1} \times \cdots \times \mu_{W_n}$. So we compute on the product space described at the start of this section, built from the $2n$ distributions $\mu_{W_1}, \dots, \mu_{W_n}, \gamma, \dots, \gamma$, on which W_i is the i -th coordinate and the last n coordinates $g_1, \dots, g_n$ are independent standard Gaussian variables. Expected values on this product space are iterated sums and integrals over its factors. They are linear and monotone, $\mathbb{E}[F(U)H(V)] = \mathbb{E}F(U)\mathbb{E}H(V)$ when U and V are functions of disjoint sets of coordinates and $F(U)$ and $H(V)$ have finite expected values, and a function of $g_1, \dots, g_n$ alone has the same expected value as on $(\mathbb{R}^n, \gamma_n)$. All three facts follow from the Fubini-Tonelli theorem, [4, Theorem 2.37], which this book uses without proof.

Step 1: the Gaussian end. Put $G_i = s_i g_i$. By example 4.1.9, $\mathbb{E}G_i = 0$ and $\mathbb{E}G_i^2 = s_i^2$. Also $\mathbb{E}|G_i|^3 = s_i^3 \int_{\mathbb{R}} |x|^3 p(x) dx = \frac{4}{\sqrt{2\pi}} s_i^3$, because $-(x^2 + 2)e^{-x^2/2}$ is an antiderivative of $x^3 e^{-x^2/2}$, so that $\int_0^X x^3 e^{-x^2/2} dx = 2 - (X^2 + 2)e^{-X^2/2}$ by [2, The Fundamental Theorem of Calculus, Second Form], which tends to 2 as $X \rightarrow \infty$ by [2, The Exponential Outgrows Every Power], and the integrand is even. By theorem 4.2.4 with $a = (s_1, \dots, s_n)$, whose length is 1, the sum $G_1 + \cdots + G_n$ has distribution $N(0, 1)$, with density p , so lemma 4.2.3 gives $\mathbb{E}\varphi(G_1 + \cdots + G_n) = \int_{\mathbb{R}} \varphi(x)p(x) dx$.

Step 2: the telescoping sum. For $1 \leq i \leq n+1$ put $T_i = G_1 + \cdots + G_{i-1} + W_i + \cdots + W_n$, so that $T_1 = W_1 + \cdots + W_n$ and $T_{n+1} = G_1 + \cdots + G_n$, and for $1 \leq i \leq n$ put $U_i = G_1 + \cdots + G_{i-1} + W_{i+1} + \cdots + W_n$, so that $T_i = U_i + W_i$ and $T_{i+1} = U_i + G_i$. Then

$$\mathbb{E}\varphi(T_1) - \mathbb{E}\varphi(T_{n+1}) = \sum_{i=1}^n (\mathbb{E}\varphi(T_i) - \mathbb{E}\varphi(T_{i+1}))$$

Step 3: one replacement. For real u and $w \neq 0$, [2, Lagrange Form of the Remainder], applied to φ at u with the Taylor polynomial of degree 2, gives a ξ between u and $u + w$ with $\varphi(u + w) = \varphi(u) + \varphi'(u)w + \frac{1}{2}\varphi''(u)w^2 + \frac{1}{6}\varphi'''(\xi)w^3$. Hence, also for $w = 0$,

$$\left| \varphi(u + w) - \varphi(u) - \varphi'(u)w - \frac{1}{2}\varphi''(u)w^2 \right| \leq \frac{c}{6}|w|^3$$

With $u = U_i$ and $w = W_i$, and then with $w = G_i$, this gives $\varphi(T_i) = \varphi(U_i) + \varphi'(U_i)W_i + \frac{1}{2}\varphi''(U_i)W_i^2 + R_i$ with $|R_i| \leq \frac{c}{6}|W_i|^3$, and $\varphi(T_{i+1}) = \varphi(U_i) + \varphi'(U_i)G_i + \frac{1}{2}\varphi''(U_i)G_i^2 + R'_i$ with $|R'_i| \leq \frac{c}{6}|G_i|^3$. The variable U_i is a function of coordinates other than those of W_i and of G_i , and φ' and φ'' are bounded, so the product rule above gives

$$\mathbb{E}[\varphi'(U_i)W_i] = \mathbb{E}\varphi'(U_i)\mathbb{E}W_i = 0 = \mathbb{E}[\varphi'(U_i)G_i]$$

and

$$\mathbb{E}[\varphi''(U_i)W_i^2] = s_i^2 \mathbb{E}\varphi''(U_i) = \mathbb{E}[\varphi''(U_i)G_i^2]$$

Taking expected values in the two expansions and subtracting, all terms cancel except the remainders, so

$$|\mathbb{E}\varphi(T_i) - \mathbb{E}\varphi(T_{i+1})| = |\mathbb{E}R_i - \mathbb{E}R'_i| \leq \frac{c}{6} \left(\mathbb{E}|W_i|^3 + \frac{4}{\sqrt{2\pi}} s_i^3 \right)$$

Summing over i in Step 2 and using Step 1 gives the claim.

Lemma 4.3.3: From Smooth Test Functions to Distribution Functions

Let $V_1, V_2, \dots$ be random variables such that $\mathbb{E}\varphi_{t,\epsilon}(V_n) \rightarrow \int_{\mathbb{R}} \varphi_{t,\epsilon}(x)p(x) dx$ as $n \rightarrow \infty$ for every $t \in \mathbb{R}$ and every $\epsilon > 0$, with $\varphi_{t,\epsilon}$ as in lemma 4.3.1. Then for every $t \in \mathbb{R}$

$$\lim_{n \rightarrow \infty} \mathbb{P}(V_n \leq t) = \int_{-\infty}^t p(x) dx$$

Proof:

Write $\Phi(t) = \int_{-\infty}^t p(x) dx$. For $t < t'$, $\Phi(t') - \Phi(t) = \int_t^{t'} p(x) dx$, and $0 \leq p \leq 1/\sqrt{2\pi}$ gives $0 \leq \Phi(t') - \Phi(t) \leq (t' - t)/\sqrt{2\pi}$ by [2, Order Bounds and Additivity over a Subdivision Point]. Fix t and $\eta > 0$, and let $\epsilon > 0$ with $\epsilon/\sqrt{2\pi} < \eta/2$. By lemma 4.3.1 at t and at $t - \epsilon$,

$$I(x \leq t - \epsilon) \leq \varphi_{t-\epsilon,\epsilon}(x) \leq I(x \leq t) \leq \varphi_{t,\epsilon}(x) \leq I(x \leq t + \epsilon)$$

for every x . Expected values preserve these inequalities: on a discrete space the defining series compare term by term ([2, Limits Preserve Weak Inequalities]), for a variable with a density the defining integrals compare by [2, Order Bounds and Additivity over a Subdivision Point], followed by [2, Limits Preserve Weak Inequalities] for the improper integrals, and on the product spaces of this section this is part of the Fubini-Tonelli theorem quoted in the proof of lemma 4.3.2. So $\mathbb{E}\varphi_{t-\epsilon,\epsilon}(V_n) \leq \mathbb{P}(V_n \leq t) \leq \mathbb{E}\varphi_{t,\epsilon}(V_n)$. Since $p \geq 0$, monotonicity of the integral ([2, Order Bounds and Additivity over a Subdivision Point], and for the improper integrals of these nonnegative integrands [2, Limits at Infinity: Algebra and Monotone Convergence]) gives

$$\Phi(t) - \frac{\eta}{2} < \Phi(t - \epsilon) \leq \int_{\mathbb{R}} \varphi_{t-\epsilon,\epsilon}(x)p(x) dx \quad \text{and} \quad \int_{\mathbb{R}} \varphi_{t,\epsilon}(x)p(x) dx \leq \Phi(t + \epsilon) < \Phi(t) + \frac{\eta}{2}$$

By the hypothesis, for all large n both $\mathbb{E}\varphi_{t,\epsilon}(V_n)$ and $\mathbb{E}\varphi_{t-\epsilon,\epsilon}(V_n)$ lie within $\frac{\eta}{2}$ of their limits, so $\Phi(t) - \eta < \mathbb{P}(V_n \leq t) < \Phi(t) + \eta$, as claimed.

Theorem 4.3.4: Central Limit Theorem (C.L.T.)

Let $X_1, X_2, \dots$ be random variables, each discrete or with a density, such that $X_1, \dots, X_n$ are independent for every n . Suppose $\mathbb{E}X_i = \mu_i$ and $\text{Var}(X_i) = \sigma_i^2$ with $0 < \sigma_i^2 < \infty$, and $\mathbb{E}|X_i - \mu_i|^3 \leq K\sigma_i^3$ for all i and some constant K . Let Z_n be the normalized sum defined above. Then for all $t \in \mathbb{R}$:

$$\lim_{n \rightarrow \infty} \mathbb{P}(Z_n \leq t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-x^2/2} dx$$

This is called *convergence in distribution*, since the distribution functions of the Z_n converge to the distribution function of $N(0, 1)$.

Proof:

Put $Y_i = (X_i - \mu_i)/\sigma_i$, so that $Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i$. By linearity of the expected value, $\mathbb{E}Y_i = 0$, $\mathbb{E}Y_i^2 = \text{Var}(X_i)/\sigma_i^2 = 1$ and $\mathbb{E}|Y_i|^3 = \mathbb{E}|X_i - \mu_i|^3/\sigma_i^3 \leq K$. Fix n and put $W_i = Y_i/\sqrt{n}$ for

$1 \leq i \leq n$. Each W_i is a function of X_i alone, so $\mathbb{P}(W_1 \in A_1, \dots, W_n \in A_n) = \mathbb{P}(X_1 \in B_1, \dots, X_n \in B_n)$ with $B_i = \{x : (x - \mu_i)/(\sigma_i\sqrt{n}) \in A_i\}$, and this factors into $\prod_i \mathbb{P}(W_i \in A_i)$ by definition 1.2.11. So $W_1, \dots, W_n$ are independent, with $\mathbb{E}W_i = 0$, $s_i^2 = \mathbb{E}W_i^2 = 1/n$ and $\mathbb{E}|W_i|^3 \leq K/n^{3/2}$, and $s_1^2 + \dots + s_n^2 = 1$.

Let $t \in \mathbb{R}$ and $\epsilon > 0$. By lemma 4.3.1, $\varphi_{t,\epsilon}$ satisfies the hypotheses of lemma 4.3.2 with $c = 210/\epsilon^3$, and $W_1 + \dots + W_n = Z_n$, so

$$\left| \mathbb{E}\varphi_{t,\epsilon}(Z_n) - \int_{\mathbb{R}} \varphi_{t,\epsilon}(x)p(x) dx \right| \leq \frac{35}{\epsilon^3} \sum_{i=1}^n \left(\frac{K}{n^{3/2}} + \frac{4}{\sqrt{2\pi}} \cdot \frac{1}{n^{3/2}} \right) = \frac{35}{\epsilon^3} \left(K + \frac{4}{\sqrt{2\pi}} \right) \frac{1}{\sqrt{n}}$$

The right side tends to 0 as $n \rightarrow \infty$, so $\mathbb{E}\varphi_{t,\epsilon}(Z_n) \rightarrow \int_{\mathbb{R}} \varphi_{t,\epsilon}(x)p(x) dx$ for every $t \in \mathbb{R}$ and every $\epsilon > 0$. Lemma 4.3.3 with $V_n = Z_n$ now gives $\mathbb{P}(Z_n \leq t) \rightarrow \int_{-\infty}^t p(x) dx$ for every t , as claimed.

For i.i.d. variables, the conclusion of theorem 4.3.4 needs only $0 < \sigma^2 < \infty$ and no third moment. This is [3, Theorem 3.4.1], whose proof uses characteristic functions, a tool this book does not develop.

Example 4.3.5: C.L.T. Use Case

Suppose all we know about the checkout time at a grocery store is that its mean per customer is $\mu = 5$ minutes and its standard deviation is $\sigma = 2$ minutes. We want to estimate the probability that a cashier checks out at least 50 customers during a 4h shift, that is, that 50 customers are served in at most 240 minutes.

Let T_i be the time it takes to serve the i th customer, and $S_n = \sum_{i=1}^n T_i$. We want to estimate $\mathbb{P}(S_{50} \leq 240)$. Treat the T_i as i.i.d. with a finite third moment, and approximate the distribution function of Z_{50} by its limit in theorem 4.3.4. Since

$$\mathbb{P}(S_{50} \leq 240) = \mathbb{P}\left(Z_{50} \leq \frac{240 - 50 \cdot 5}{2\sqrt{50}}\right) \quad \text{and} \quad \frac{240 - 250}{2\sqrt{50}} \approx -0.71$$

this gives

$$\mathbb{P}(S_{50} \leq 240) \approx \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{-0.71} e^{-\frac{x^2}{2}} dx \approx 0.24$$

so there is approximately a 24% chance that 50 customers are served.

In a common variation of the C.L.T. the distributions also depend on n . For each n there are independent random variables $X_{n1}, \dots, X_{nn}$, each discrete or with a density, and the list may change with n . Such a family is called a *triangular array*. Write

$$\mu_{ni} = \mathbb{E}X_{ni}, \quad \sigma_{ni}^2 = \text{Var}(X_{ni}) < \infty, \quad D_n^2 = \sum_{i=1}^n \sigma_{ni}^2, \quad S_n = \sum_{i=1}^n X_{ni}$$

and assume $D_n > 0$ for all large n . In place of the earlier normalization, consider, for those n ,

$$Z_n = \frac{S_n - \mathbb{E}S_n}{\sqrt{\text{Var}(S_n)}} = \frac{1}{D_n} \sum_{i=1}^n (X_{ni} - \mu_{ni})$$

where $\text{Var}(S_n) = D_n^2$ because independent variables are uncorrelated (example 2.1.5) and the variance of a sum of uncorrelated variables is the sum of their variances. When $X_{ni} = X_i$ for an i.i.d. sequence

$X_1, X_2, \dots$, this is the earlier Z_n . Again $\mathbb{E}Z_n = 0$ and $\text{Var}(Z_n) = 1$, but the variance σ_{ni}^2/D_n^2 of the i -th summand need not be $1/n$. A sufficient condition for the conclusion of the C.L.T. is

$$\lim_{n \rightarrow \infty} \frac{1}{D_n^3} \sum_{i=1}^n \mathbb{E}|X_{ni} - \mu_{ni}|^3 = 0$$

which is called the *Lyapunov condition*.

Theorem 4.3.6: C.L.T. For Triangular Arrays

For each n let $X_{n1}, \dots, X_{nn}$ be independent random variables, each discrete or with a density, with $\mathbb{E}|X_{ni} - \mu_{ni}|^3 < \infty$, and with $D_n > 0$ for all large n . If the Lyapunov condition holds, then for all $t \in \mathbb{R}$

$$\lim_{n \rightarrow \infty} \mathbb{P}(Z_n \leq t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-x^2/2} dx$$

Proof:

Fix n with $D_n > 0$ and put $W_i = (X_{ni} - \mu_{ni})/D_n$ for $1 \leq i \leq n$. Each W_i is a function of X_{ni} alone, so $\mathbb{P}(W_1 \in A_1, \dots, W_n \in A_n)$ equals $\mathbb{P}(X_{n1} \in B_1, \dots, X_{nn} \in B_n)$ with $B_i = \{x : (x - \mu_{ni})/D_n \in A_i\}$, which factors into $\prod_i \mathbb{P}(W_i \in A_i)$ by definition 1.2.11. So $W_1, \dots, W_n$ are independent. By linearity of the expected value, $\mathbb{E}W_i = 0$ and $s_i^2 = \mathbb{E}W_i^2 = \sigma_{ni}^2/D_n^2$, so $s_1^2 + \dots + s_n^2 = 1$, and $\mathbb{E}|W_i|^3 = \mathbb{E}|X_{ni} - \mu_{ni}|^3/D_n^3 < \infty$. Put $L_n = \sum_{i=1}^n \mathbb{E}|W_i|^3$, which tends to 0 by the Lyapunov condition.

Step 1: $\sum_i s_i^3 \rightarrow 0$. Let $a > 0$. If $|W_i| \leq a$ then $W_i^2 \leq a^2$, and if $|W_i| > a$ then $W_i^2 = |W_i|^3/|W_i| < |W_i|^3/a$, so $W_i^2 \leq a^2 + |W_i|^3/a$ everywhere. Expected values preserve this inequality, by term-by-term comparison of the defining series for a discrete X_{ni} ([2, Limits Preserve Weak Inequalities]) and by [2, Order Bounds and Additivity over a Subdivision Point], followed by [2, Limits Preserve Weak Inequalities] for the improper integrals, for one with a density. With proposition 1.3.6 in the discrete case, and [2, Linearity of the Integral] followed by the limit of the improper integrals in the density case, $s_i^2 \leq a^2 + \mathbb{E}|W_i|^3/a \leq a^2 + L_n/a$ for every i . Since $L_n \rightarrow 0$, $\max_i s_i^2 < 2a^2$ for all large n , and $a > 0$ was arbitrary, so $\max_i s_i \rightarrow 0$. Then

$$\sum_{i=1}^n s_i^3 \leq \left(\max_{1 \leq i \leq n} s_i \right) \sum_{i=1}^n s_i^2 = \max_{1 \leq i \leq n} s_i \rightarrow 0$$

Step 2: conclusion. Let $t \in \mathbb{R}$ and $\epsilon > 0$. By lemma 4.3.1, $\varphi_{t,\epsilon}$ satisfies the hypotheses of lemma 4.3.2 with $c = 210/\epsilon^3$, and $W_1 + \dots + W_n = Z_n$, so

$$\left| \mathbb{E}\varphi_{t,\epsilon}(Z_n) - \int_{\mathbb{R}} \varphi_{t,\epsilon}(x)p(x) dx \right| \leq \frac{35}{\epsilon^3} \left(L_n + \frac{4}{\sqrt{2\pi}} \sum_{i=1}^n s_i^3 \right)$$

which tends to 0 by the Lyapunov condition and Step 1. Lemma 4.3.3 with $V_n = Z_n$ gives the claim.

The Lyapunov condition implies the *Lindeberg condition*

$$\lim_{n \rightarrow \infty} \frac{1}{D_n^2} \sum_{i=1}^n \mathbb{E}[(X_{ni} - \mu_{ni})^2 I(|X_{ni} - \mu_{ni}| > \epsilon D_n)] = 0 \quad \text{for every } \epsilon > 0$$

since $x^2 I(|x| > \epsilon D_n) \leq |x|^3 / (\epsilon D_n)$ for every real x . The Lindeberg condition needs no third moments, and the conclusion of theorem 4.3.6 holds under it. This is the Lindeberg-Feller theorem, [3, Theorem 3.4.10].

Example 4.3.7: Records In A Random Permutation

Let $\pi = (\pi_1, \dots, \pi_n)$ be a uniformly random permutation of $\{1, \dots, n\}$, so that each of the $n!$ permutations has probability $1/n!$. We say that π_k is a *record* if $\pi_j < \pi_k$ for every $j < k$. Let I_k be the indicator that π_k is a record, and let $R_n = I_1 + \dots + I_n$ be the number of records.

For $1 \leq k \leq n$ let $r_k = \#\{j \leq k : \pi_j \leq \pi_k\}$ be the rank of π_k among $\pi_1, \dots, \pi_k$. Then $r_k \in \{1, \dots, k\}$ and $I_k = I(r_k = k)$. The map $\pi \mapsto (r_1, \dots, r_n)$ from permutations to $\{1\} \times \{1, 2\} \times \dots \times \{1, \dots, n\}$ is injective, because for $k = n, n-1, \dots, 1$ in turn, π_k is the r_k -th smallest element of $\{1, \dots, n\} \setminus \{\pi_{k+1}, \dots, \pi_n\}$. Both sets have $n!$ elements, so the map is a bijection, and

$$\mathbb{P}(r_1 = c_1, \dots, r_n = c_n) = \frac{1}{n!} = \prod_{k=1}^n \frac{1}{k}$$

for every point $(c_1, \dots, c_n)$ of the product set. Summing over all coordinates but the k -th gives $\mathbb{P}(r_k = c_k) = 1/k$, so the joint probabilities factor, and $r_1, \dots, r_n$ are independent. By lemma 1.5.10, $I_1, \dots, I_n$ are independent, with $\mathbb{P}(I_k = 1) = 1/k$.

Since $I_k^2 = I_k$, $\text{Var}(I_k) = \frac{1}{k} - \frac{1}{k^2}$. So R_n has $\mathbb{E}R_n = \sum_{k=1}^n \frac{1}{k}$, and the triangular array $X_{nk} = I_k$, $1 \leq k \leq n$, has

$$D_n^2 = \sum_{k=1}^n \frac{1}{k} \left(1 - \frac{1}{k}\right) \geq \frac{1}{2} \sum_{k=2}^n \frac{1}{k}$$

because $1 - \frac{1}{k} \geq \frac{1}{2}$ for $k \geq 2$. The right side tends to ∞ by [2, The Harmonic Series], so $D_n \rightarrow \infty$. Since $|I_k - \frac{1}{k}| \leq 1$, $\mathbb{E}|I_k - \frac{1}{k}|^3 \leq \mathbb{E}(I_k - \frac{1}{k})^2 = \text{Var}(I_k)$, and hence

$$\frac{1}{D_n^3} \sum_{k=1}^n \mathbb{E} \left| I_k - \frac{1}{k} \right|^3 \leq \frac{D_n^2}{D_n^3} = \frac{1}{D_n} \rightarrow 0$$

This is the Lyapunov condition, so theorem 4.3.6 gives, for every $t \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{R_n - \sum_{k=1}^n \frac{1}{k}}{D_n} \leq t \right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-x^2/2} dx$$

4.4 Distributions Related to Gaussian

In this section, we study many distributions related to the Gaussian distribution:

1. Gamma distribution
2. chi-squared distribution
3. F -distribution
4. Student's t -distribution

These will be the focus of attention for simple linear regression. Linear regression was not covered in my year, but these still have interesting properties worth studying for their own right.

Definition 4.4.1: Gamma Function

For all $\alpha > 0$:

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$$

The integral converges for every $\alpha > 0$, and $\Gamma(\alpha) > 0$. Its integrand is continuous and positive on $(0, \infty)$. On $(0, 1]$ the integrand is at most $x^{\alpha-1}$, and $\int_\delta^1 x^{\alpha-1} dx = (1 - \delta^\alpha)/\alpha \leq 1/\alpha$ for $0 < \delta < 1$, by [2, The Fundamental Theorem of Calculus, Second Form] and [2, The Derivative of a Real Power]. On $[1, \infty)$, fix an integer $k \geq \alpha - 1$. Then $x^{\alpha-1} \leq x^k$ for $x \geq 1$, and $x^k e^{-x/2} = 2^k (x/2)^k e^{-x/2}$ tends to 0 as $x \rightarrow \infty$ by [2, The Exponential Outgrows Every Power]. So it is at most 1 for x beyond some M , and it is bounded on $[1, M]$ by [2, A Continuous Function on a Closed Bounded Interval Is Bounded]. With K a bound on $[1, \infty)$, the integrand $x^{\alpha-1} e^{-x} \leq x^k e^{-x/2} \cdot e^{-x/2}$ is at most $K e^{-x/2}$ there, and $\int_1^X e^{-x/2} dx \leq 2e^{-1/2}$. Parts (1), (2) and (4) of [2, Convergence Tests for Improper Integrals] give convergence at both ends, and $\Gamma(\alpha) \geq \int_1^2 x^{\alpha-1} e^{-x} dx > 0$: the integrand is nonnegative, and it is continuous and positive on $[1, 2]$, so by [2, The Extreme Value Theorem] it has a positive minimum m there, and part (2) of [2, Order Bounds and Additivity over a Subdivision Point] gives $\int_1^2 x^{\alpha-1} e^{-x} dx \geq m > 0$.

To make this a distribution, we can divide by $\Gamma(\alpha)$ on both sides. If we further do the change of variables of $x = \beta y$ for some $\beta > 0$, that is, apply theorem 0.4.1 with $\varphi(y) = \beta y$ from $(0, \infty)$ onto $(0, \infty)$, we get:

$$1 = \int_0^\infty \frac{1}{\Gamma(\alpha)} x^{\alpha-1} e^{-x} dx = \int_0^\infty \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y} dy$$

Labeling the integrand on the right as $f_{\alpha, \beta}$, we give it a name:

Definition 4.4.2: Gamma Distribution

Let $f_{\alpha, \beta}$ be defined as above. Then it defines the *Gamma distribution with parameters α and β* , usually denoted $\Gamma(\alpha, \beta)$ and it is the density of this distribution.

For $0 < \delta < X$, [2, Integration by Parts] with $F(x) = x^\alpha$ and $G(x) = -e^{-x}$ gives

$$\int_\delta^X x^\alpha e^{-x} dx = \delta^\alpha e^{-\delta} - X^\alpha e^{-X} + \alpha \int_\delta^X x^{\alpha-1} e^{-x} dx$$

As $\delta \rightarrow 0^+$ and $X \rightarrow \infty$, the boundary terms tend to 0, the second by [2, The Exponential Outgrows Every Power], since $X^\alpha \leq X^{k+1}$ for $X \geq 1$. So $\Gamma(\alpha + 1) = \alpha \Gamma(\alpha)$. Since $\Gamma(1) = \int_0^\infty e^{-x} dx = 1$, induction gives, for any $n \in \mathbb{N}_{>0}$,

$$\Gamma(n) = (n-1)!$$

Example 4.4.3: Moments Of Gamma Distribution

Show that if $X \sim \Gamma(\alpha, \beta)$,

$$\mathbb{E}X^k = \frac{(\alpha + k - 1) \cdots \alpha}{\beta^k}$$

In particular, the important values to remember are :

$$\mathbb{E}X = \frac{\alpha}{\beta} \quad \mathbb{E}X^2 = \frac{(\alpha + 1)\alpha}{\beta^2} \quad \text{Var}(X) = \frac{\alpha}{\beta^2}$$

Proposition 4.4.4: Density Of Sums Of Random Variables

Let X_1 and X_2 be independent random variables with densities $f(x)$ and $g(x)$. Then $X_1 + X_2$ has density

$$h(x) = \int_{-\infty}^{\infty} f(x - y)g(y)dy$$

For those who know some analysis, h is the convolution of f and g : $h = f * g$.

Proof:

We compute the cumulative distribution of $X_1 + X_2$. By definition 1.2.11, the joint distribution of (X_1, X_2) is the product of the distributions of X_1 and X_2 , which have densities f and g , so it has the product density $f(x)g(y)$ on $\mathbb{R}^2$. Computing its integrals as iterated integrals, and exchanging the order of integration in the fourth line, is the Fubini-Tonelli theorem, [4, Theorem 2.37], which this book uses without proof. The third line is theorem 0.4.1 in the inner integral, with $\varphi(z) = z - y$ from $(-\infty, t)$ onto $(-\infty, t - y)$, whose derivative is 1:

$$\begin{aligned} \mathbb{P}(X_1 + X_2 \leq t) &= \int_{\mathbb{R}} \int_{\mathbb{R}} I(x + y \leq t) f(x)g(y) dx dy \\ &= \int_{\mathbb{R}} \left[\int_{-\infty}^{t-y} f(x)g(y) dx \right] dy \\ &= \int_{\mathbb{R}} \left[\int_{-\infty}^t f(z - y)g(y) dz \right] dy \\ &= \int_{-\infty}^t \left[\int_{\mathbb{R}} f(z - y)g(y) dy \right] dz \\ &= \int_{-\infty}^t h(z) dz \end{aligned}$$

For $a < b$, subtracting the identity at $t = a$ from the one at $t = b$ gives $\mathbb{P}(a < X_1 + X_2 \leq b) = \int_a^b h(z) dz$. A single point c has probability at most $\mathbb{P}(c - \eta < X_1 + X_2 \leq c) = \int_{c-\eta}^c h(z) dz$ for every $\eta > 0$, which tends to 0 as $\eta \rightarrow 0^+$ because $\int_{-\infty}^c h(z) dz$ converges. So adding or removing endpoints does not change these probabilities. The same Tonelli computation with 1 in place of $I(x + y \leq t)$ gives $\int_{\mathbb{R}} h(z) dz = (\int f) (\int g) = 1$, which covers the unbounded intervals. Hence $\mathbb{P}(X_1 + X_2 \in A) = \int_A h(z) dz$ for every interval A , and h is a density of $X_1 + X_2$ in the sense of definition 4.1.4, as claimed.

Corollary 4.4.5: Sums of Independent Gamma Variables

Let $\beta > 0$ and $\alpha_1, \dots, \alpha_n > 0$, and let $X_1, \dots, X_n$ be independent random variables with $X_i \sim \Gamma(\alpha_i, \beta)$ for each i . Then

$$X_1 + \dots + X_n \sim \Gamma(\alpha_1 + \dots + \alpha_n, \beta)$$

Proof:

For $\alpha, \beta > 0$ write $f_{\alpha, \beta}(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$ for $x > 0$ and $f_{\alpha, \beta}(x) = 0$ for $x \leq 0$, the density of $\Gamma(\alpha, \beta)$ from definition 4.4.2. The display before that definition, which is theorem 0.4.1 with $\varphi(y) = \beta y$ on $(0, \infty)$, says that for all $c, \beta > 0$

$$\int_0^\infty x^{c-1} e^{-\beta x} dx = \frac{\Gamma(c)}{\beta^c}$$

Put $S_k = X_1 + \dots + X_k$ and $A_k = \alpha_1 + \dots + \alpha_k$. The claim is the case $k = n$ of the statement that S_k has density $f_{A_k, \beta}$, which we prove by induction on k . For $k = 1$ it is the hypothesis on X_1 .

Let $2 \leq k \leq n$ and suppose that S_{k-1} has density $f_{A_{k-1}, \beta}$. The variables S_{k-1} and X_k are independent, in the sense of definition 1.2.11, by [4, Proposition 10.2]: functions of disjoint groups of independent random variables are independent. That proposition is applied to the variables $X_1, \dots, X_k$, which are independent because $X_1, \dots, X_n$ are, and to the continuous, hence Borel measurable, functions $f_1(x_1, \dots, x_{k-1}) = x_1 + \dots + x_{k-1}$ and $f_2(x_k) = x_k$. Its proof uses measure theory, which this book does not develop. So proposition 4.4.4 shows that $S_k = S_{k-1} + X_k$ has density

$$h(x) = \int_{-\infty}^\infty f_{a, \beta}(x-y) f_{b, \beta}(y) dy$$

where $a = A_{k-1}$ and $b = \alpha_k$. The integrand vanishes unless $0 < y < x$, so $h(x) = 0$ for $x \leq 0$. For $x > 0$, since $e^{-\beta(x-y)} e^{-\beta y} = e^{-\beta x}$,

$$h(x) = \frac{\beta^{a+b} e^{-\beta x}}{\Gamma(a)\Gamma(b)} \int_0^x (x-y)^{a-1} y^{b-1} dy = \frac{\beta^{a+b} \kappa}{\Gamma(a)\Gamma(b)} x^{a+b-1} e^{-\beta x}$$

where $\kappa = \int_0^1 (1-u)^{a-1} u^{b-1} du$, an integral with continuous positive integrand on $(0, 1)$ that is improper at both ends. It converges. On $(0, \frac{1}{2}]$ the integrand is at most $\max(1, 2^{1-a}) u^{b-1}$, and $\int_\delta^{1/2} u^{b-1} du = (2^{-b} - \delta^b)/b \leq 2^{-b}/b$ for $0 < \delta < \frac{1}{2}$, by [2, The Fundamental Theorem of Calculus, Second Form] and [2, The Derivative of a Real Power]. So part (4) of [2, Convergence Tests for Improper Integrals] gives convergence at 0. For $0 < \delta < \frac{1}{2}$, [2, Substitution] with $u = 1 - v$ gives $\int_{1/2}^{1-\delta} (1-u)^{a-1} u^{b-1} du = \int_\delta^{1/2} v^{a-1} (1-v)^{b-1} dv$. Here the integrand is at most $\max(1, 2^{1-b}) v^{a-1}$ and $\int_\delta^{1/2} v^{a-1} dv \leq 2^{-a}/a$, so the right-hand side converges as $\delta \rightarrow 0^+$ by the same part of the convergence tests, and the integral converges at 1 as well. The second equality in the display above is then theorem 0.4.1 applied to $u \mapsto (1-u)^{a-1} u^{b-1}$ on $(0, 1)$ with $\varphi(y) = y/x$ from $(0, x)$ onto $(0, 1)$, because $(1-y/x)^{a-1} (y/x)^{b-1} \cdot \frac{1}{x} = x^{-(a+b-1)} (x-y)^{a-1} y^{b-1}$. Since h is a density of S_k , definition 4.1.4 with the event $\mathbb{R}$ gives $\int_{\mathbb{R}} h = \mathbb{P}(S_k \in \mathbb{R}) = 1$. By the display above and the first display of the proof, $\int_{\mathbb{R}} h = \frac{\beta^{a+b} \kappa}{\Gamma(a)\Gamma(b)} \cdot \frac{\Gamma(a+b)}{\beta^{a+b}}$. Hence $\frac{\kappa}{\Gamma(a)\Gamma(b)} = \frac{1}{\Gamma(a+b)}$, so $h = f_{a+b, \beta} = f_{A_k, \beta}$, completing the induction.

5

Markov Chains

Definition 5.0.1: Random Variable Sequence

Let $\{X_i\}$ be some collection of random variables. Then if we put them in an order, we will say they form a *sequence*.

When $n = 1$ or $\mathbb{P}(X_1 = x_1, \dots, X_{n-1} = x_{n-1}) > 0$, proposition 1.5.3 applied to the events $\{X_k = x_k\}$ gives

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \prod_{k=0}^{n-1} \mathbb{P}(X_{k+1} = x_{k+1} | X_1 = x_1, \dots, X_k = x_k)$$

where for $k = 0$ we have $\mathbb{P}(X_1 = x_1)$.

Definition 5.0.2: Markov Chain

Let $X_1, \dots, X_n, \dots$ be a sequence of random variables. Then this sequence is called a *Markov chain* if, for every $k \geq 1$ and all $x_1, \dots, x_{k+1}$ with $\mathbb{P}(X_1 = x_1, \dots, X_k = x_k) > 0$,

$$\mathbb{P}(X_{k+1} = x_{k+1} | X_1 = x_1, \dots, X_k = x_k) = \mathbb{P}(X_{k+1} = x_{k+1} | X_k = x_k) \quad (5.1)$$

that is, the conditional probability only depends on the last random variable.

The property demonstrated in equation 5.1 is sometimes called the *Markov property* or *memoryless property*. In this chapter, we will limit ourselves to *finite state* Markov chains, that is Markov chains where all the X_i 's take values from a finite set:

$$X_i \in S = \{s_1, \dots, s_m\}$$

A

Glossary

A.1 Distributions

1. Bernoulli Distribution: $X \sim B(p)$, $\mathbb{P}(X = 0) = (1 - p)$, $\mathbb{P}(X = 1) = p$. A fair Bernoulli distribution has $p = 1/2$

$$\mathbb{E}X = p \quad \mathbb{E}X^2 = p \quad \text{Var}(X) = p(1 - p) \quad \mathbb{E}X^k = p \quad \mathbb{E}e^{tX} = (1 - p) + pe^t$$

2. Rademacher distribution: $X \sim \epsilon$, $\mathbb{P}(X = -1) = \mathbb{P}(X = 1) = 1/2$

$$\mathbb{E}X = 0 \quad \mathbb{E}X^2 = 1 \quad \text{Var}(X) = 1 \quad \mathbb{E}X^k = \begin{cases} 1 & k \text{ is even} \\ 0 & k \text{ is odd} \end{cases} \quad \mathbb{E}e^{tX} = \cosh(t)$$

3. Binomial Distribution: $X \sim B(n, p)$. $\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$

$$\mathbb{E}X = np \quad \mathbb{E}X^2 = np(1 - p) + n^2 p^2 \quad \text{Var}(X) = np(1 - p)$$

$$\mathbb{E}X^k = \sum_{i=1}^k \left\{ \begin{matrix} k \\ i \end{matrix} \right\} n^i p^i \quad \mathbb{E}e^{tX} = ((1 - p) + pe^t)^n$$

where $\left\{ \begin{matrix} k \\ i \end{matrix} \right\}$ is the *stirling number* and $n^{\underline{i}} = n(n-1) \cdots (n-i+1)$ Note: $X = \sum_{i=1}^n X_i$ where the X_i 's are i.i.d. Bernoulli random variables, giving us easy $\mathbb{E}X$ and $\text{Var}(X)$

4. Geometric Distribution: $X \sim \text{Geo}(p)$ $\mathbb{P}(X = n) = (1 - p)^{n-1}p$

$$\begin{aligned}\mathbb{E}X &= \frac{1}{p} & \mathbb{E}X^2 &= \frac{2-p}{p^2} \\ \text{Var}(X) &= \frac{1-p}{p^2} & \mathbb{E}e^{tX} &= \frac{pe^t}{1-(1-p)e^t} \\ \mathbb{E}X^n &= \sum_{k=1}^{\infty} (1-p)^{k-1}p \cdot k^n\end{aligned}$$

where $t < -\ln(1-p)$ for the M.G.F. Note that higher moments of the geometric distribution do not seem to admit a nice simplification.

5. Poisson Distribution: $X \sim \text{Poiss}(\lambda)$ $\mathbb{P}(X = k) = \frac{\lambda^k}{k!}e^{-\lambda}$

$$\mathbb{E}X = \lambda \quad \mathbb{E}X^2 = \lambda + \lambda^2 \quad \text{Var}(X) = \lambda \quad \mathbb{E}X^k = \sum_{i=1}^k \lambda^i \left\{ \begin{matrix} k \\ i \end{matrix} \right\} \quad \mathbb{E}e^{tX} = \exp(\lambda(e^t - 1))$$

Note that $\lim_{n \rightarrow \infty} B(n, \frac{\lambda}{n}) = \text{Poiss}(\lambda)$, that the sum of two independent Poisson with λ_1 and λ_2 is Poisson with $\lambda_1 + \lambda_2$

6. Gaussian Distribution: Given $p(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}$ and $\mathbb{P}(A) = \int_A p(x)dx = \gamma(A)$ then $X \sim g$, $g(x) = x$, then:

$$\mathbb{P}(g \in A) = \gamma(\{x \mid x \in A\}) = \gamma(A)$$

Other notation: $X \sim N(0, 1)$. More generally, if $P_X(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$, then $X \sim N(\mu, \sigma^2)$

$$\mathbb{P}(X \leq t) = \frac{1}{2} \left(1 + \text{erf} \left(\frac{t-\mu}{\sigma\sqrt{2}} \right) \right) \quad \mathbb{E}X = \mu \quad \text{Var}(X) = \sigma^2 \quad \mathbb{E}e^{tX} = \exp \left(\mu t + \frac{\sigma^2 t^2}{2} \right)$$

where $\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$

7. Uniform Distribution: Let $p_X(x) = \frac{1}{b-a}I(a \leq x \leq b)$ be the density. Then $\mathbb{P}(X \in A)$ is called the uniform distribution.

$$\mathbb{P}(X \leq t) = \begin{cases} 0 & t < a \\ \frac{t-a}{b-a} & a \leq t \leq b \\ 1 & t > b \end{cases} \quad \mathbb{E}X = \frac{1}{2}(a+b) \quad \mathbb{E}X^2 = \frac{b^3 - a^3}{3b - 3a} \quad \text{Var}(X) = \frac{1}{12}(b-a)^2$$

$$\mathbb{E}X^n = \frac{b^{n+1} - a^{n+1}}{(n+1)(b-a)} = \frac{1}{n+1} \sum_{k=0}^n a^k b^{n-k} \quad \mathbb{E}e^{tX} = \begin{cases} \frac{e^{tb} - e^{ta}}{t(b-a)} & t \neq 0 \\ 1 & t = 0 \end{cases}$$

When $a = -b$, the M.G.F. reduces to $\mathbb{E}e^{tX} = \frac{\sinh(bt)}{bt}$ for $t \neq 0$.

8. Exponential Distribution: Let $p_X(x) = \lambda e^{-\lambda x}I(x \geq 0)$ be the density. Then $\mathbb{P}(X \in A)$ is called the exponential distribution

$$\mathbb{P}(X \leq t) = 1 - e^{-\lambda t} \quad (t \geq 0) \quad \mathbb{E}X = \frac{1}{\lambda} \quad \mathbb{E}X^2 = \frac{2}{\lambda^2} \quad \text{Var}(X) = \frac{1}{\lambda^2} \quad \mathbb{E}X^n = \frac{n!}{\lambda^n}$$

9. Gamma Distribution: Let $f_{\alpha,\beta}(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} I(x > 0)$ be the density. Then $\mathbb{P}(X \in A)$ is called the Gamma distribution

$$\begin{aligned} \mathbb{E}X &= \frac{\alpha}{\beta} & \mathbb{E}X^2 &= \frac{(\alpha+1)\alpha}{\beta^2} & \text{Var}(X) &= \frac{\alpha}{\beta^2} \\ \mathbb{E}X^k &= \frac{(\alpha+k-1)\cdots\alpha}{\beta^k} & \mathbb{E}e^{tX} &= \left(1 - \frac{t}{\beta}\right)^{-\alpha} \quad (t < \beta) \end{aligned}$$

Index

- Bernoulli Distribution, 22
 - Product, 23
 - Random Variable, 22
- Bernstein Polynomials, 58
- Big-O Notation, v
- Binomial Distribution, 25

- Chain Rule of Probability, 33
- Conditional Distribution, 41
- Conditional Probability, 33
- Conditional Random Variable, 35
- Covariance, 55
- Covariance Matrix, 101
- Cumulative Distribution Function, 93

- Density, 92

- Expected Value
 - Countable, 15
 - Finite, 6
 - Gaussian, 94
 - Gaussian on $\mathbb{R}^n$, 99

- Gamma Distribution, 111
- Gamma Function, 111
- Gaussian Distribution, 89
 - on $\mathbb{R}^n$, 97
- Geometric Distribution, 26

- Image Measure, 12
- Independence
 - Discrete, 8
 - General, 9
 - of Events, 33
 - of Multiple Events, 34
 - of Random Variables, 34

- Jacobian Matrix, xi
- Joint Distribution, 42
 - Measure, 8

- Kullback-Leibler Divergence, 72

- Lindeberg Replacement, 105

- Marginal Distribution, 42
- Markov Chain, 115
- Mean, 6
- Multinomial Coefficient, iii

- Nice Function, 91

- Pinsker's Inequality
 - Bernoulli, 76
- Poisson Distribution, 27
- Probability Measure, 3
- Product Space, 35

- Random Variable, 5
 - Sequence, 115
- Random Vector, 23

- Smooth Indicator, 104

- Tail Probability, 19

- Variance, 53

Bibliography

- [1] B. Bollobás. “The chromatic number of random graphs”. In: *Combinatorica* 8.1 (Mar. 1988), pp. 49–55. DOI: [10.1007/bf02122551](https://doi.org/10.1007/bf02122551). URL: <https://doi.org/10.1007/bf02122551>.
- [2] Nathanael Chwojko-Srawley. *Everything You Need To Know About Elementary Analysis*. 2024. URL: <https://nathanaelsrawley.com/assets/pdfs/books/EYNTKA-elementary-analysis.pdf>.
- [3] Rick Durrett. *Probability: Theory and Examples*. 5th ed. Cambridge Series in Statistical and Probabilistic Mathematics 49. Cambridge University Press, Apr. 5, 2019. ISBN: 9781108591034. DOI: [10.1017/9781108591034](https://doi.org/10.1017/9781108591034). URL: <https://doi.org/10.1017/9781108591034>.
- [4] Gerald B. Folland. *Real Analysis: Modern Techniques and Their Applications*. 2nd ed. Wiley, 1999. ISBN: 978-0-471-31716-6.
- [5] Walter Rudin. *Real and Complex Analysis*. 3rd ed. New York: McGraw-Hill, 1987. ISBN: 0-07-054234-1.